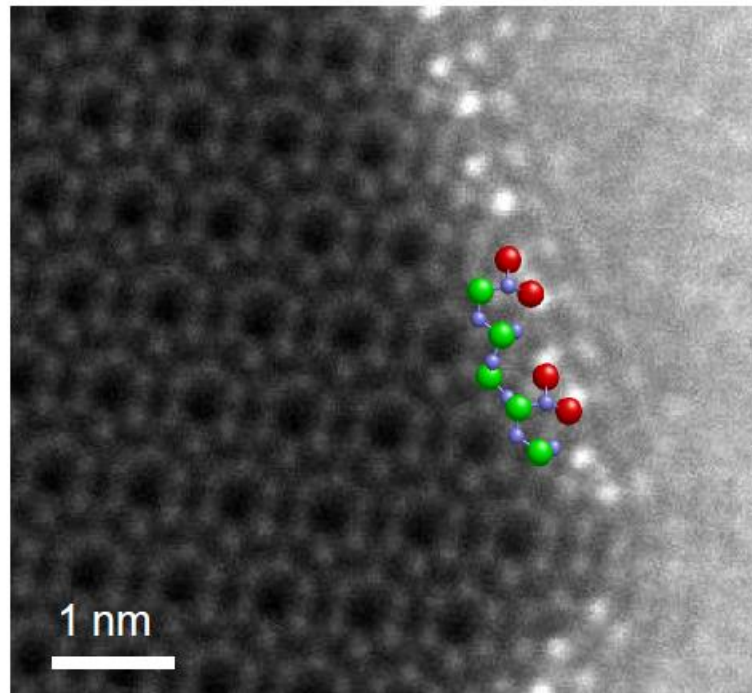
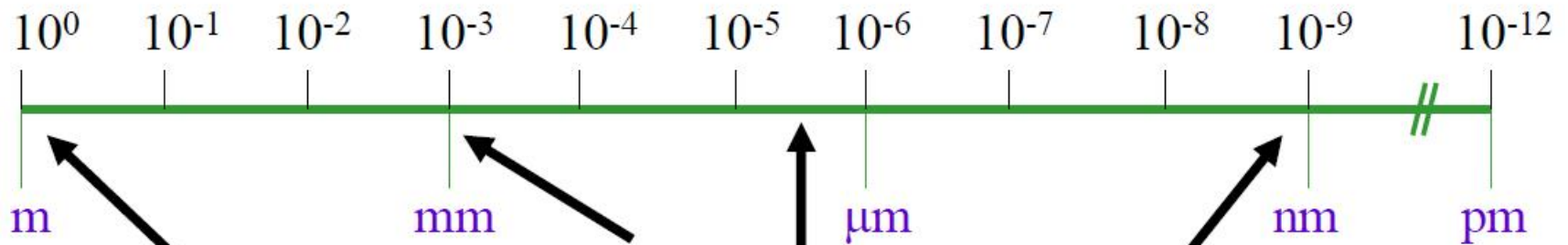
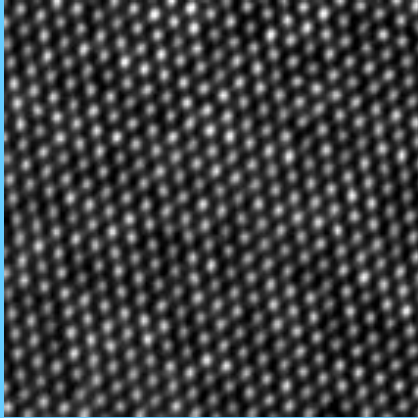


Transmission Electron Microscopy (TEM)

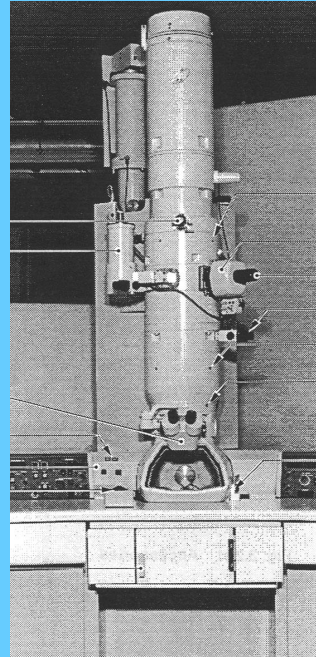
Length-scales



Transmission Electron Microscopy (TEM)



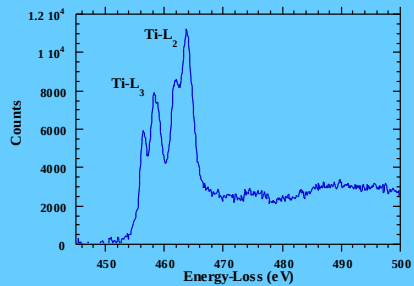
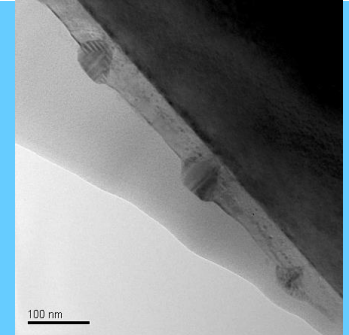
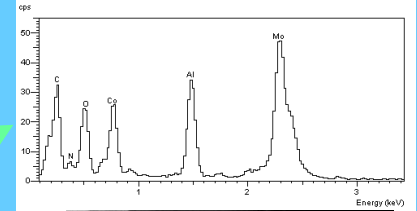
HRTEM



EDXS

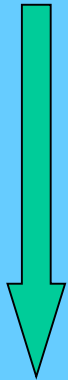
Imaging

Diffraction



Why so many terminology !

TEM
Bright-field
Dark-field
SAED
CBED



Imaging
Morphology
Structure
Defect

AEM
EDXS
EELS
Image Filter
EDX mapping



Composition
Chemical state
Spatial distribution

HRTEM



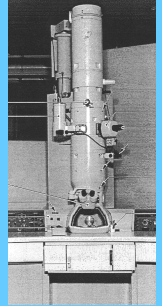
Structural image

STEM
ADF
HAAD



Image
Z-contrast

Transmission Electron Microscopy



Point of history

Principle----contrast mechanisms

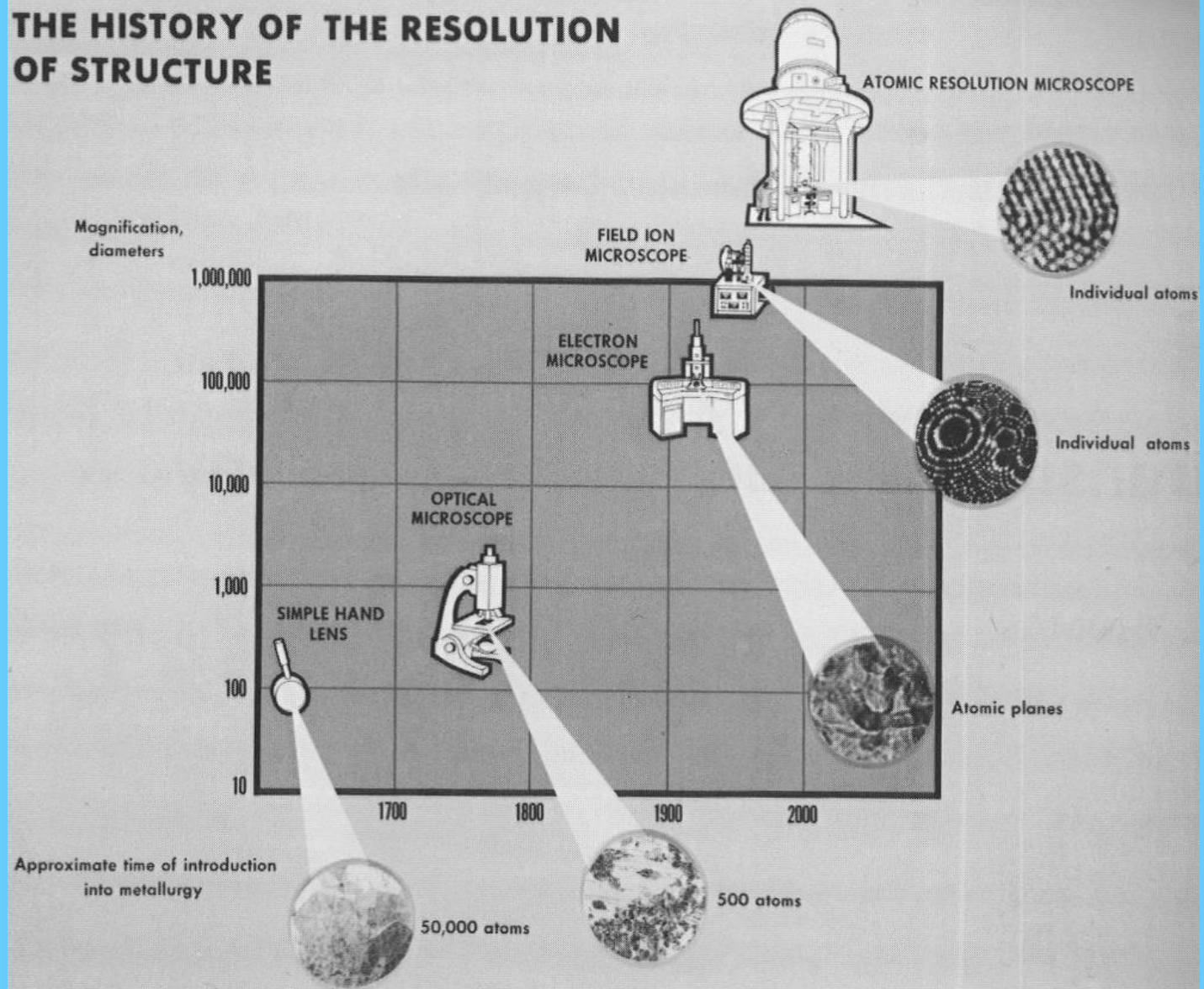
Components

Specimen preparations

Working examples

Summary

THE HISTORY OF THE RESOLUTION OF STRUCTURE

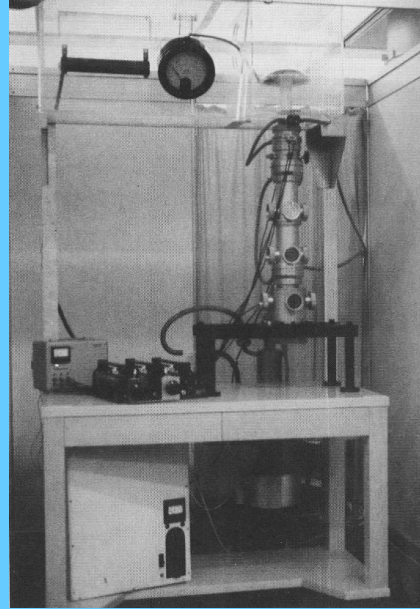


(Courtesy G. Thomas, in Image of Materials, Oxford Press, 1991)

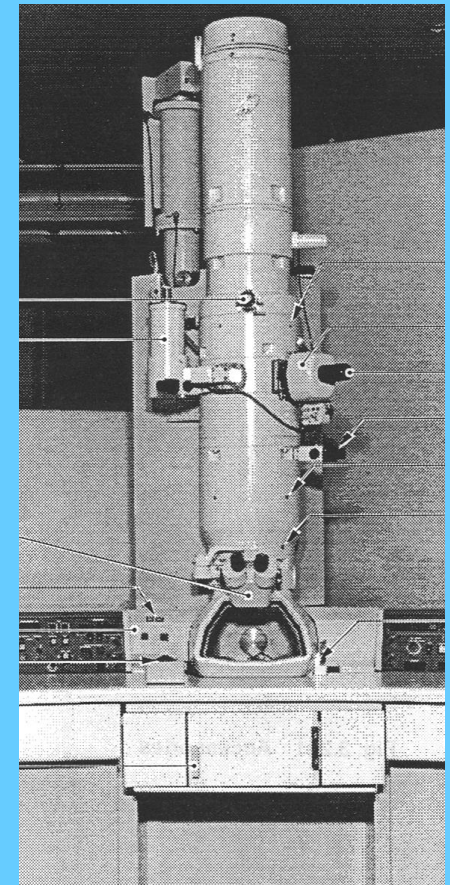


~ \$ 500

Optical microscope



**The 1932 operational microscope built by von
Ruska and Colleagues, Berlin.**



~ \$ 1,000,000

TEM

Nobel Prize to Ernst von Ruska (TEM) and Binnig and Rohrer (for STM)(1986)

Resolution for using light



Rayleigh's Criterion

$$R = 0.61\lambda/\mu\sin\alpha$$

λ :the wavelength of light

μ :the refractive index

α : the semi-angle that the microscope aperture subtends at the specimen

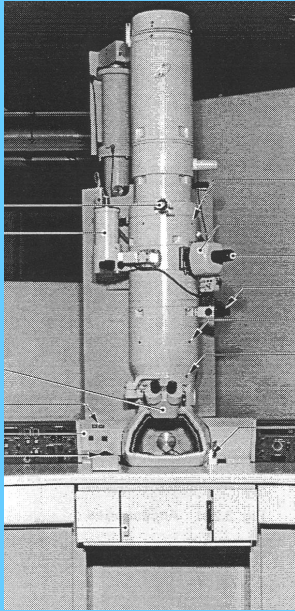
Green light: $\lambda \sim 400 \text{ nm}$

$\mu \sim 1.7$ (oil immersion)

$\sin\alpha \sim 1$ (make it large)

$$R \sim 150 \text{ nm}$$

Resolution for Using Electron



de Broglie wavelength:

$$\lambda = h/p$$

p : electron momentum

h : Planck's constant

$$\lambda = \frac{h}{[2m_0eV(1 + eV/2m_0c^2)]^{1/2}}$$

Rayleigh's Criterion

$$R = 0.61\lambda/\mu\sin\alpha$$

200 kV: $\lambda \sim 0.0025$ nm

$\mu \sim 1$ (vacuum)

$\sin\alpha \sim \alpha \sim 0.1$ radians

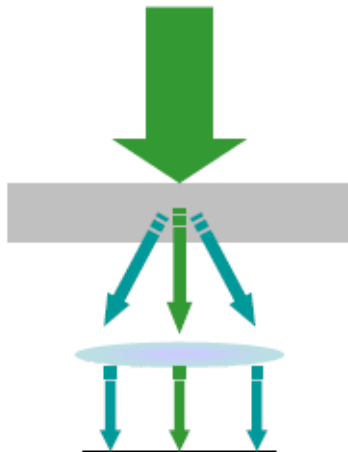
$R \sim 0.02$ nm, 1/10 of bond length

Unrealistic! ??

TEM vs. SEM

Transmission EM

- Projection view
- Volume sensitive
- Diffraction effects
- Higher spatial resolution
- Analytical use (STEM)

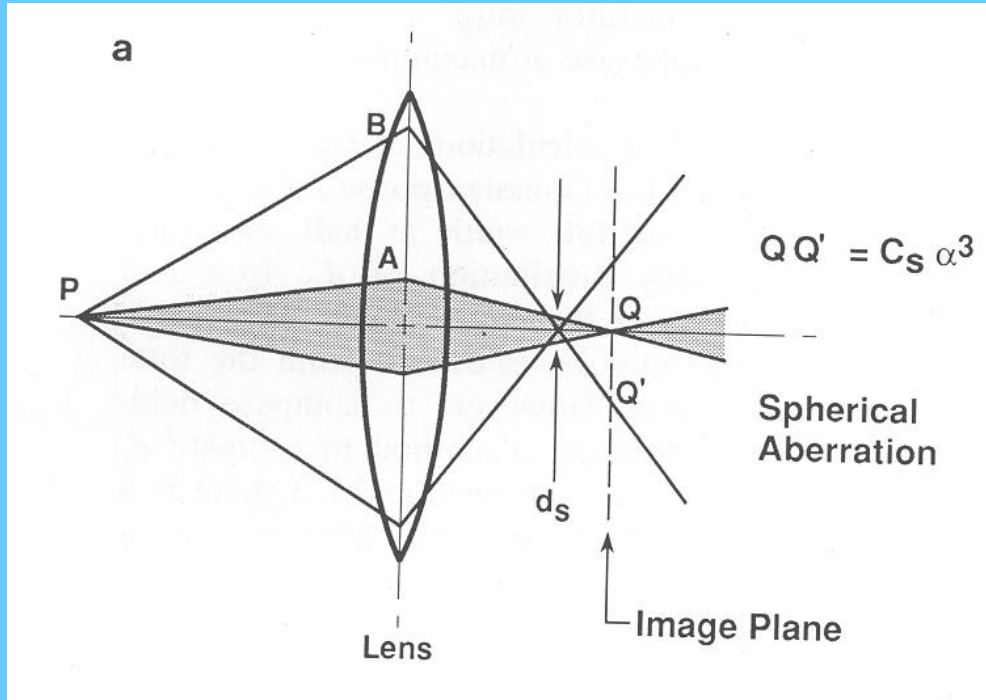


Scanning EM

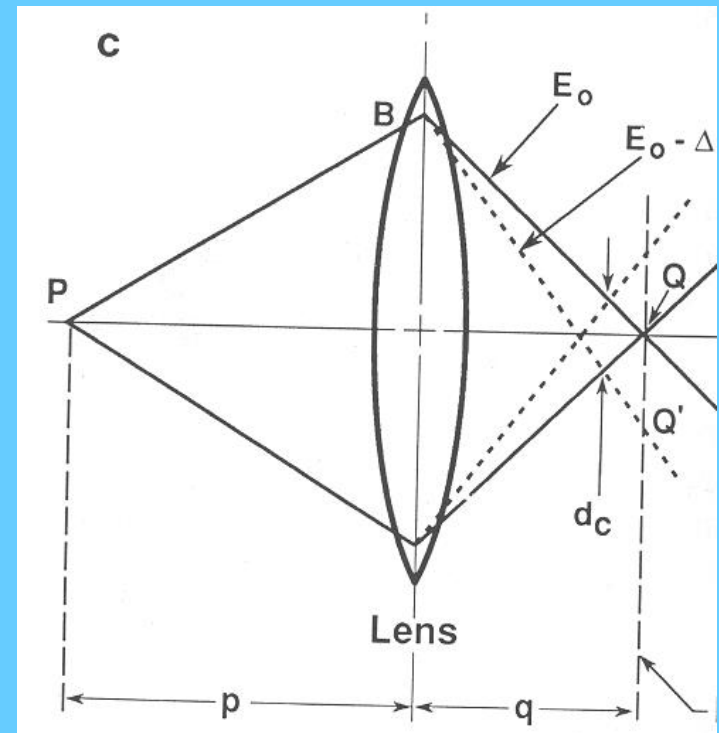
- Topography effects
- backscattering
- Fewer optical elements
- Ease of use
- Analytical use (scanning mode)



Resolution is limited by lens aberrations



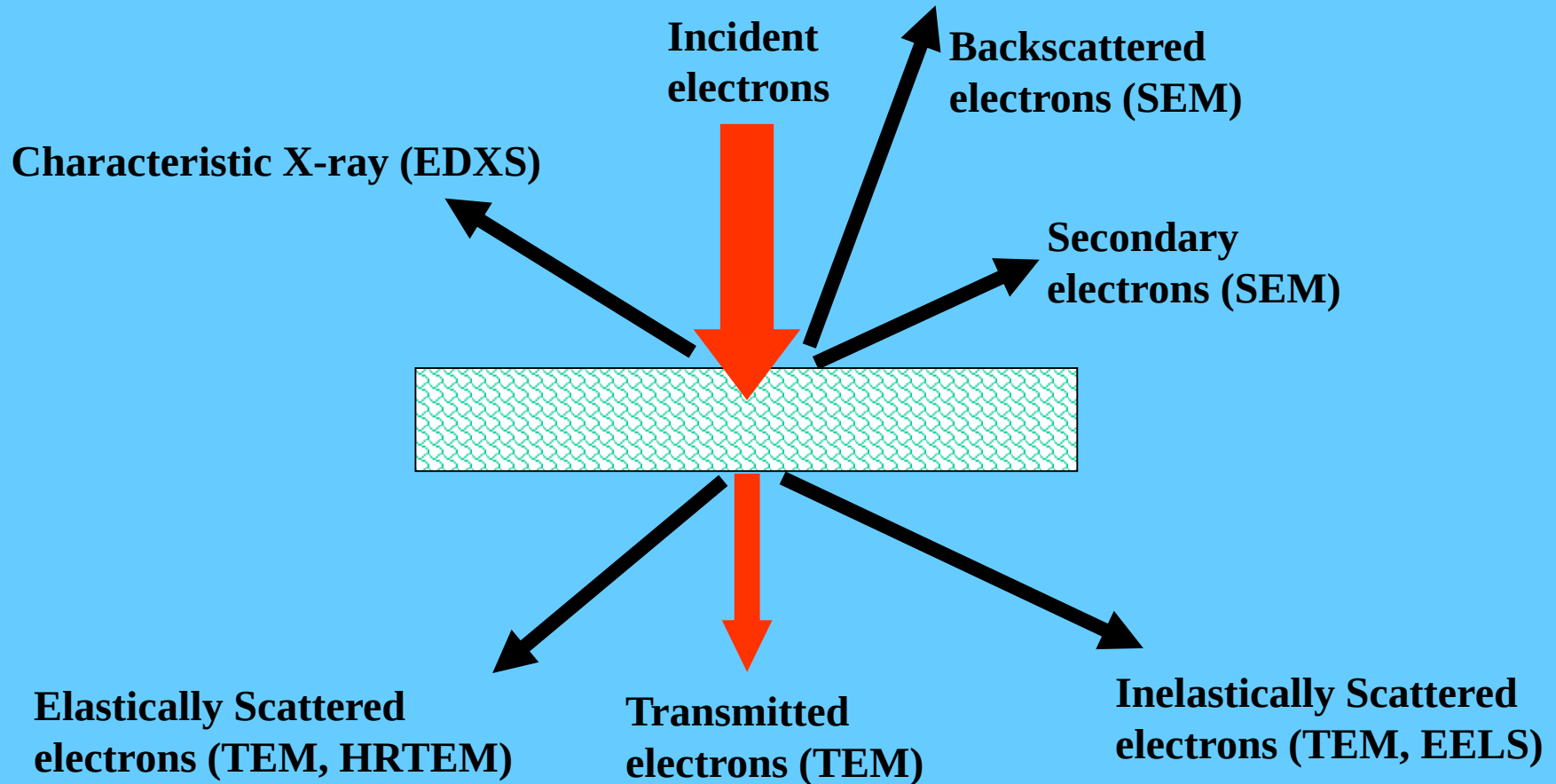
Spherical Aberration



Chromatic Aberration

Highest resolution obtained to date is 0.086 nm (example)

Interaction of electron with specimen

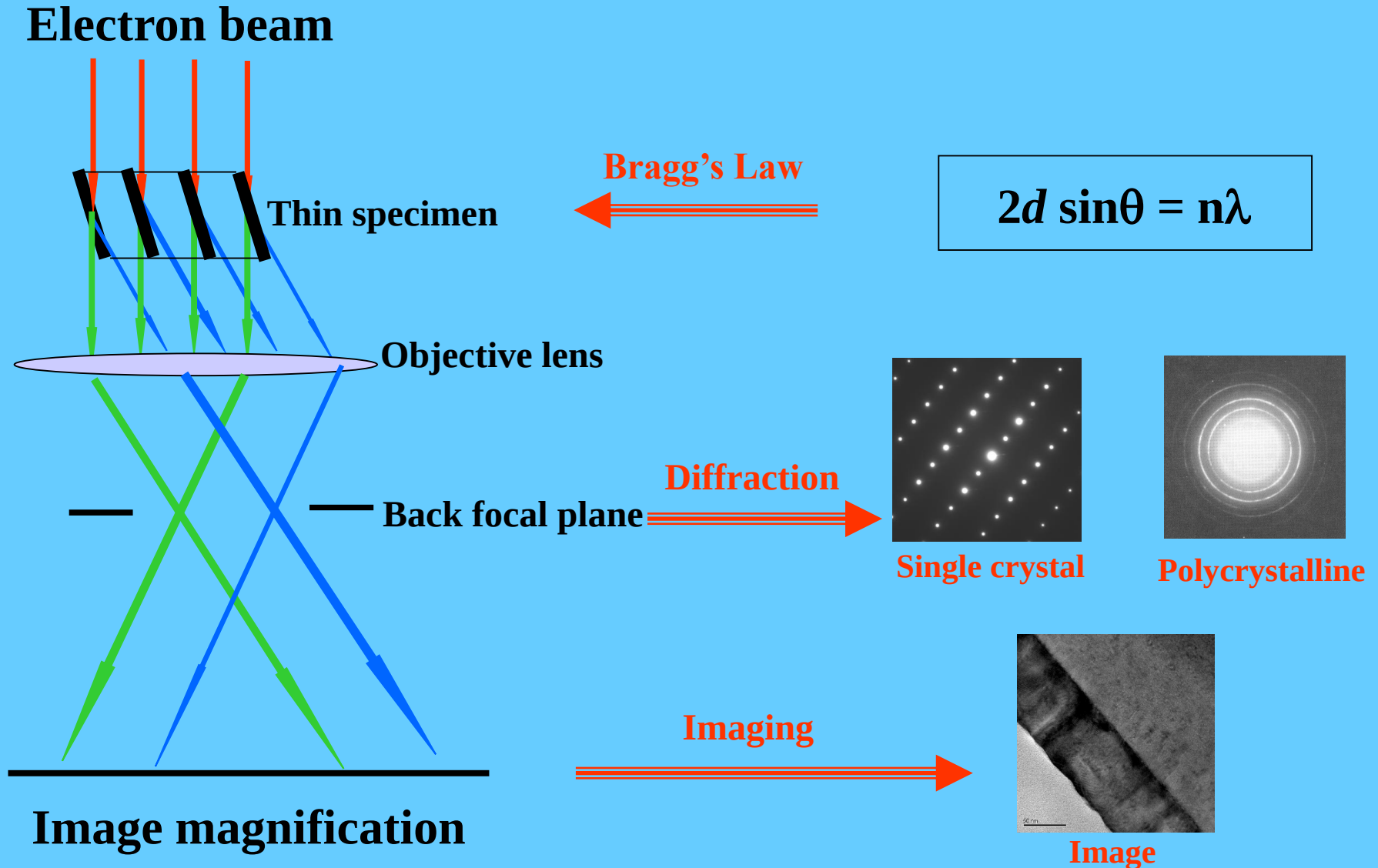


**Specimen must be transparent for electron
ranging from atomic size to 1 μm**

TEM vs. SEM



Electron Diffraction and Imaging



Limitations of TEM

- Charging of the sample
→ increase conductivity
- Beam damaging (80-3,000 keV electrons)
→ cooling of the sample
- Specimen preparation (electron transparency)
→ specimen thickness $\approx$ 10-500 nm
- Projection view
- Environment (mechanical, magnetic, electrical)

Beam damage

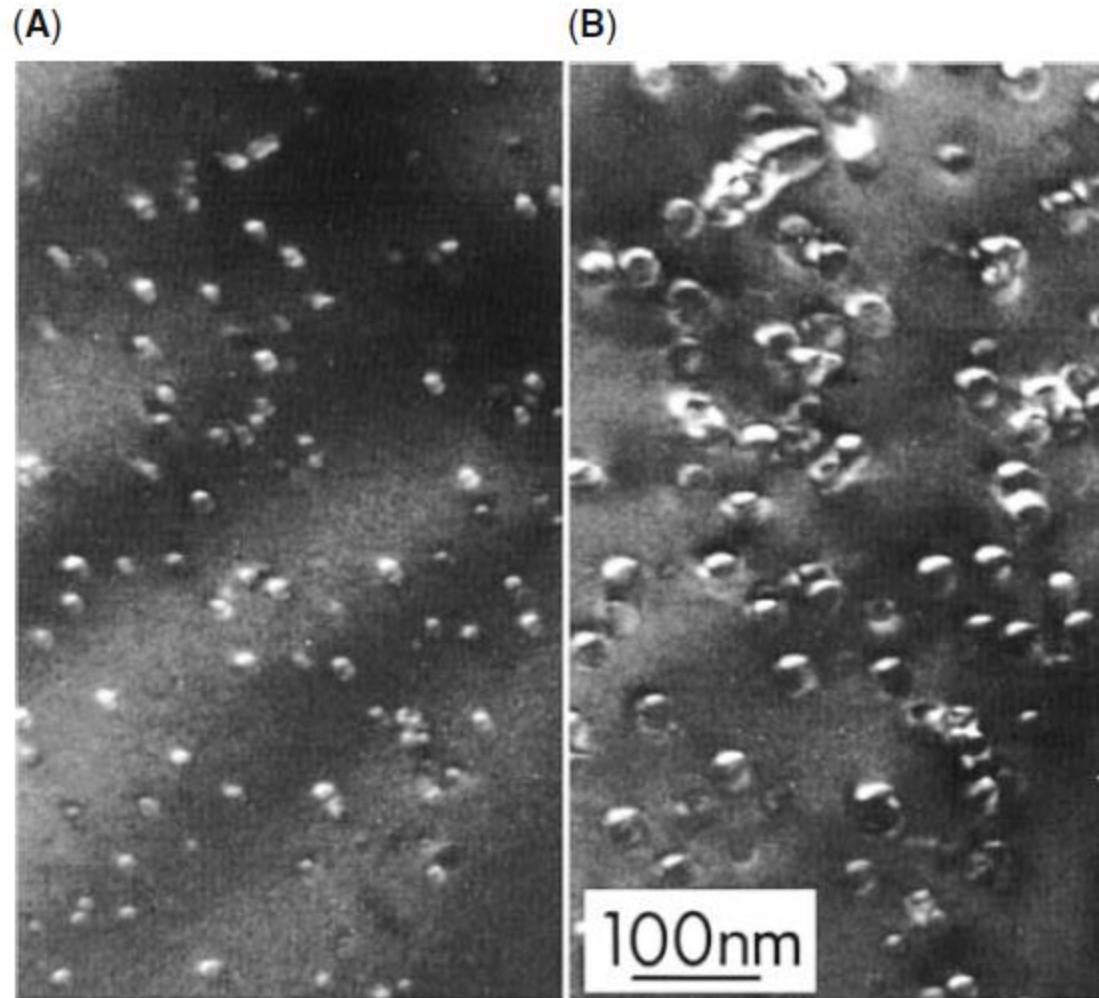
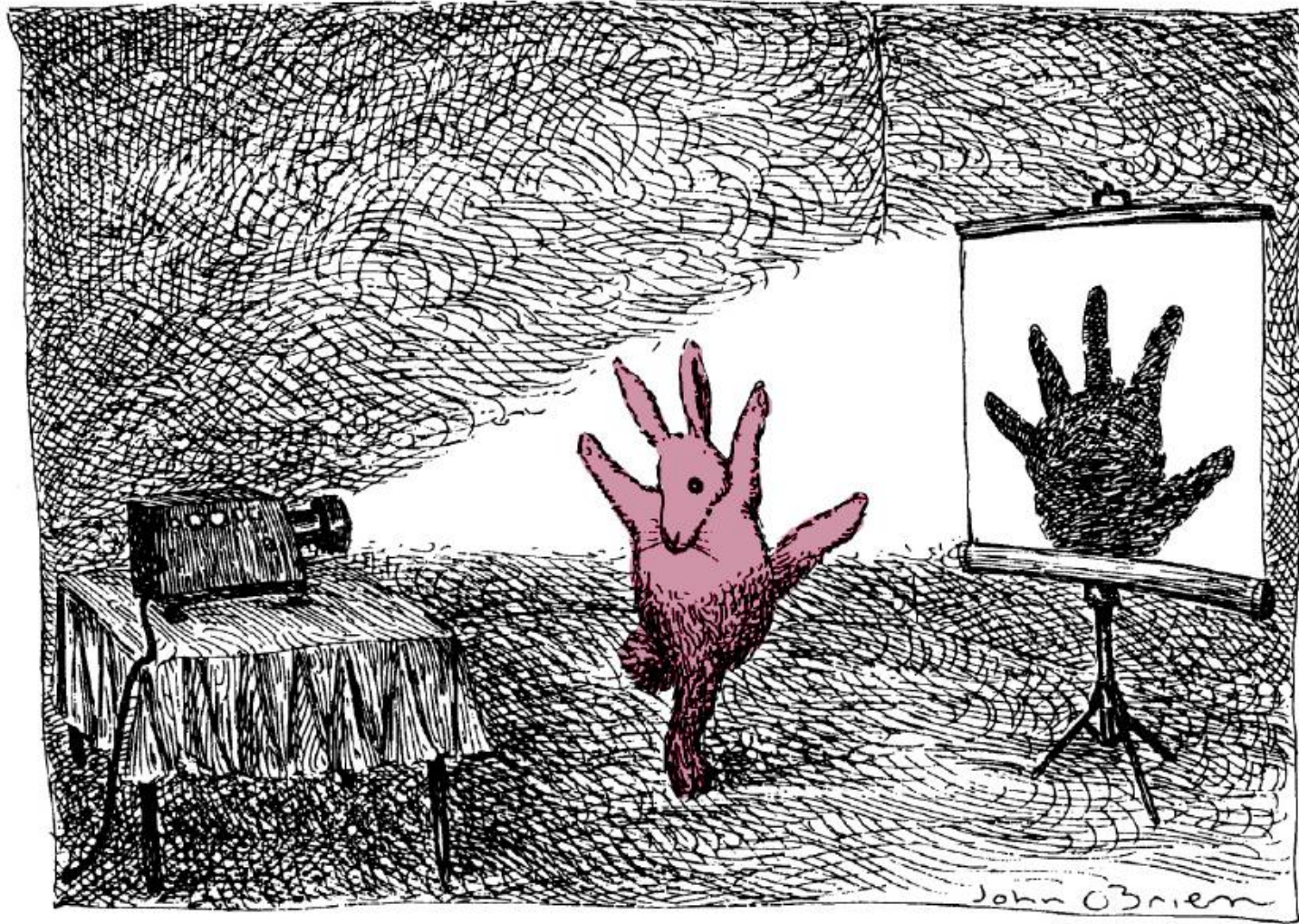


FIGURE 1.8. Beam damage (bright bubble-like regions) in quartz after bombardment with 125 keV electrons. With increasing time from (A) to (B) the damaged regions increase in size.

Limitations of TEM

- Charging of the sample
→ increase conductivity
- Beam damaging (80-3,000 keV electrons)
→ cooling of the sample
- Specimen preparation (electron transparency)
→ specimen thickness $\approx$ 10-500 nm
- Projection view
- Environment (mechanical, magnetic, electrical)

Projection view



A single projection image is plainly insufficient to infer the structure of an object.
Drawing by John O'Brien; © 1991 The New Yorker Magazine

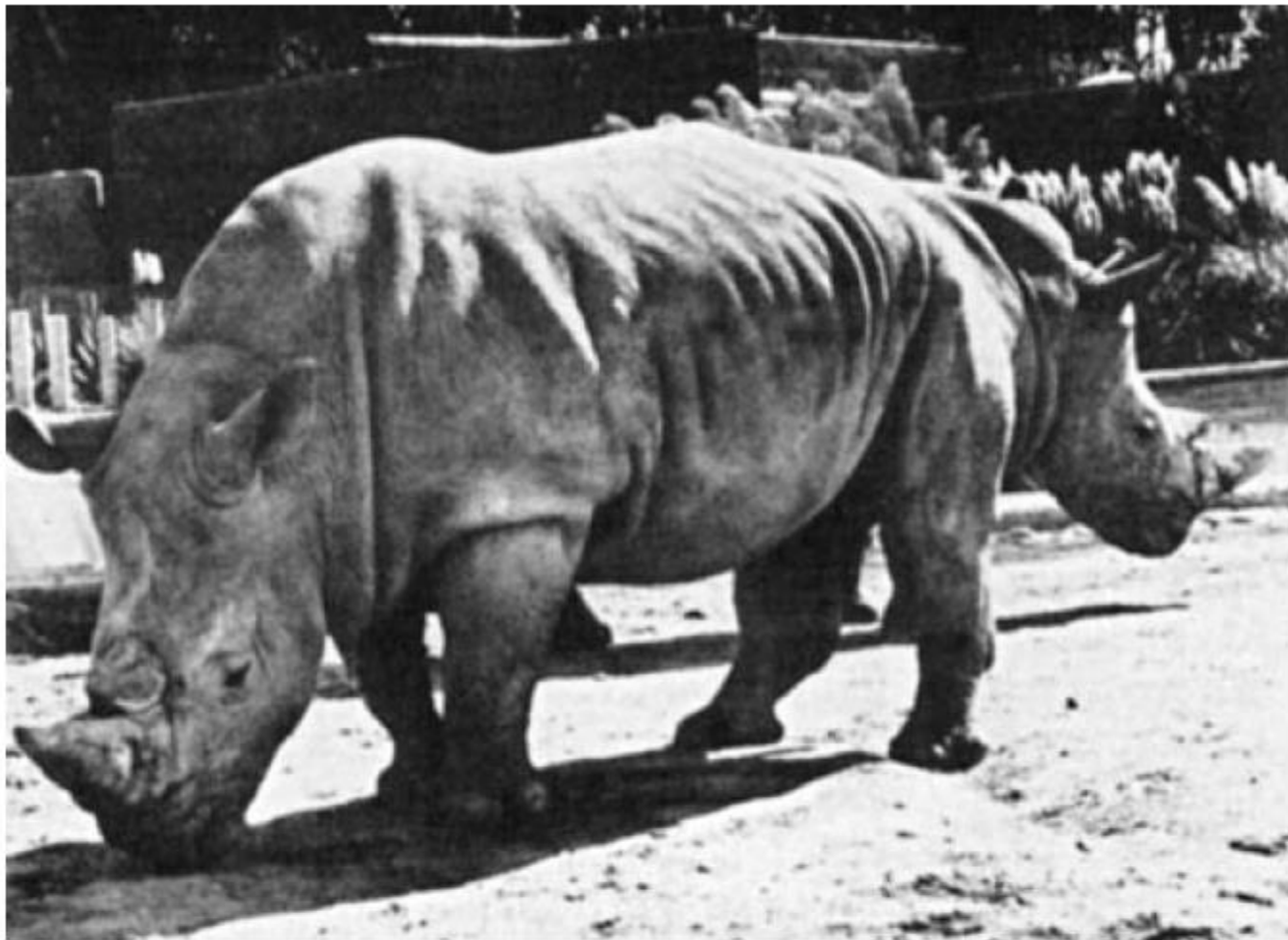
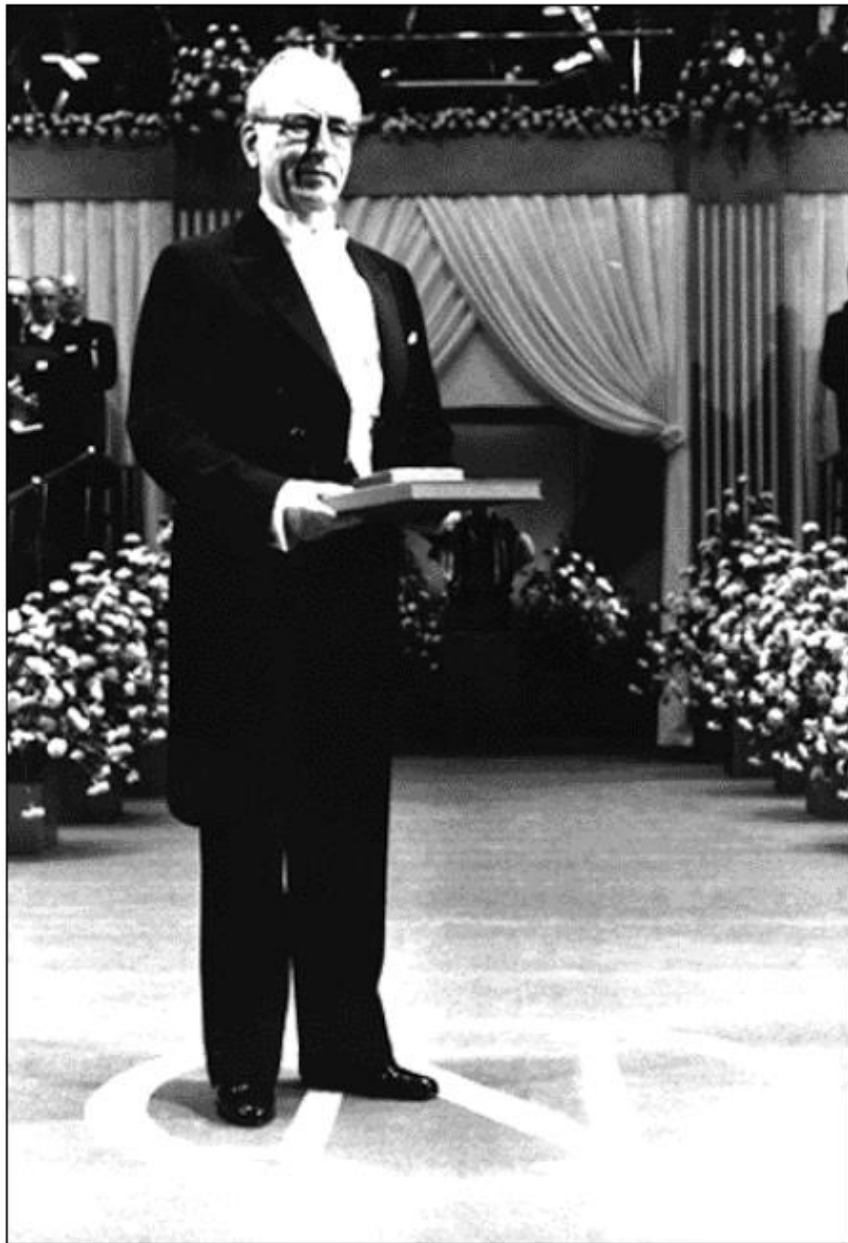


FIGURE 1.7. Photograph of two rhinos taken so that, in projection, they appear as one two-headed beast. Such projection artifacts in reflected-light images are easily discernible to the human eye but similar artifacts in TEM images are easily mistaken for ‘real’ features.





- Nobel prize in Physics
Dec 10, 1986



First commercial Instruments

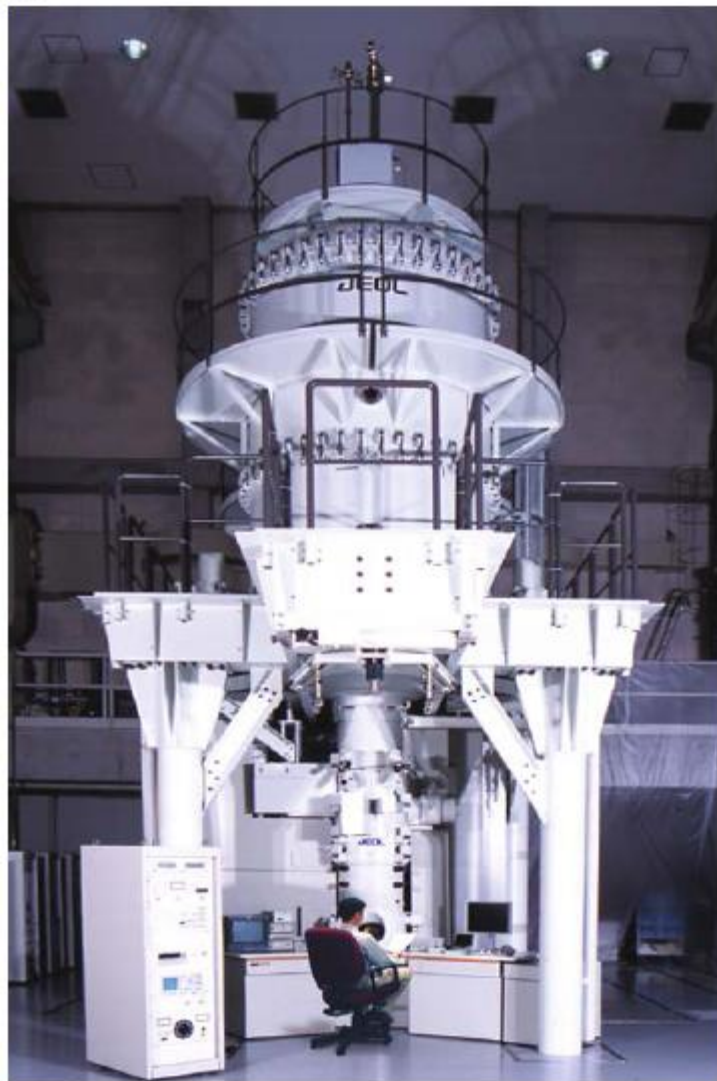


- B. v. Borries, E. Ruska,
Ein Uebermikroskop fuer
Forschungsindustrie
(An Ultramicroscope for
industrial research.)
Naturwissenschaften 27, 577-582
(1939)

State-of-the art instruments (1)



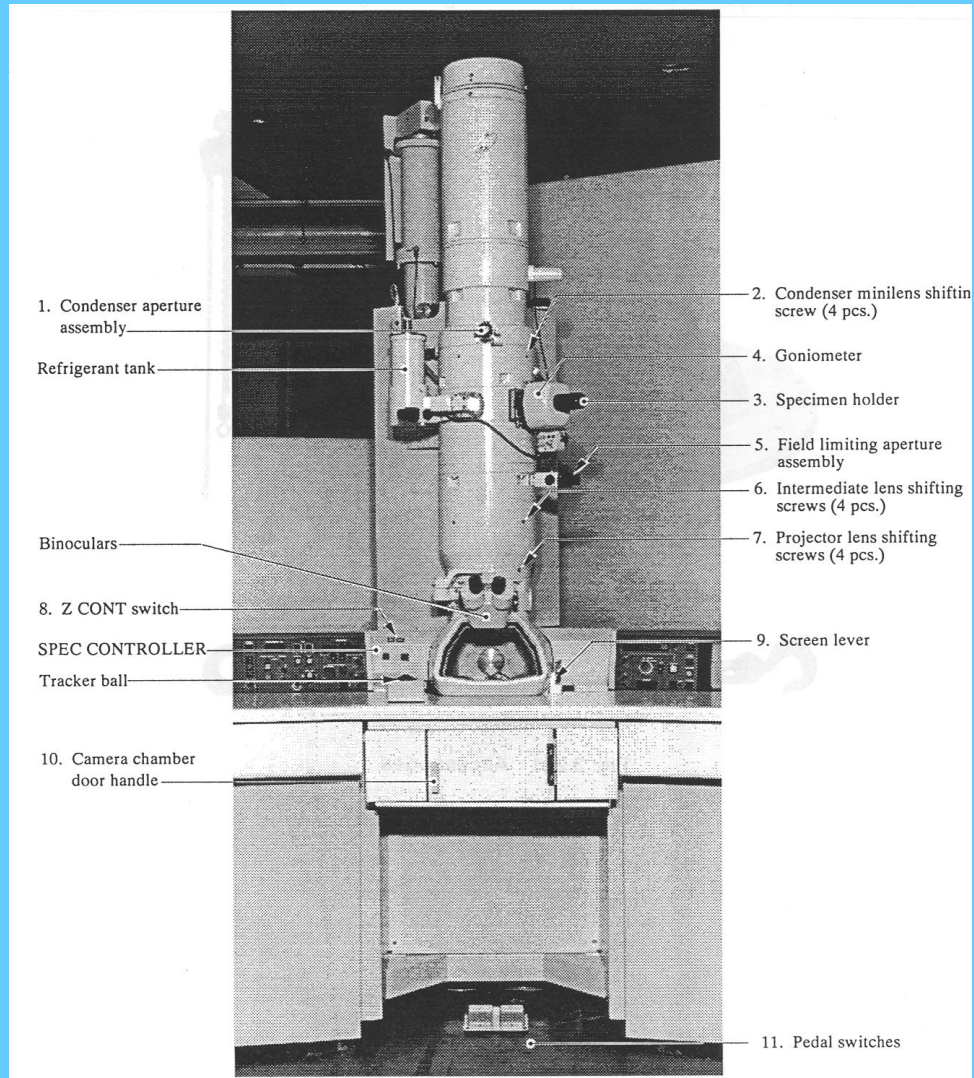
(A)



(B)

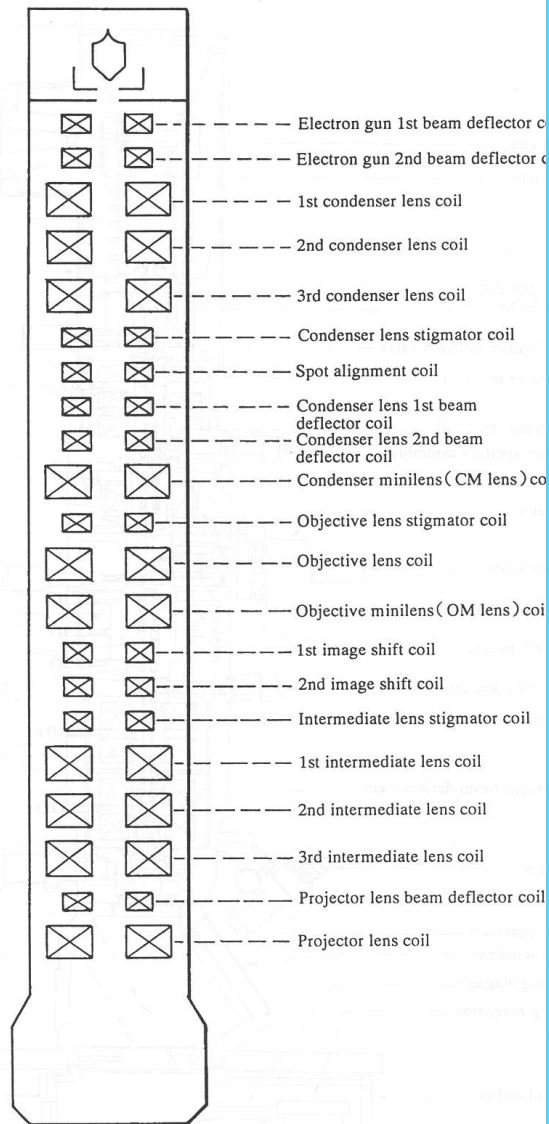
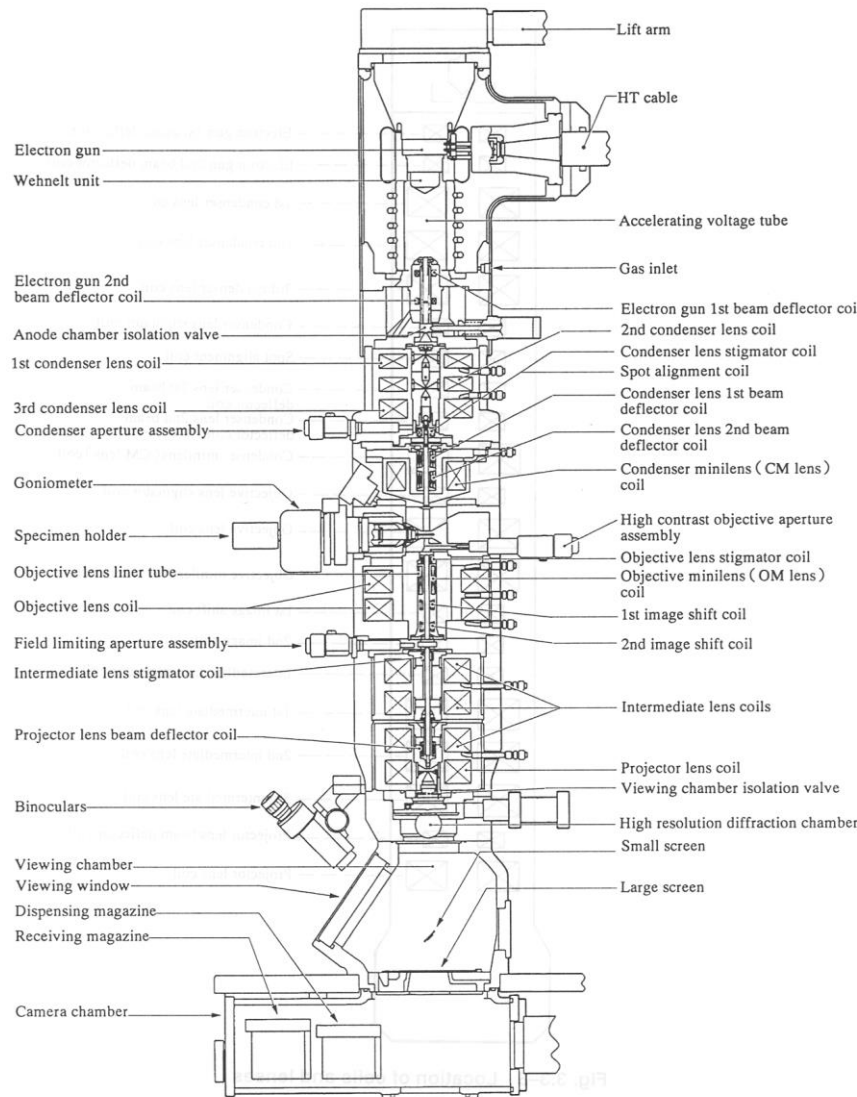


A real microscope: JEOL 2010

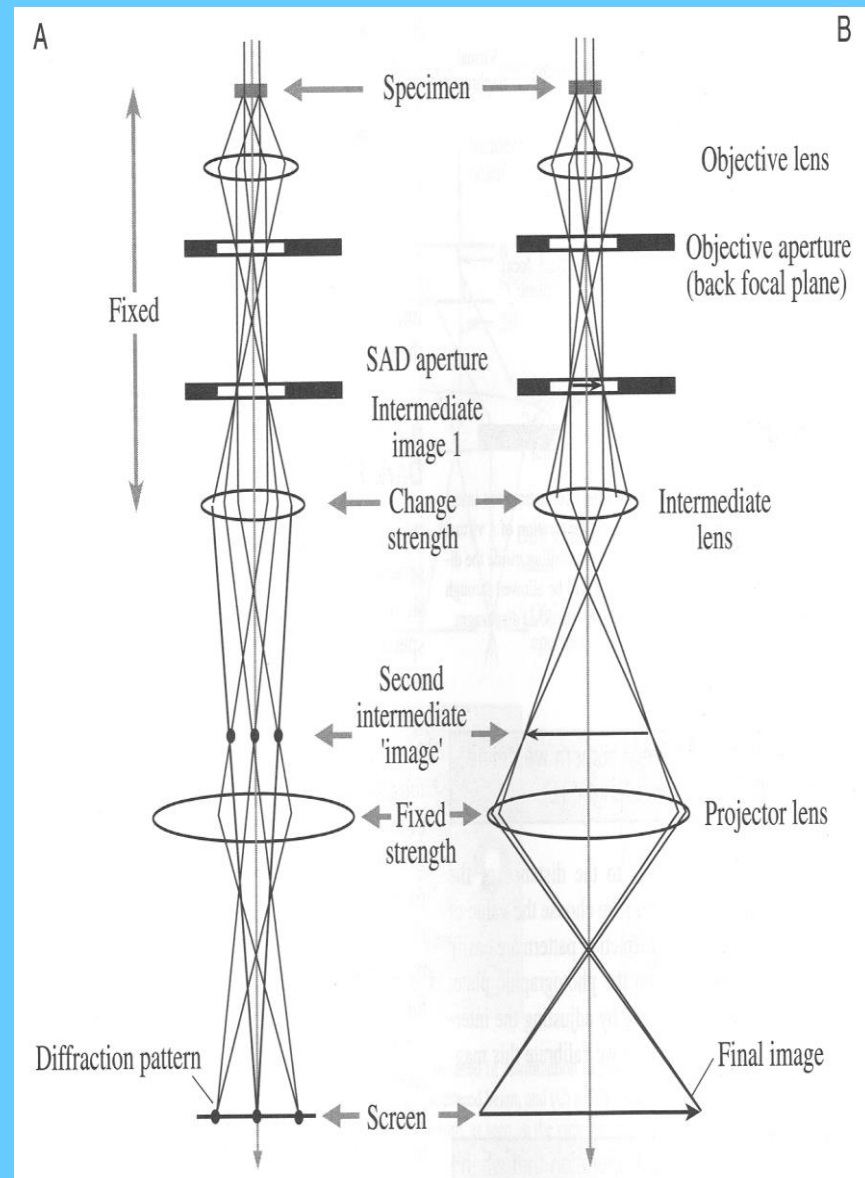
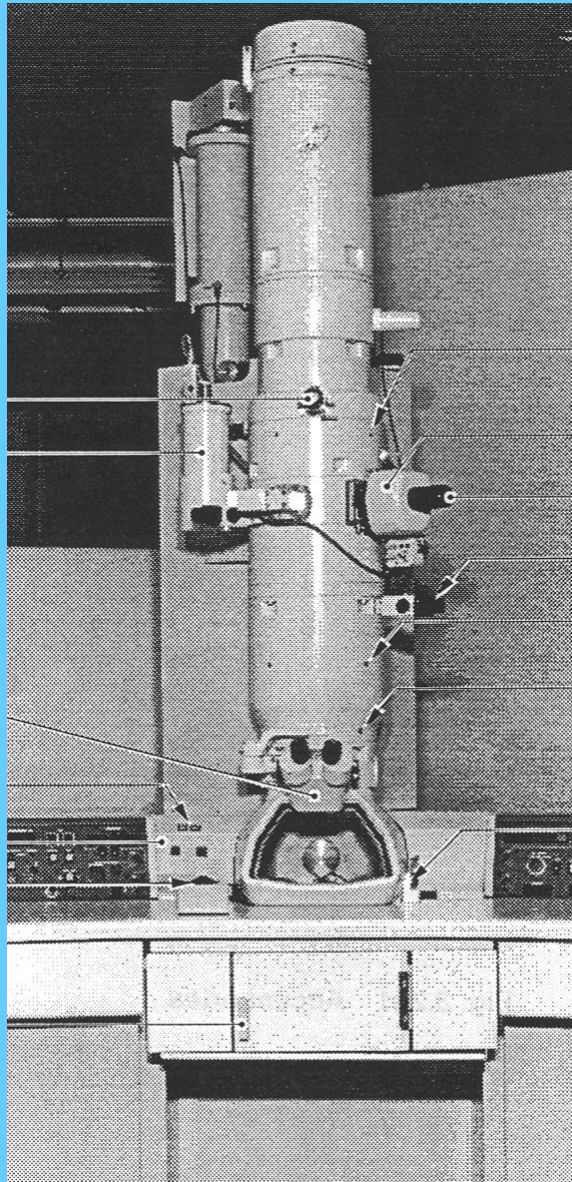


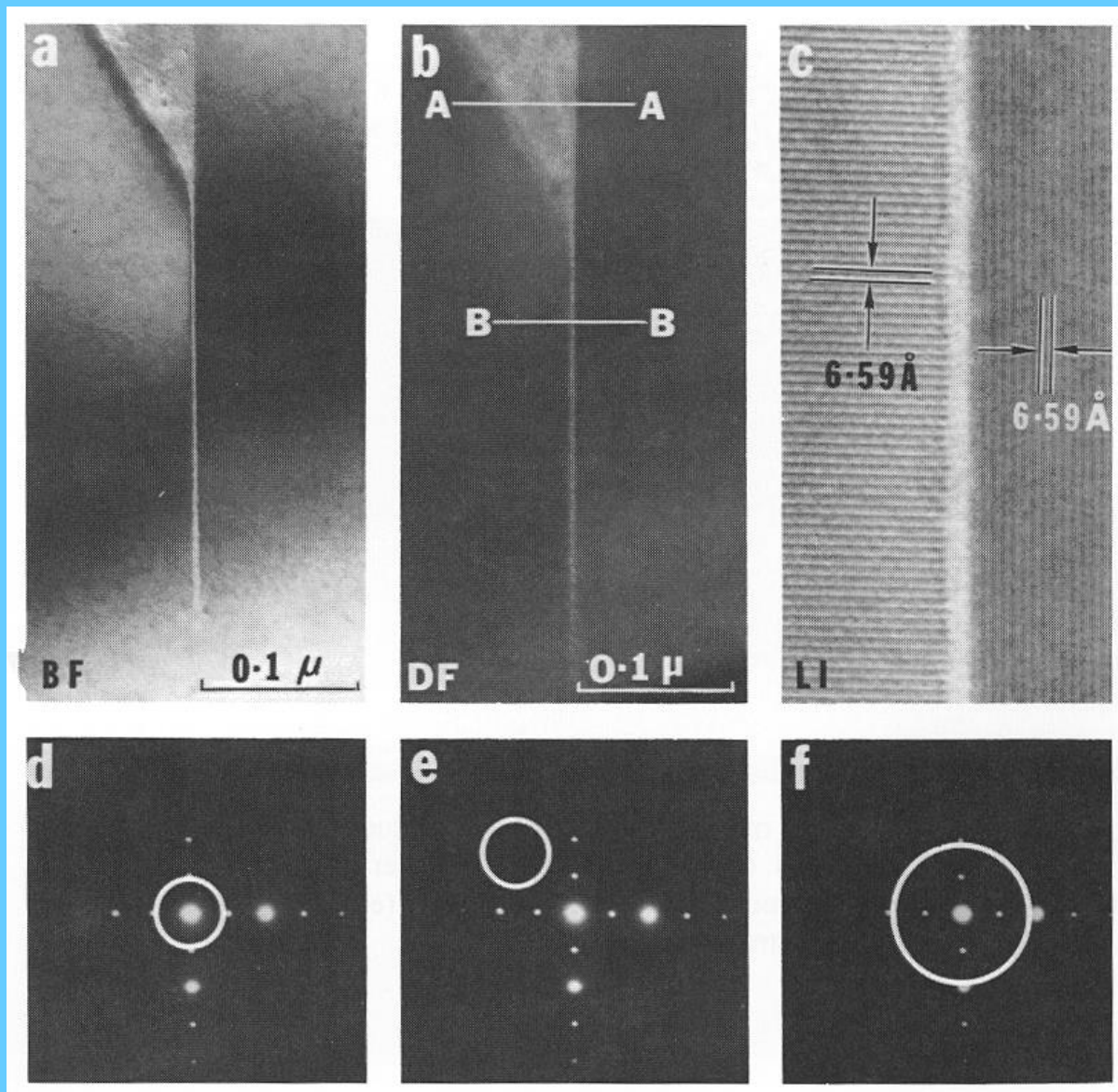
Inside of Jeol 2010

The condenser minilens coil should be deleted in the HC configuration.



How to manipulate it to get contrast



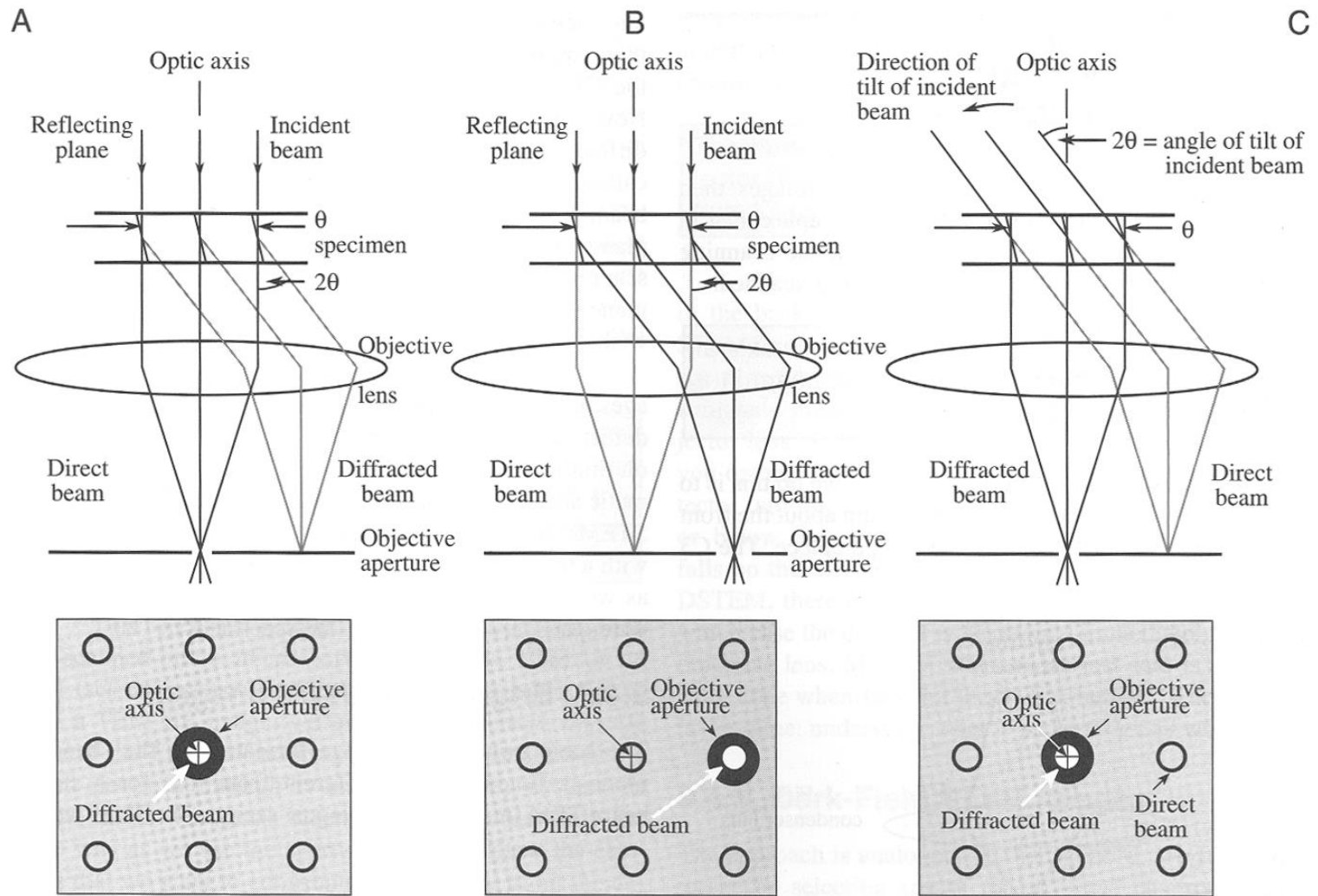


Bright field

Dark field

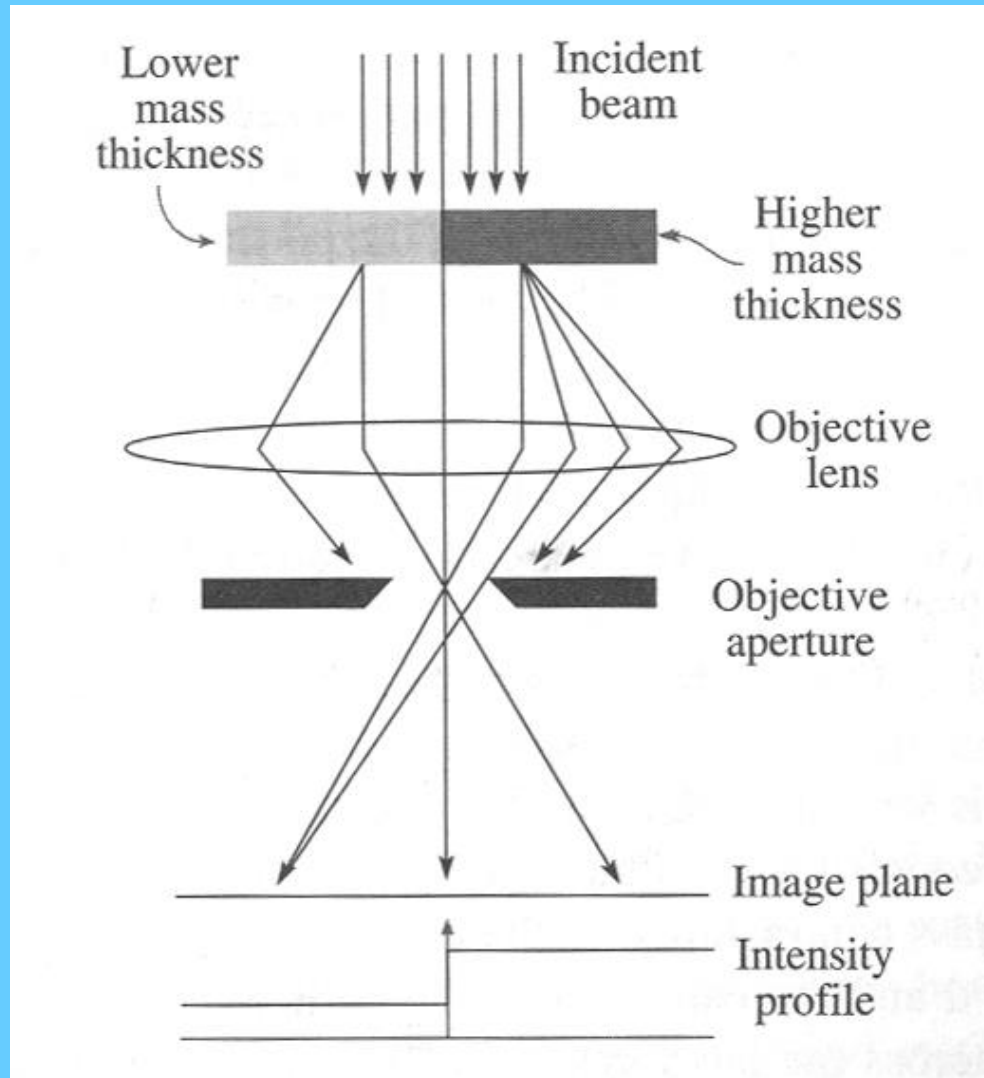
HRTEM Imaging

(Courtesy G. Thomas, in *Image of Materials*, Oxford Press, 1991)



Positioning of Objective aperture to get contrast

Mechanism of Mass-Thickness contrast

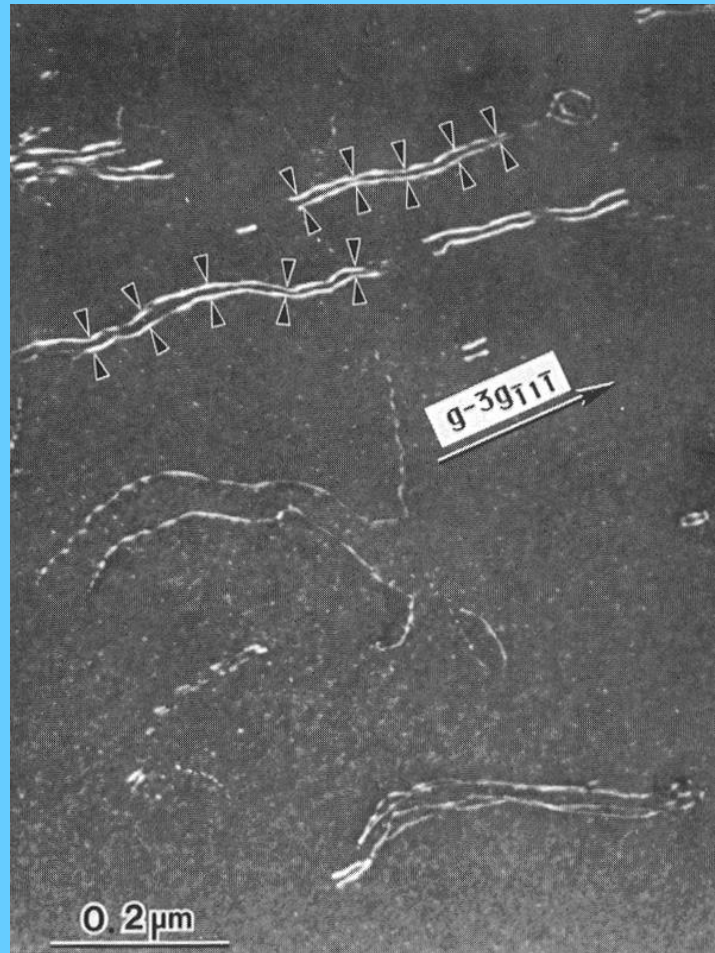


Contrast Mechanisms

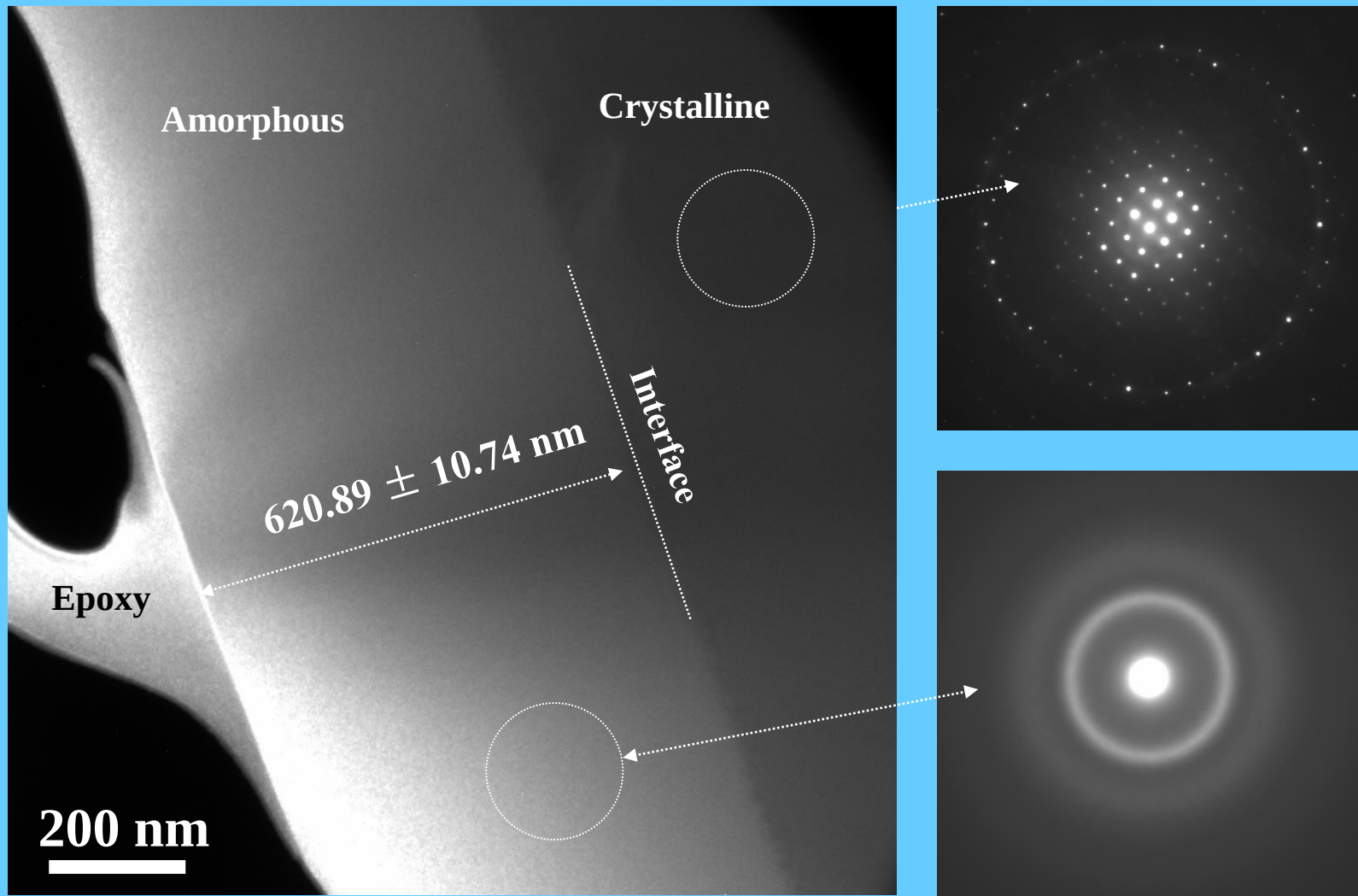
Three Contrast Mechanisms

- ➡ **Diffraction contrast: bright field and dark field**
- ➡ **Phase contrast (HRTEM)**
- ➡ **Mass thickness contrast**

Conventional TEM image showing dislocations



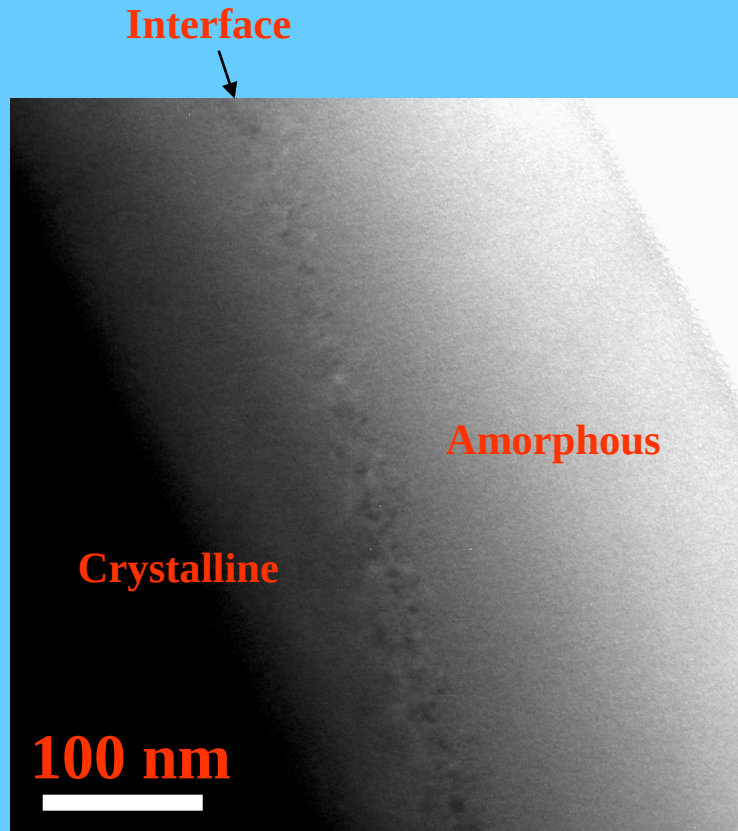
Imaging and diffraction



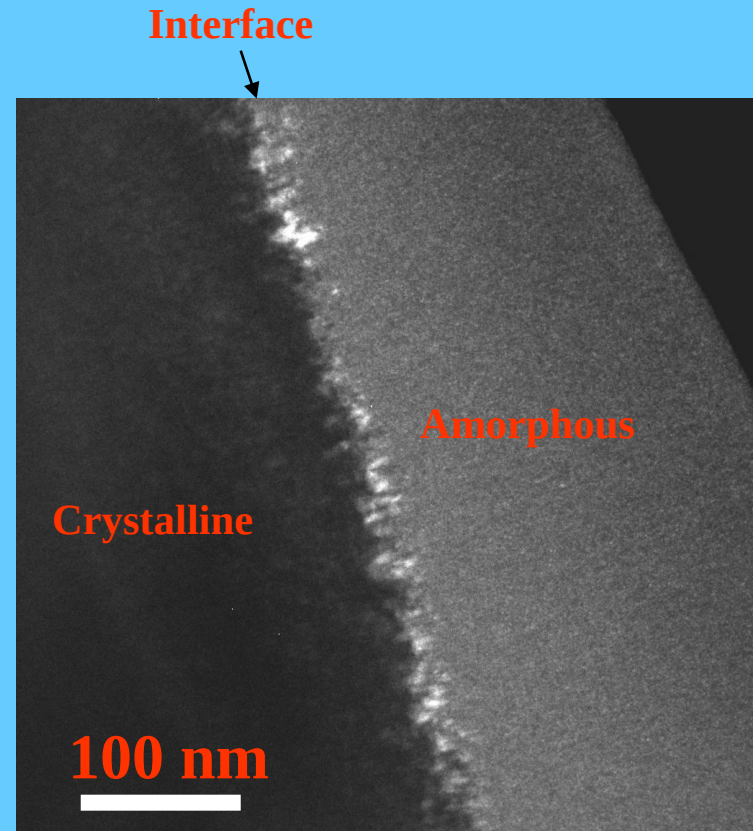
The thickness is an average measured on 4 different regions on the diffuse dark field image. One of these regions is shown above. The standard deviation is regardless the transitional layer thickness of 20 nm as revealed in the following pictures.

Conventional TEM imaging

Bright-field image

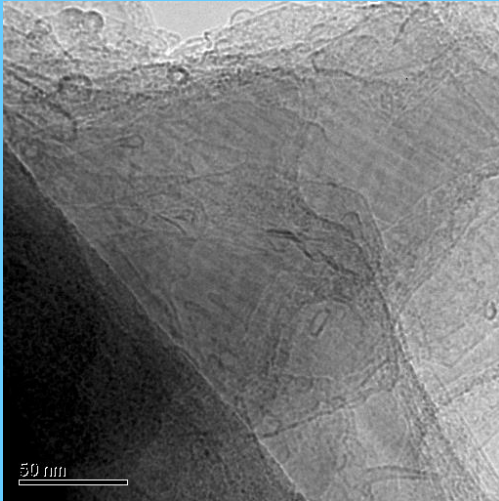


Weak beam dark-field image

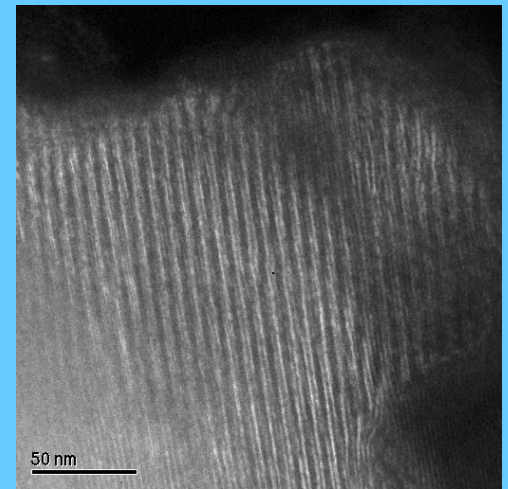
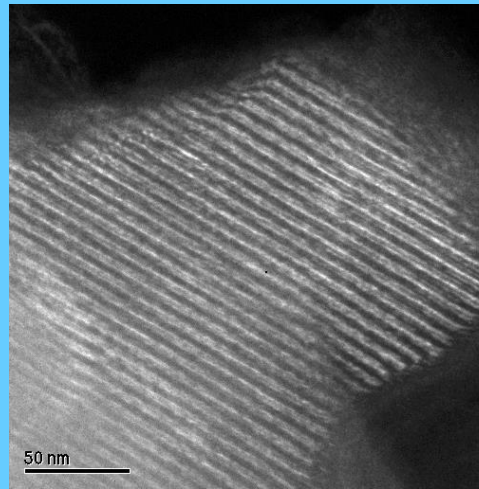


Conventional TEM imaging

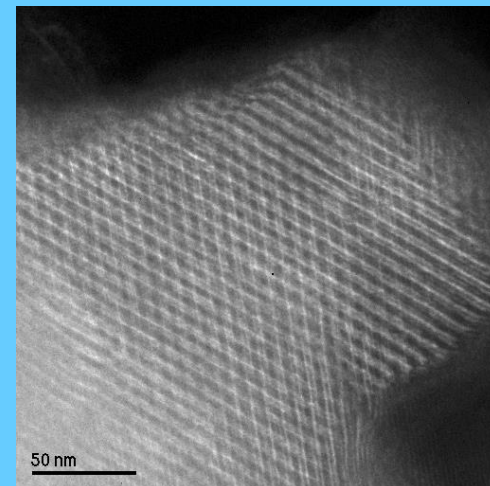
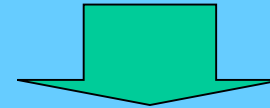
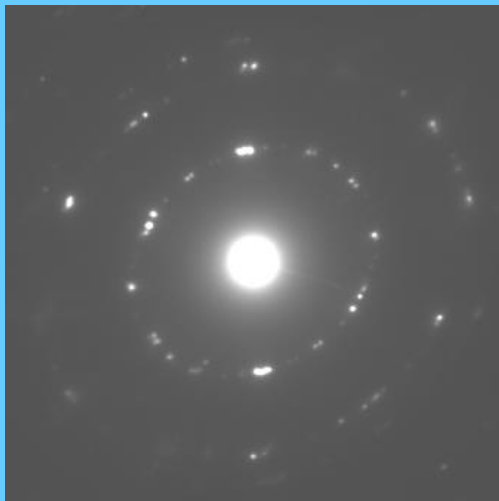
Bright field



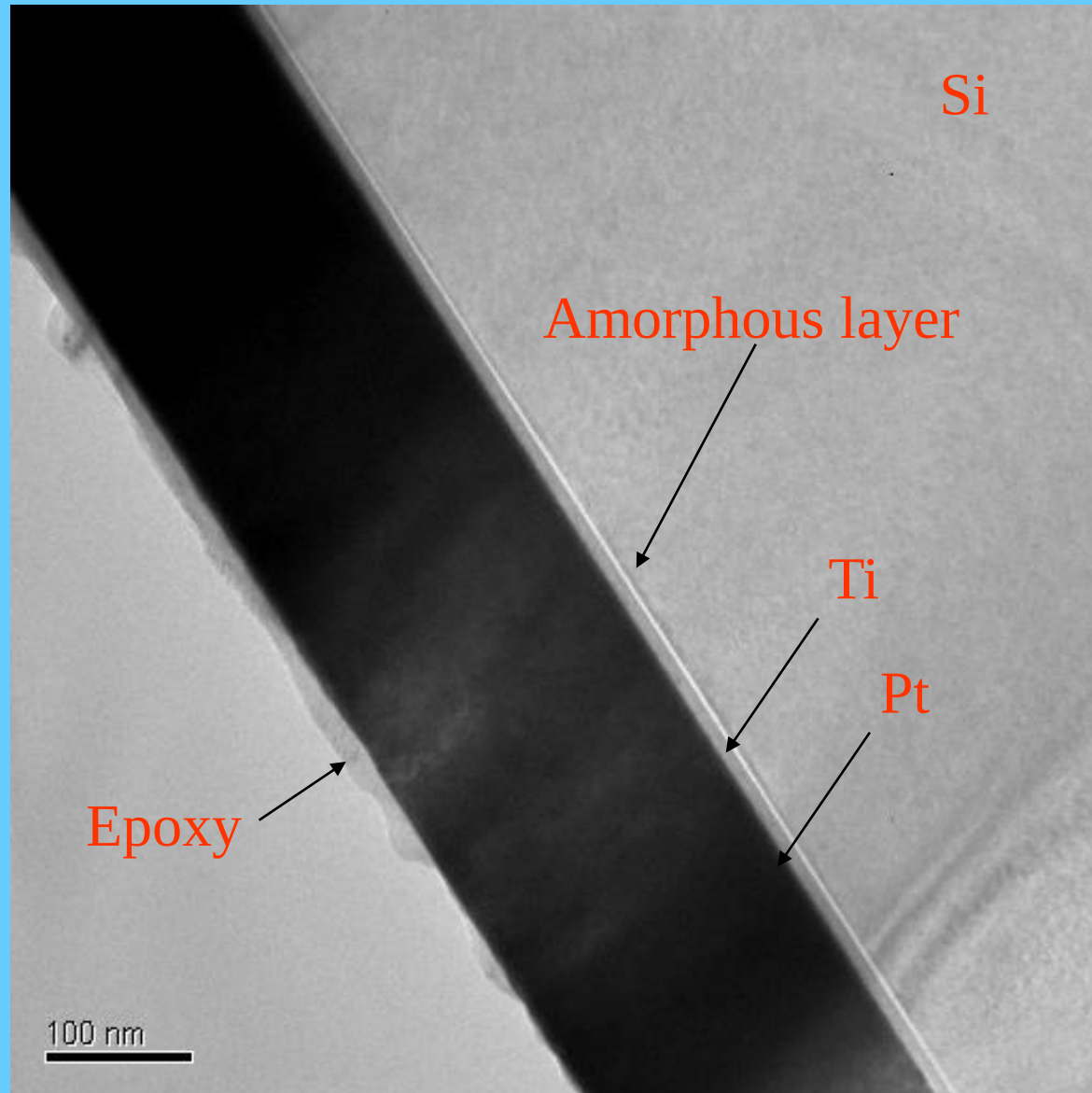
Dark filed



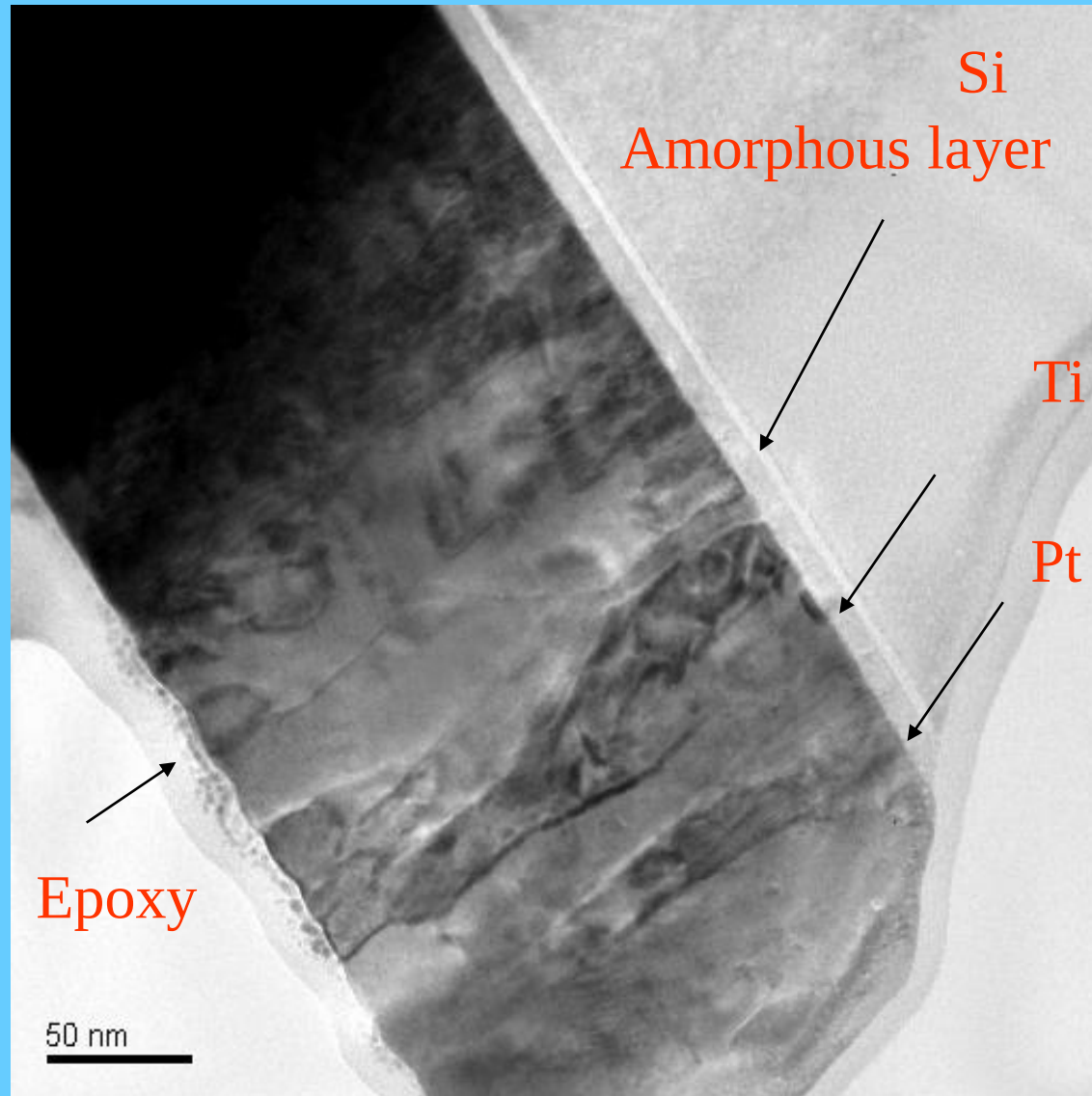
Diffraction



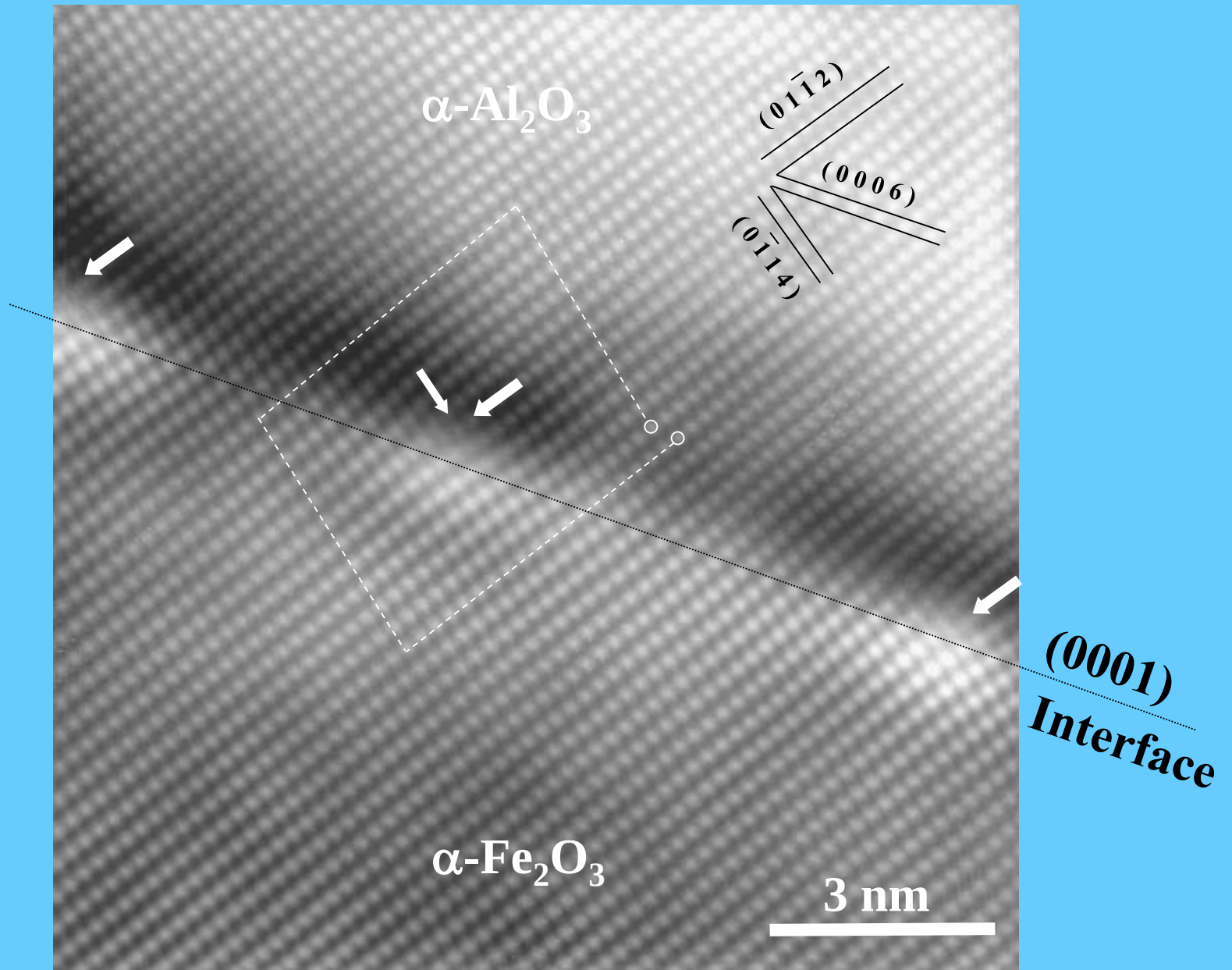
Cross sectional TEM image of Pt/Ti on Si



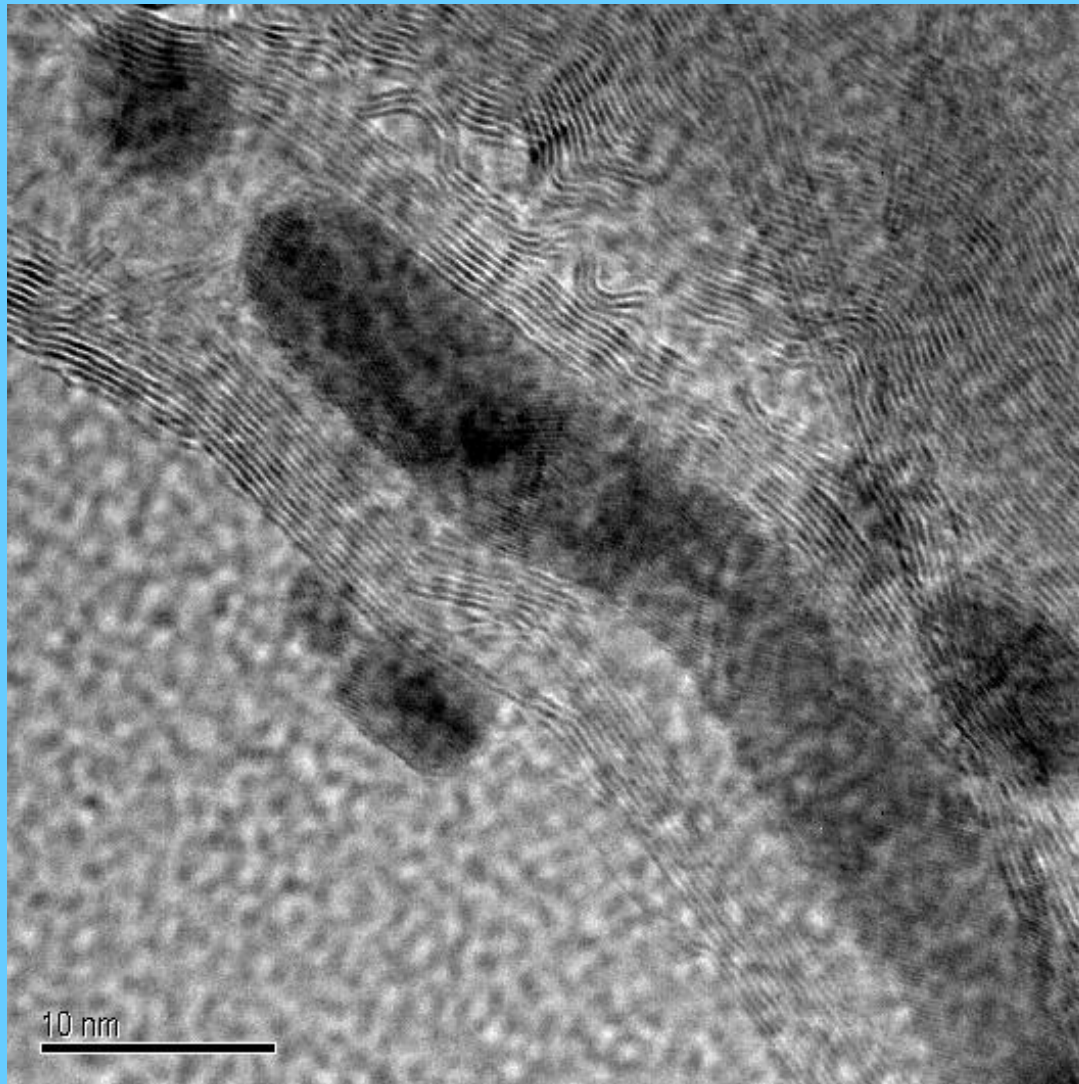
Cross sectional TEM image of Pt/Ti on Si



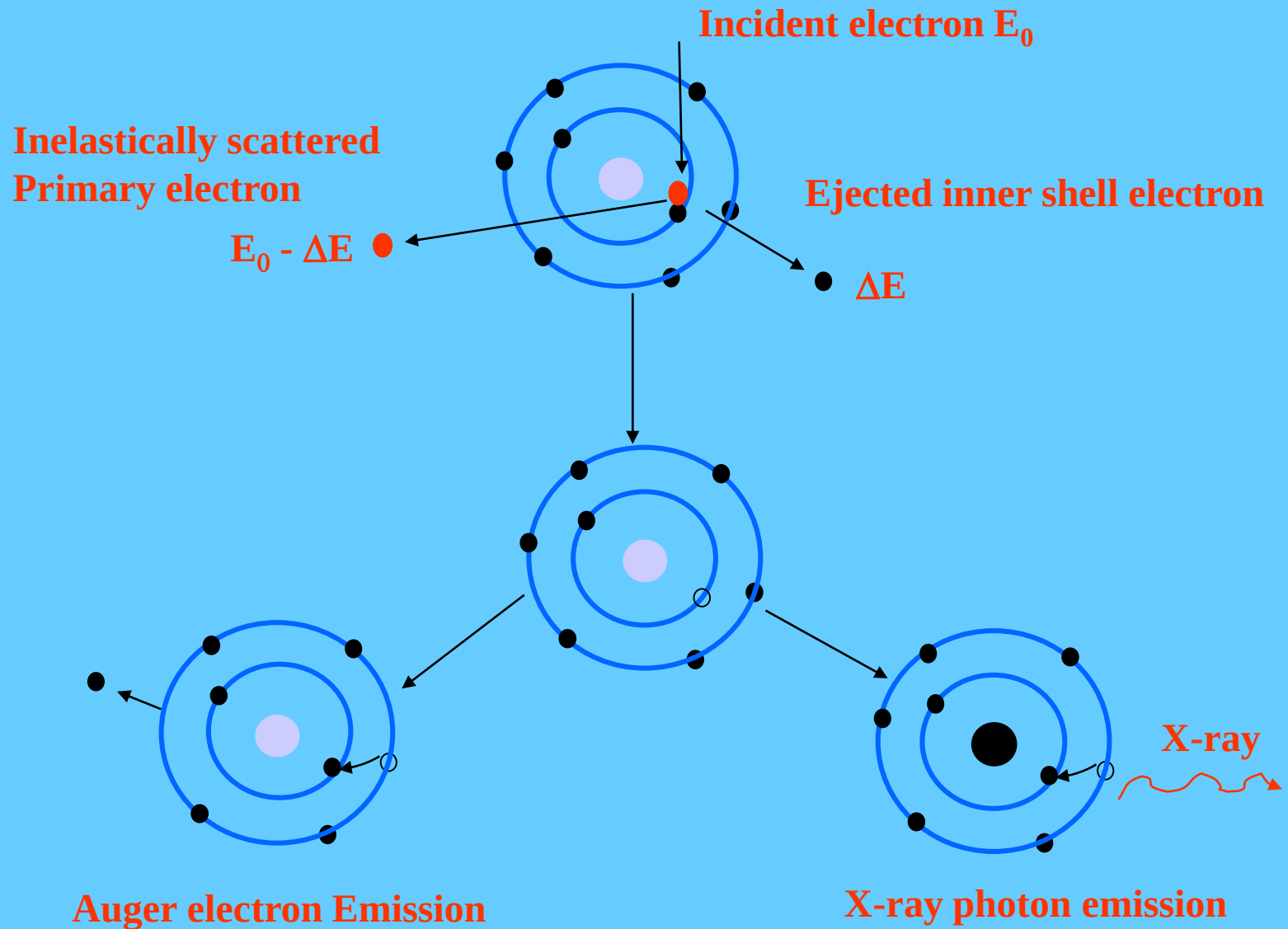
Interface misfit dislocations



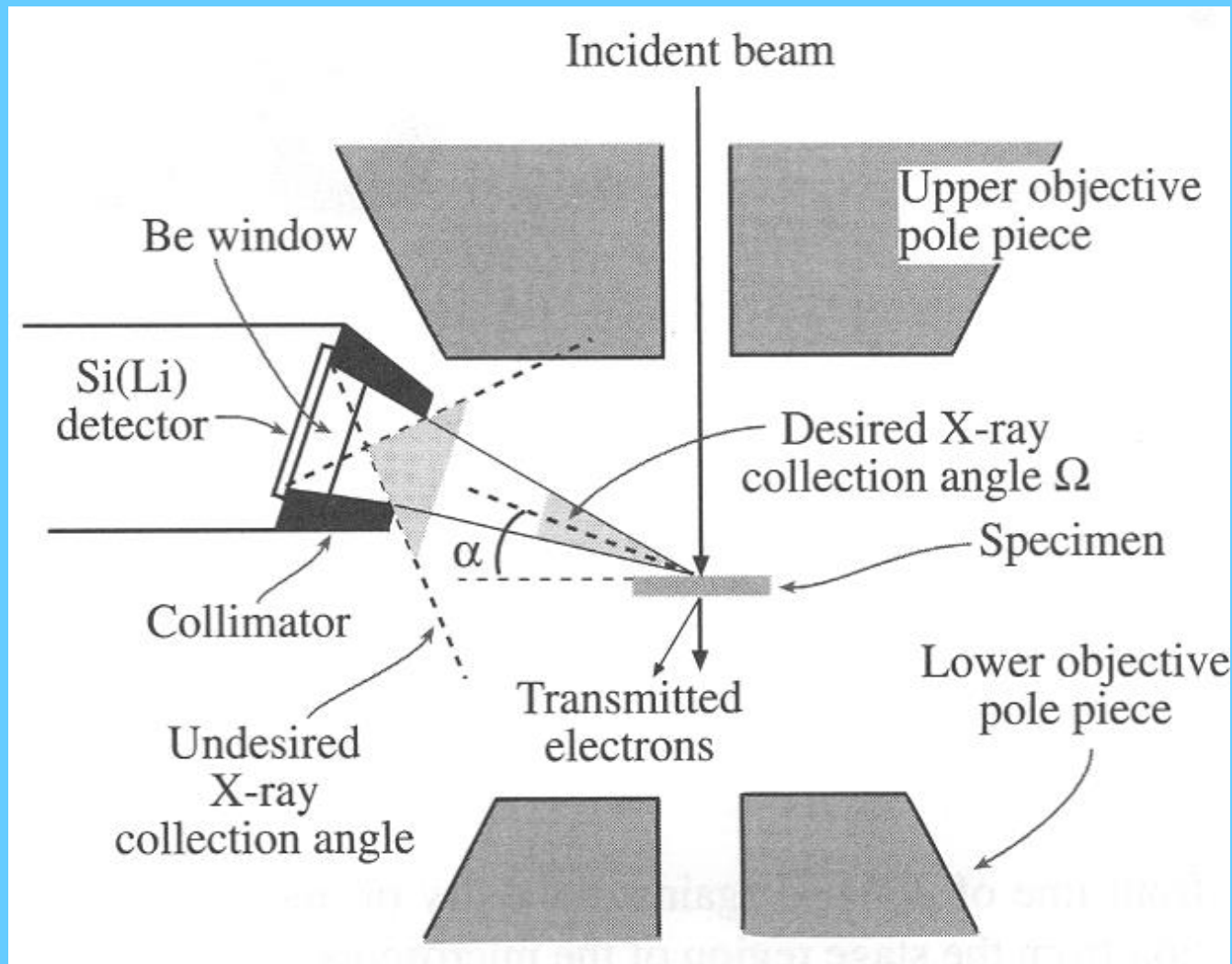
Metal filled carbon nano-tube



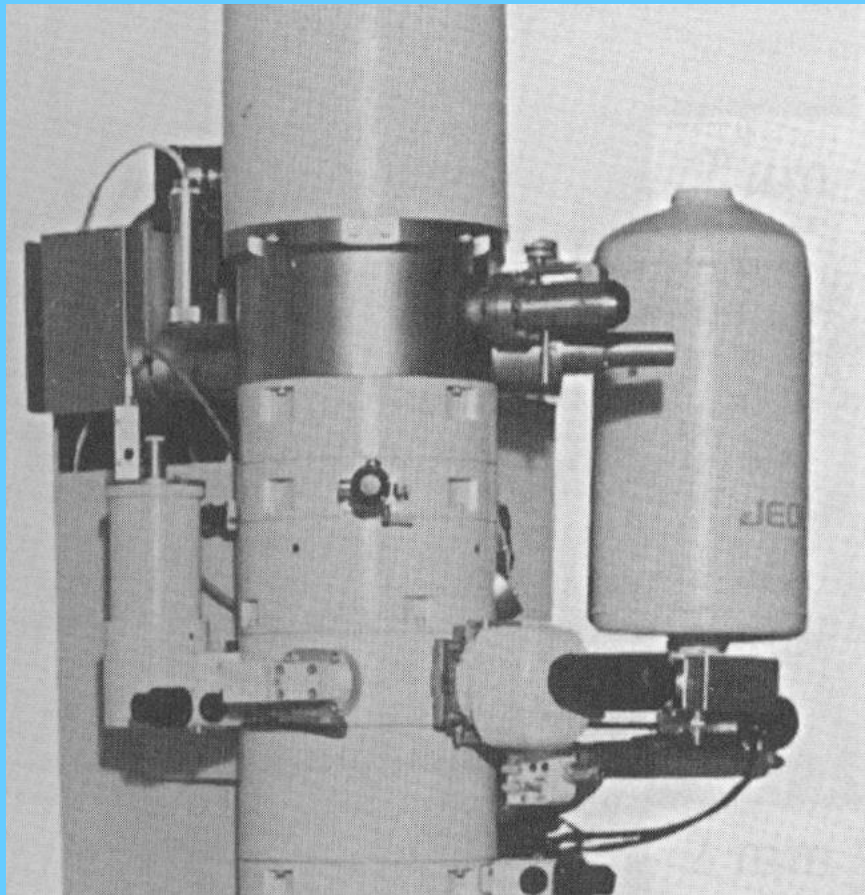
Inelastic scattering



Energy dispersive x-ray spectrometer



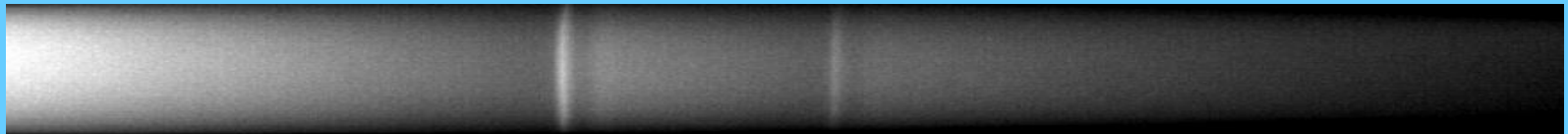
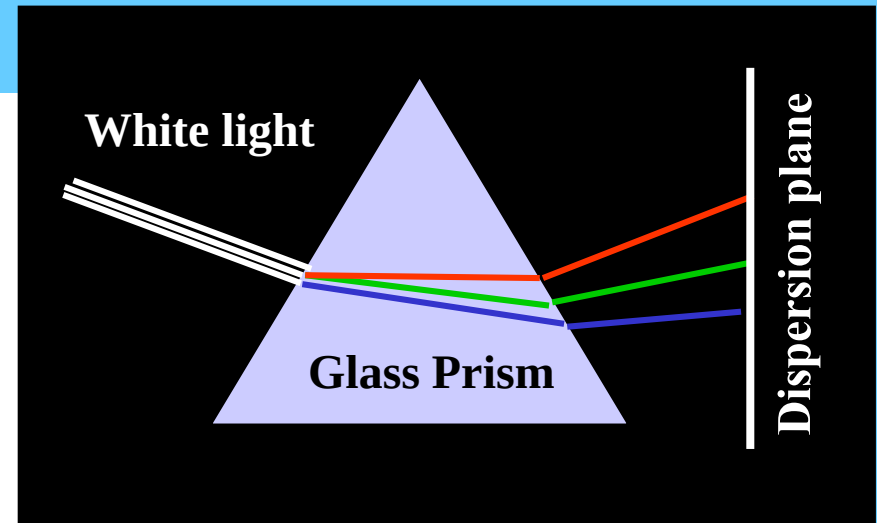
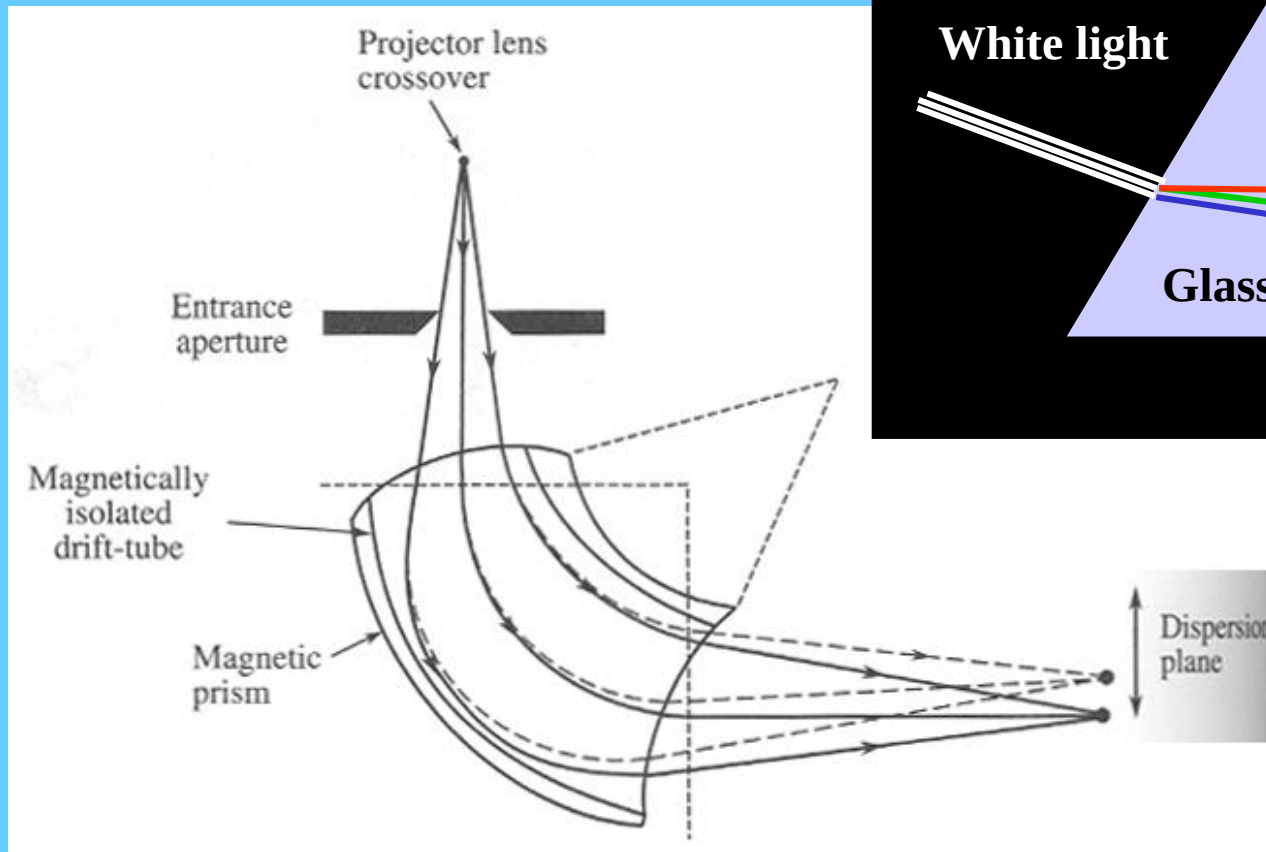
Energy dispersive x-ray spectrometer



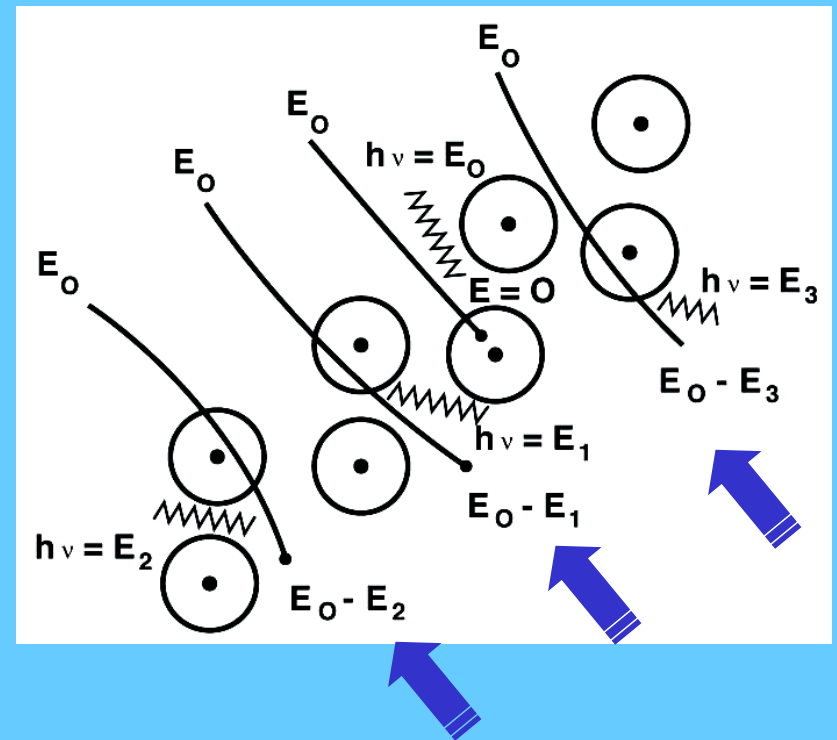
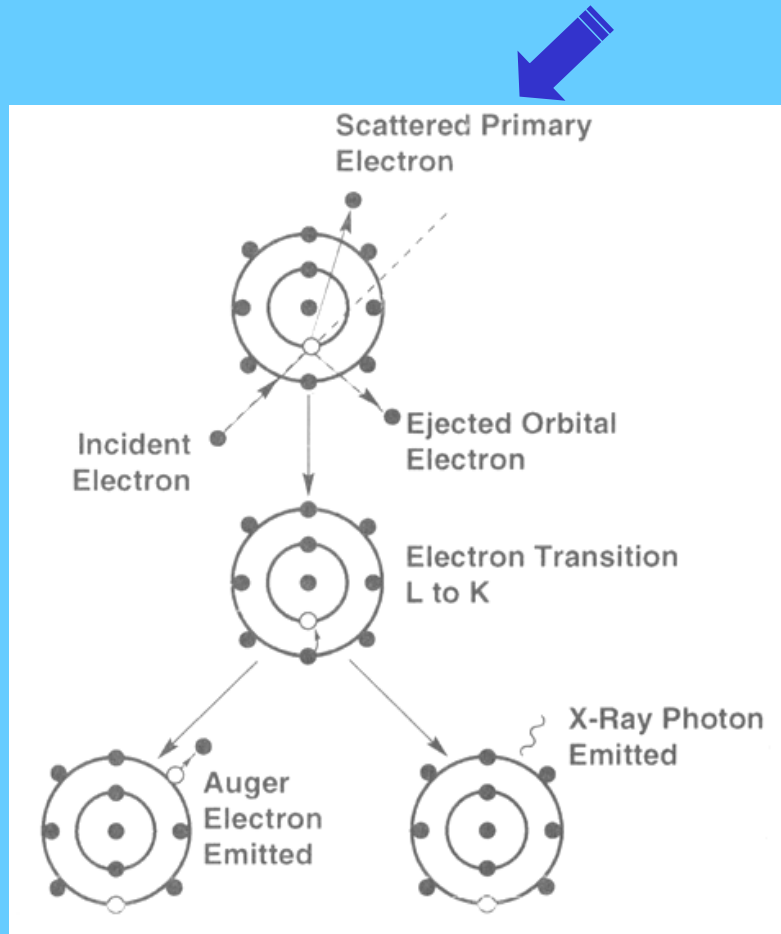
Sensitivity: $\sim 0.1\%$

Energy resolution: 126 eV

Electron Energy-Loss Spectroscopy(EELS)



Electron Energy-Loss Spectrometry (EELS)



What happens to those “inelastically” scattered electrons?!

EELS in TEM/STEM

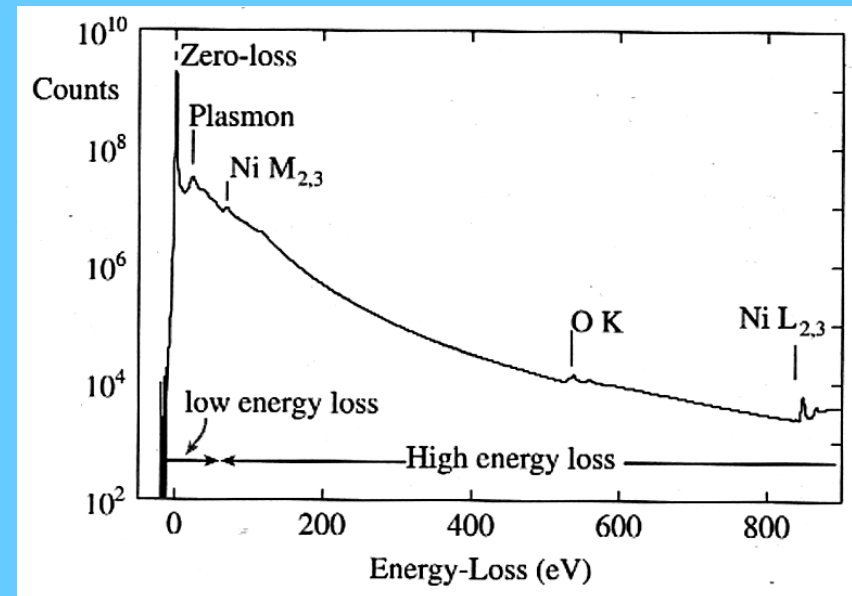
Analysis of the energy loss (and sometimes momentum-transfer) of electrons transmitted through the specimen

•Key Advantages (compare w/EDS):

- Spatial resolution in STEM $\sim d$, the electron beam size
- Detectability $\sim$ complementary and competitive to EDS
- Any solid/gas/liquid – in principle
- Qualitative analysis of any element of $Z > 1$!!
- Quantitative analysis by inner-shell ionization edges of many elements
- Absolute mass/composition determination
- Rich signal includes chemical bonding, valence, dielectric function, plasmon and free carriers, effective mass etc.

•Challenges:

- Need very thin specimen: $t < \sim 30$ nm
- Stability/quirky experimental settings High intrinsic BG
- Intensity weak for energy losses $\Delta E > 1000$ eV
- L- and M- edges not very obvious for some elements
- Quantitative analysis/interpretation

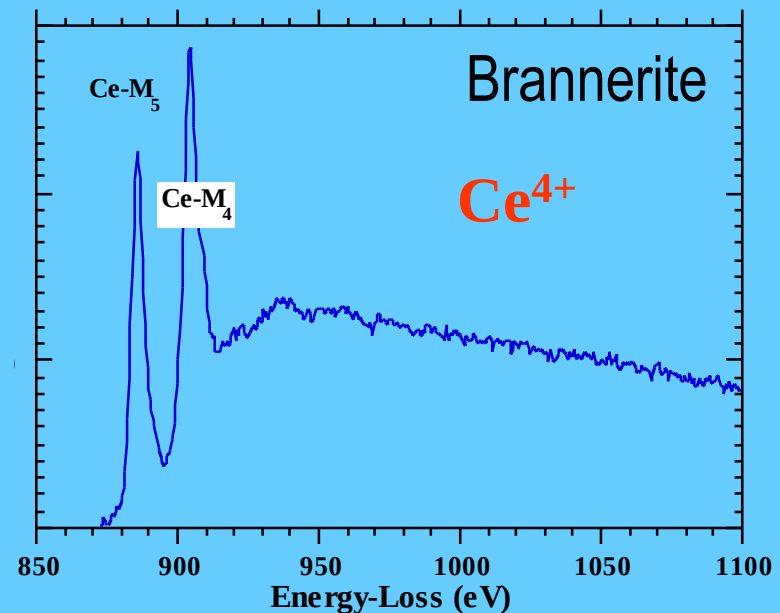
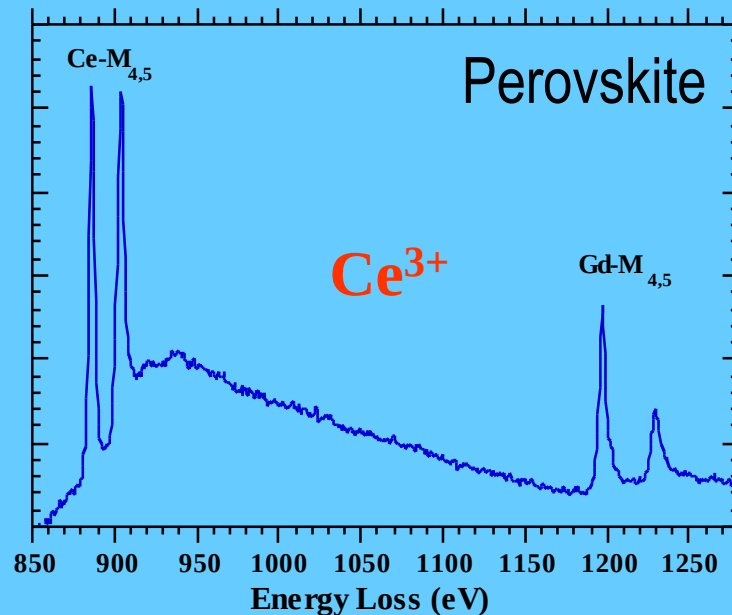


Homework

**Compare with EDS and EELS:
similarities and differences**

Oxidation State of Ce in Mn precipitate

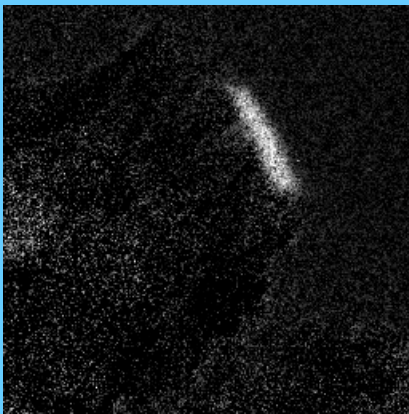
The Ce-M₄/M₅ Ratio can be used to determine the oxidation state of the cerium. The two available standards used were a cerium (III) perovskite and a cerium (IV) brannerite. These had M₄/M₅ ratios of 0.84 and 1.04, respectively. The sample containing trace cerium had a M₄/M₅ ratio of 0.79 ± 0.01 , suggesting the Ce was in the (III) oxidation state



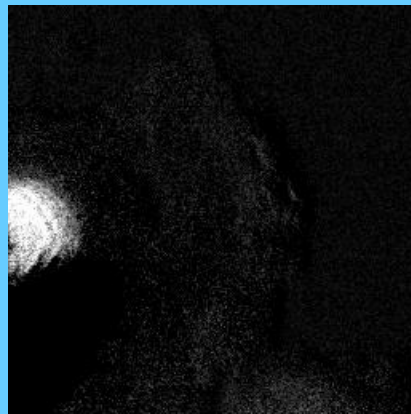
Courtesy E. Buck

Energy Filtered Imaging

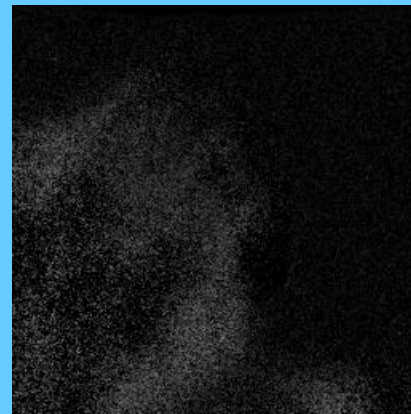
Fe-Map



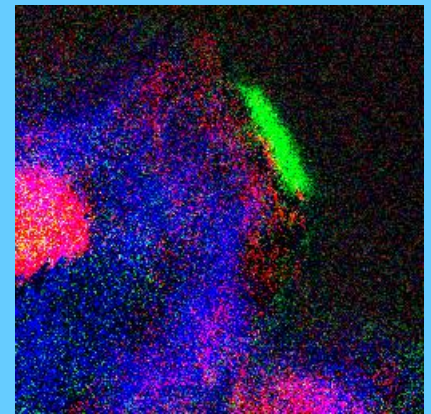
Sr-Map



Mn-Map



Combination

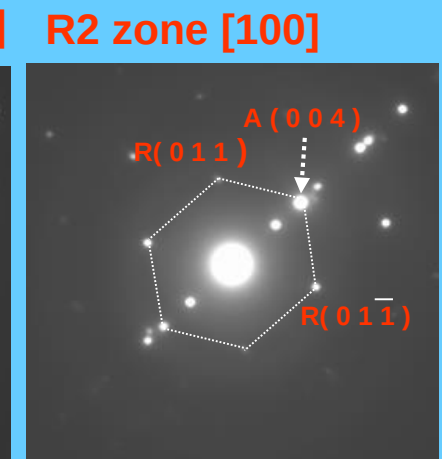
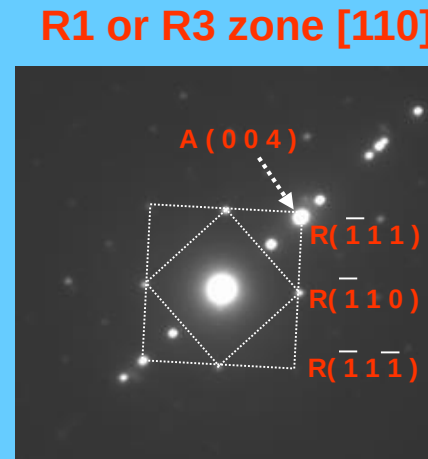
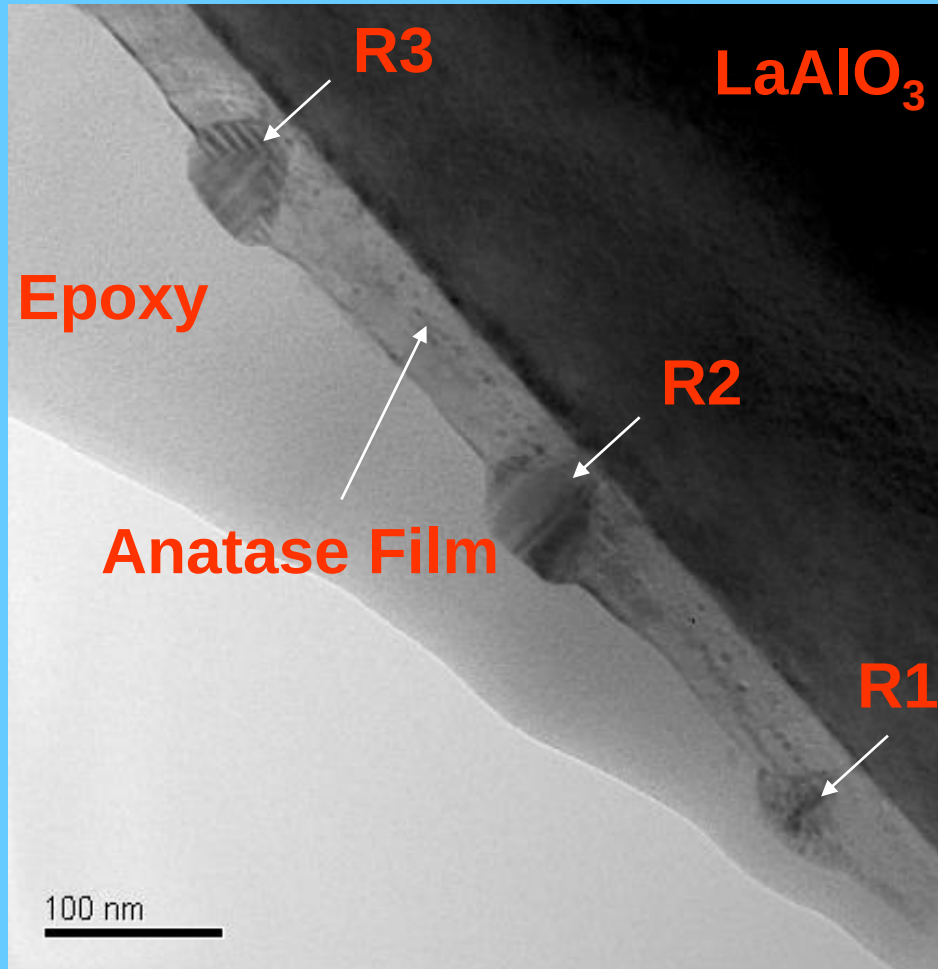



100 nm

EFTEM Imaging allows compositional mapping

Courtesy E. Buck

Pure TiO_2 , high density of precipitates

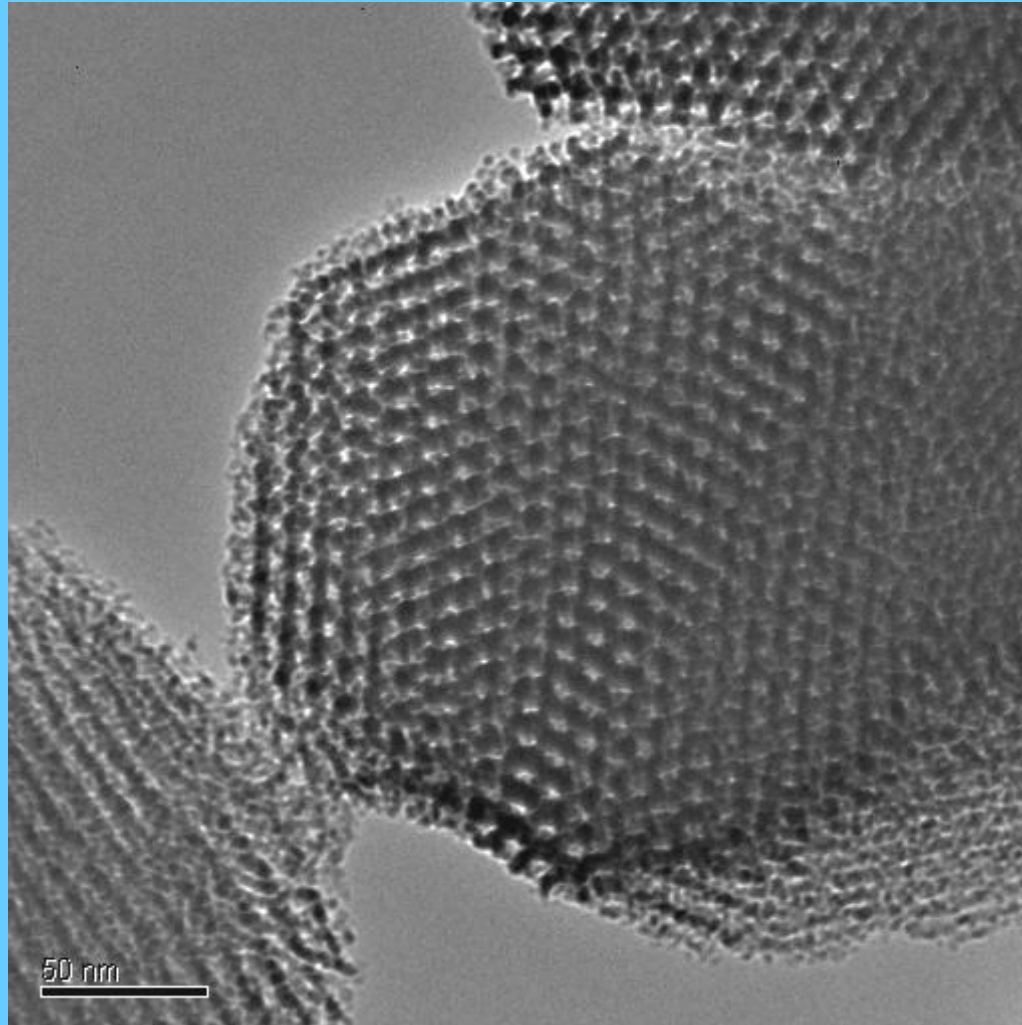


L: LaAlO_3

R: Rutile

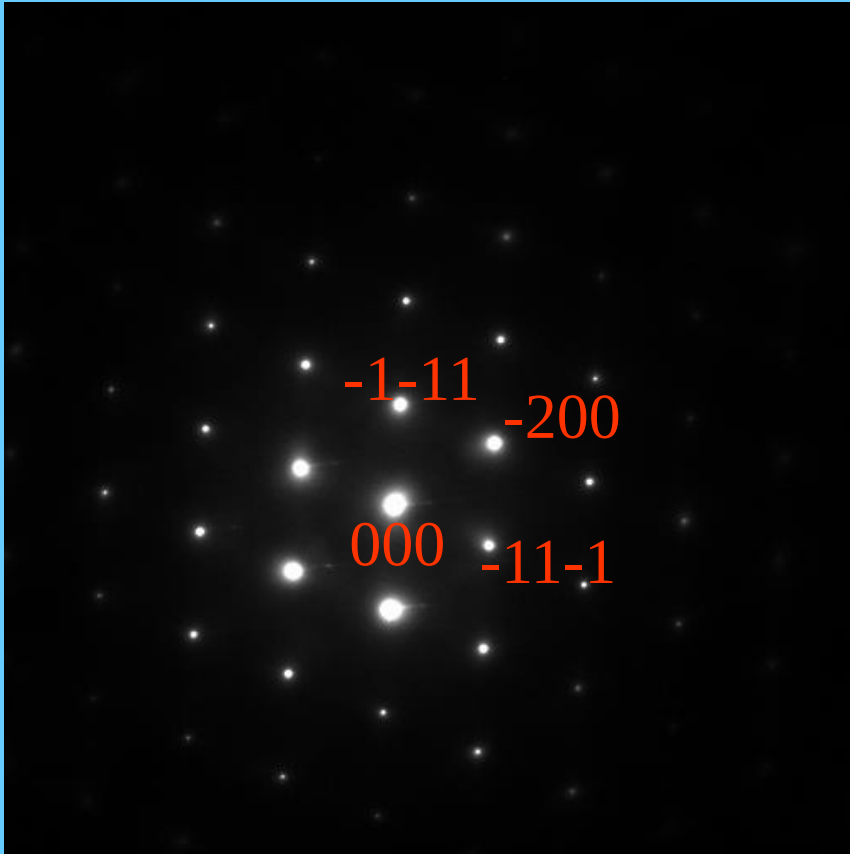
A: Anatase

Meso porous silica

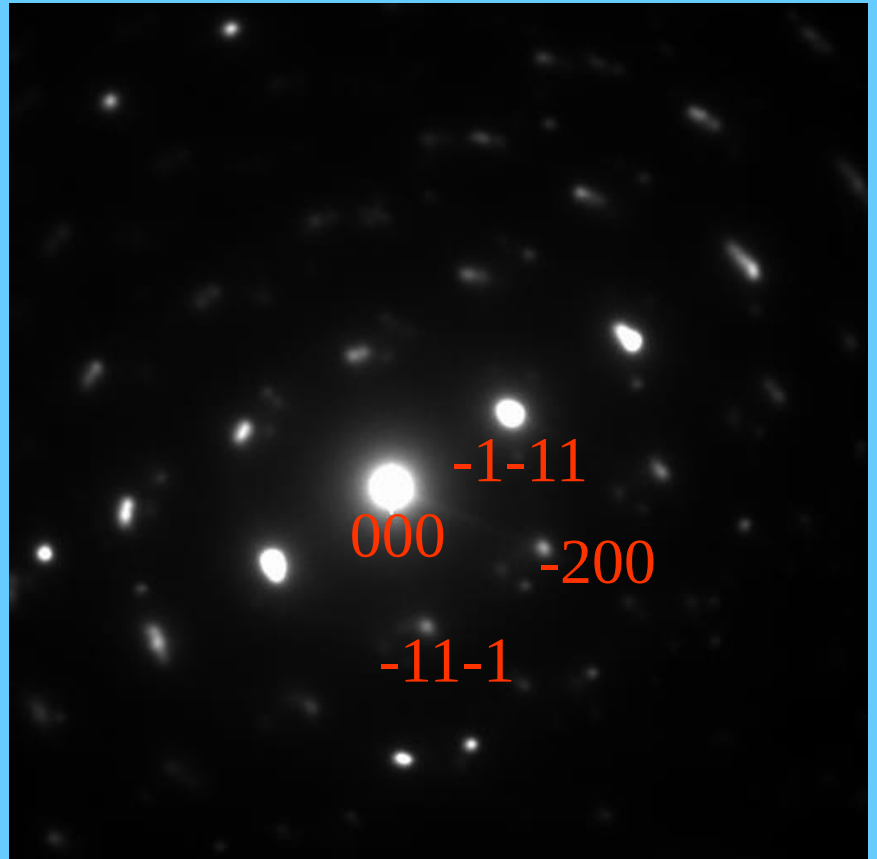


Orientation determination by electron diffraction

Si [110]

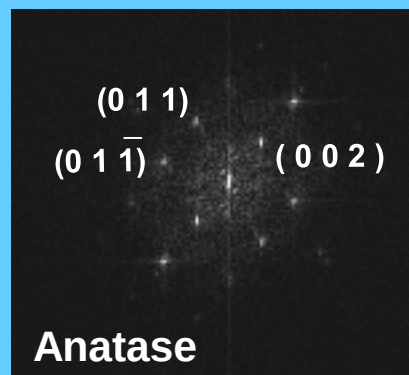
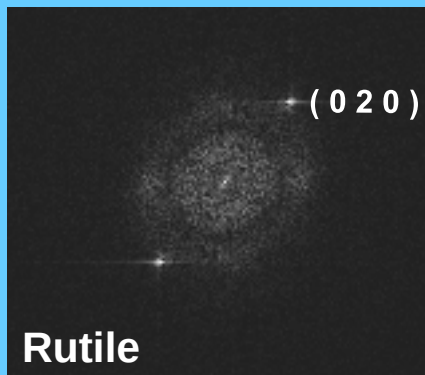
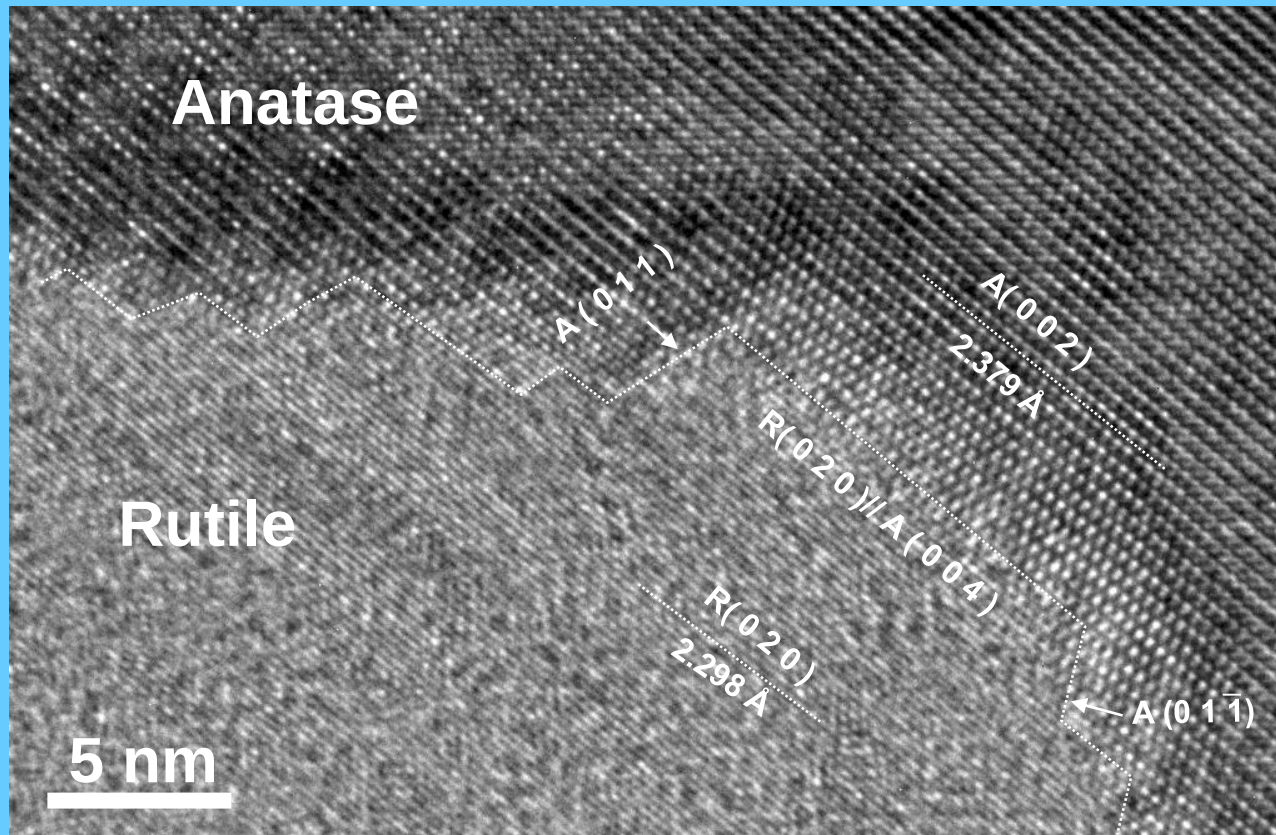


Pt [110]



Orientations: $\langle 110 \rangle \text{Si} // \langle 110 \rangle \text{Pt}$
 $\{100\} \text{Si} // \{111\} \text{Pt}$

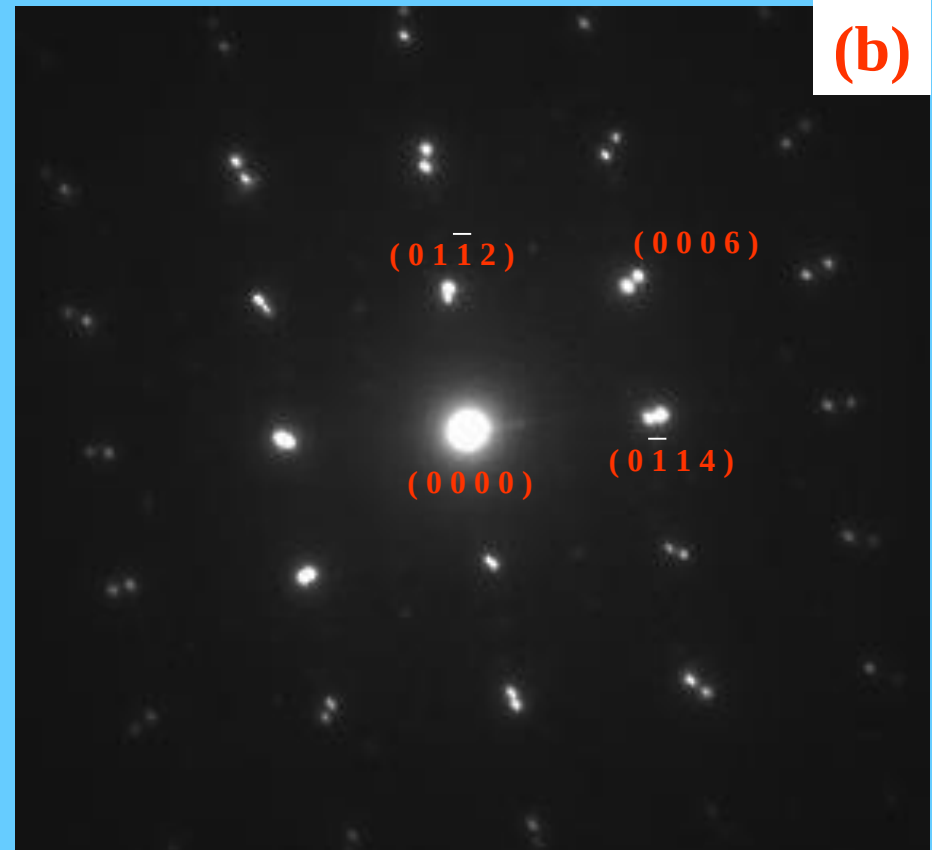
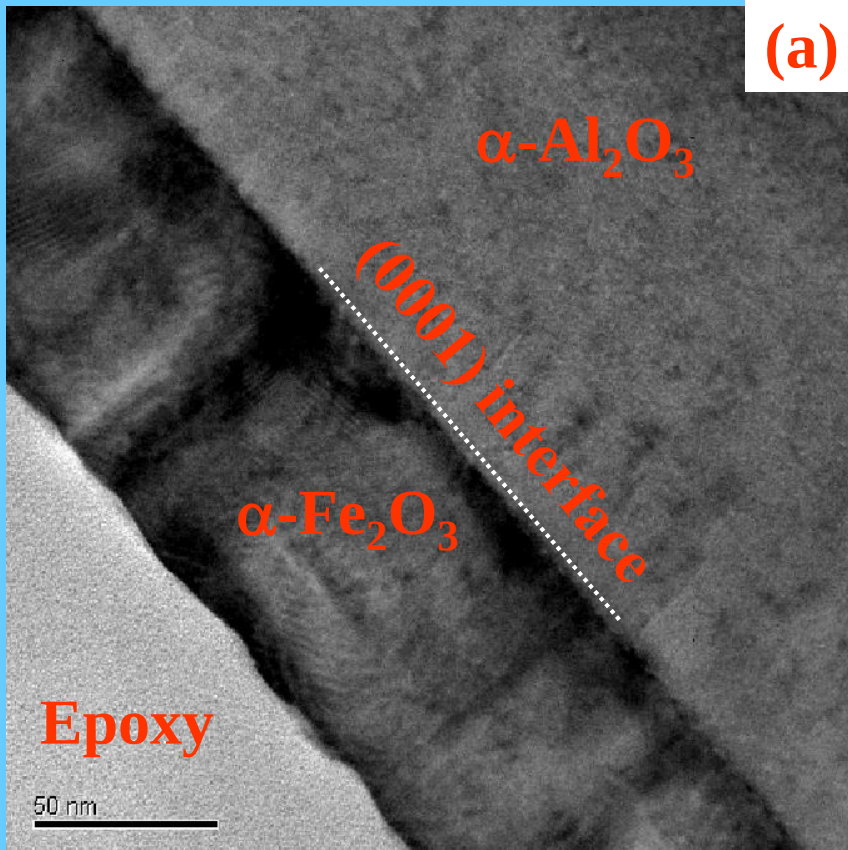
Anatase Rutile Orientation Relationship



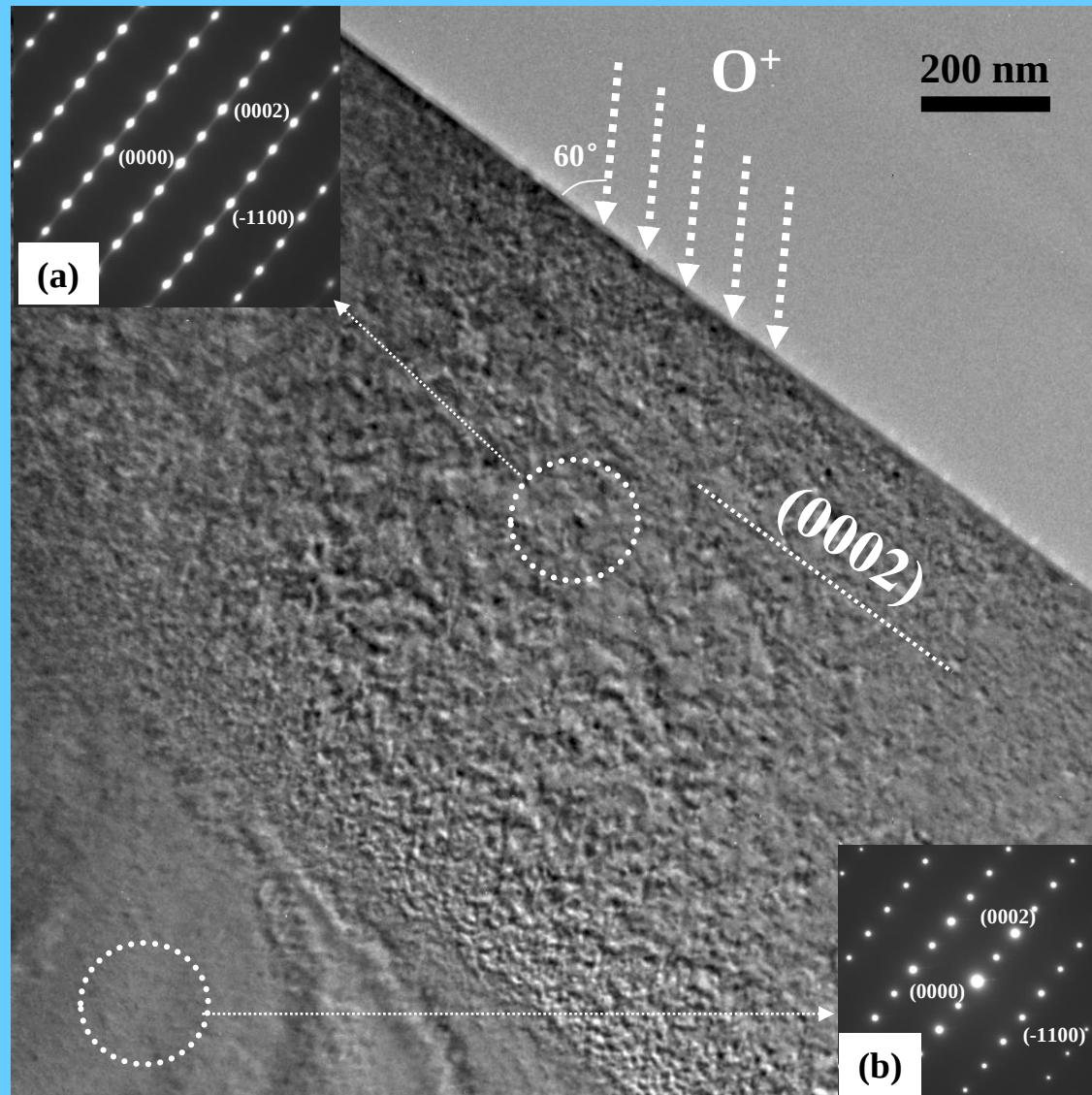
A: Anatase
R: Rutile

Cross Sectional TEM of thin film

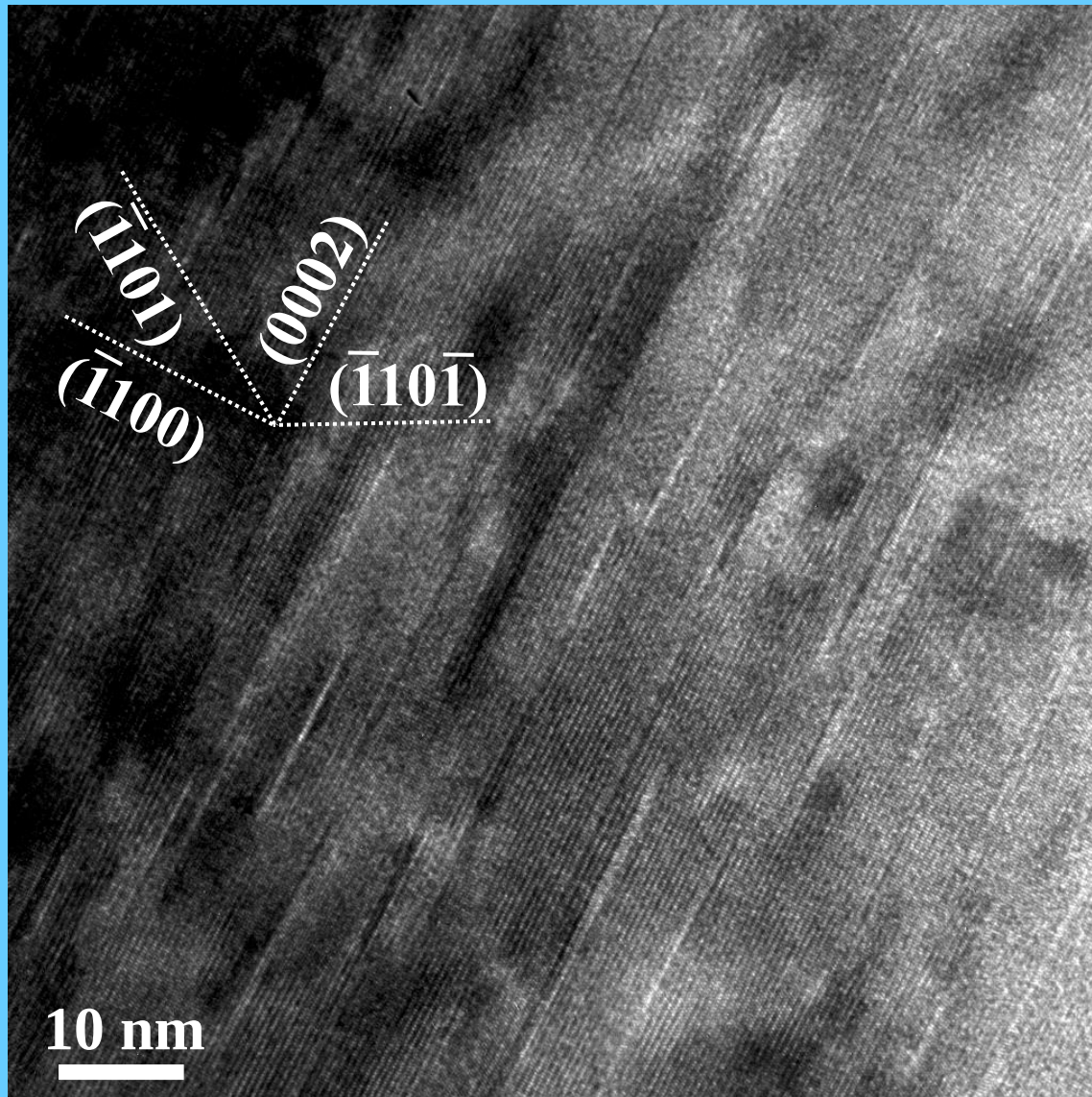
Beam direction $[\bar{2} 1 1 0]$



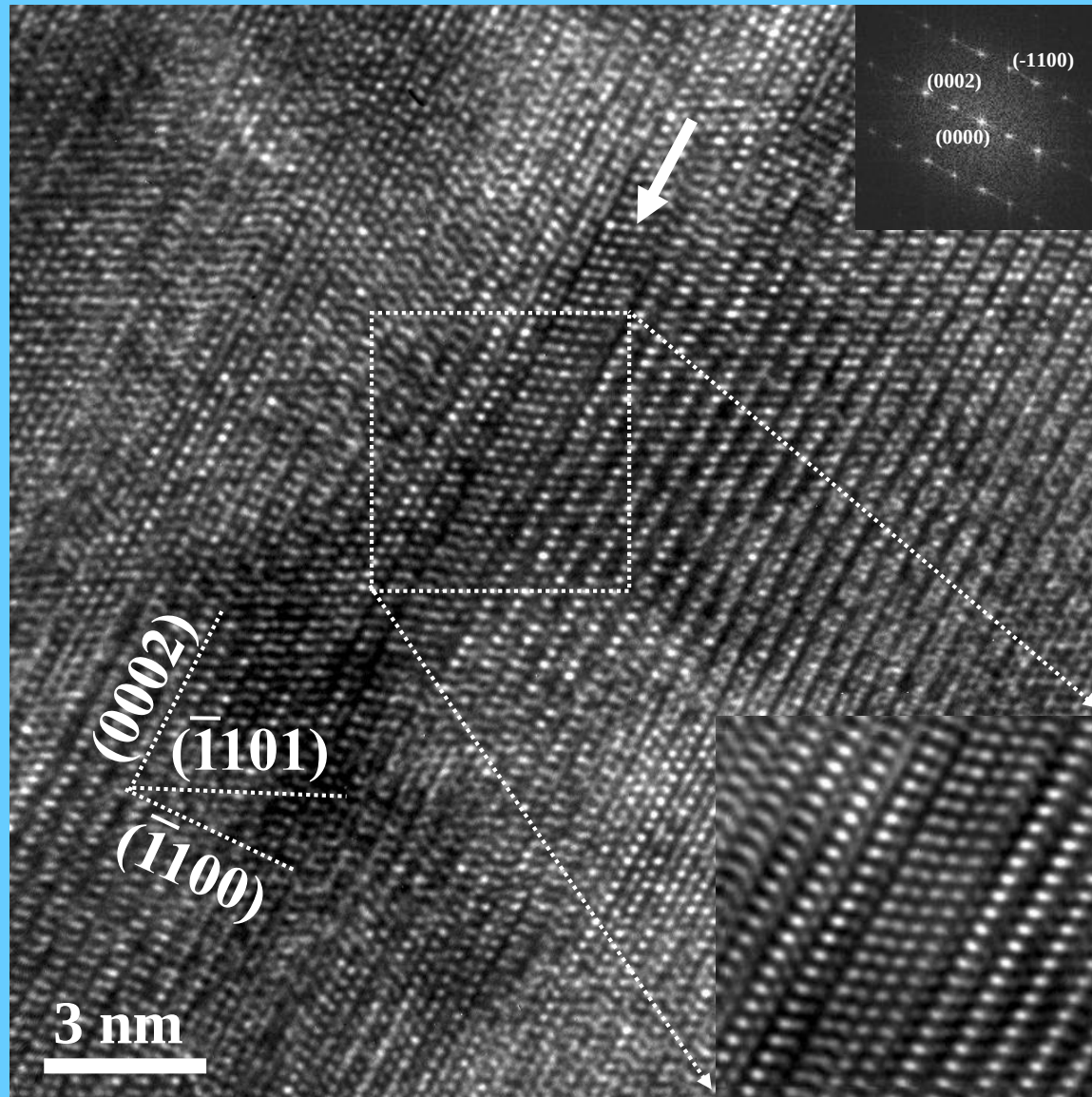
Cross section TEM image of O⁺ irradiated GaN



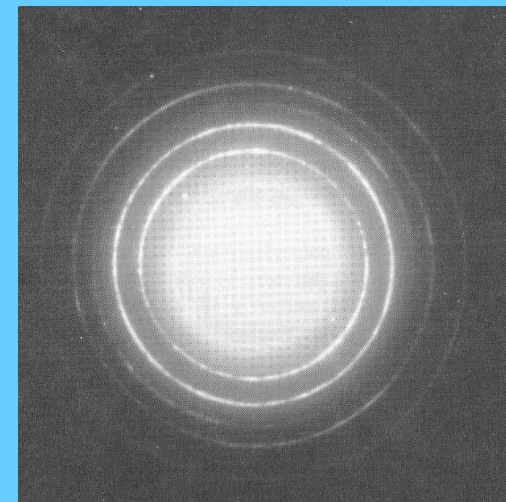
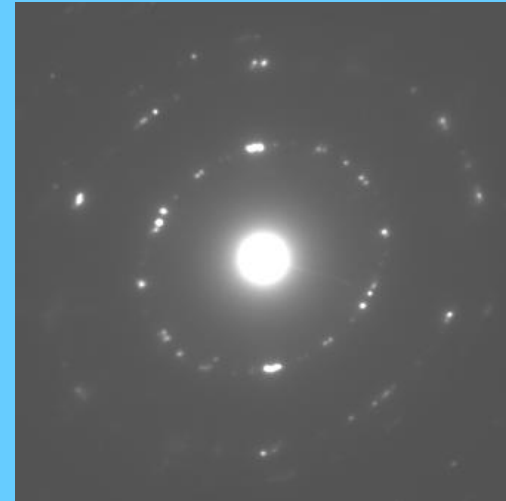
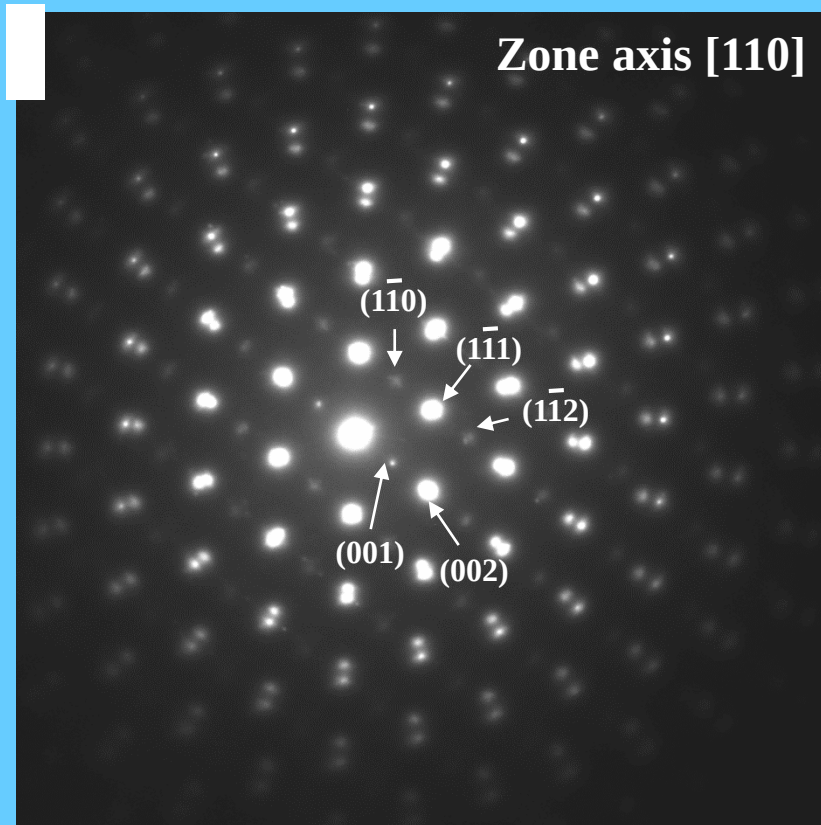
Stacking faults in GaN



Stacking fault in GaN



Diffraction patterns, why they are looks so different!

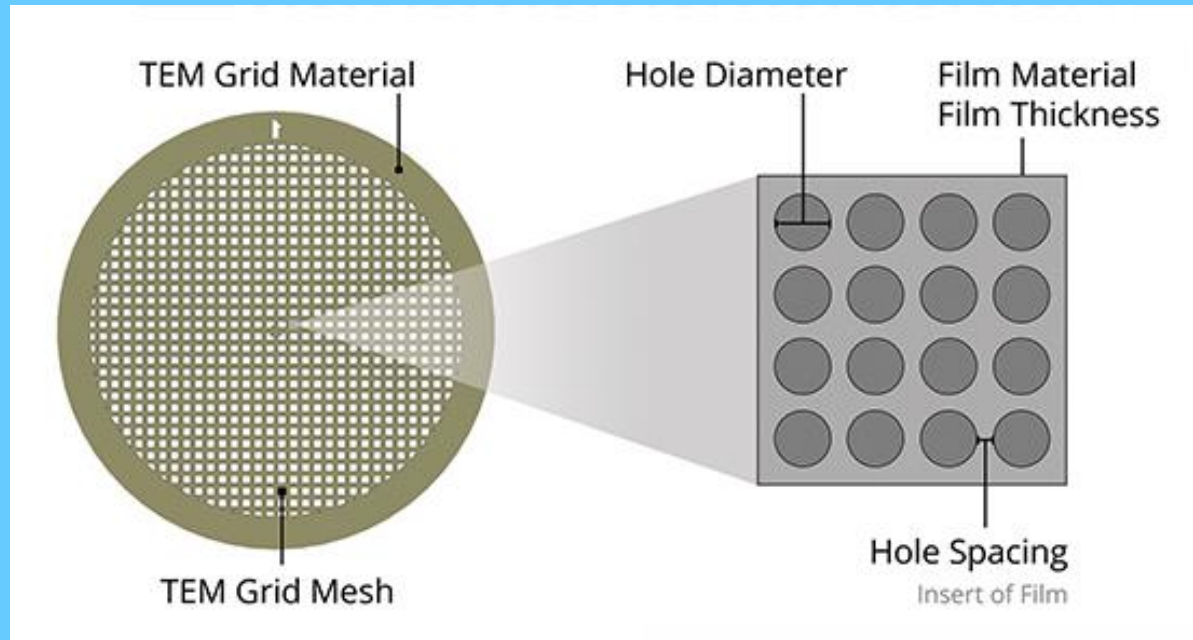


Summary/Resolution

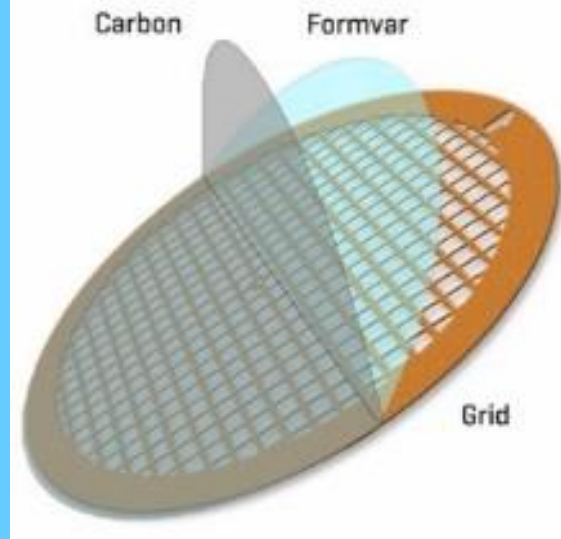
	Resolution (nm)	Energy (eV)	Sensitivity (%)
O.M.	200		
TEM/HRTEM	0.086		
STEM	0.10		
WDS	50	10	0.10
EELS	0.2	0.5	0.10
EDXS	0.2	126	0.1

TEM Sample Preparation

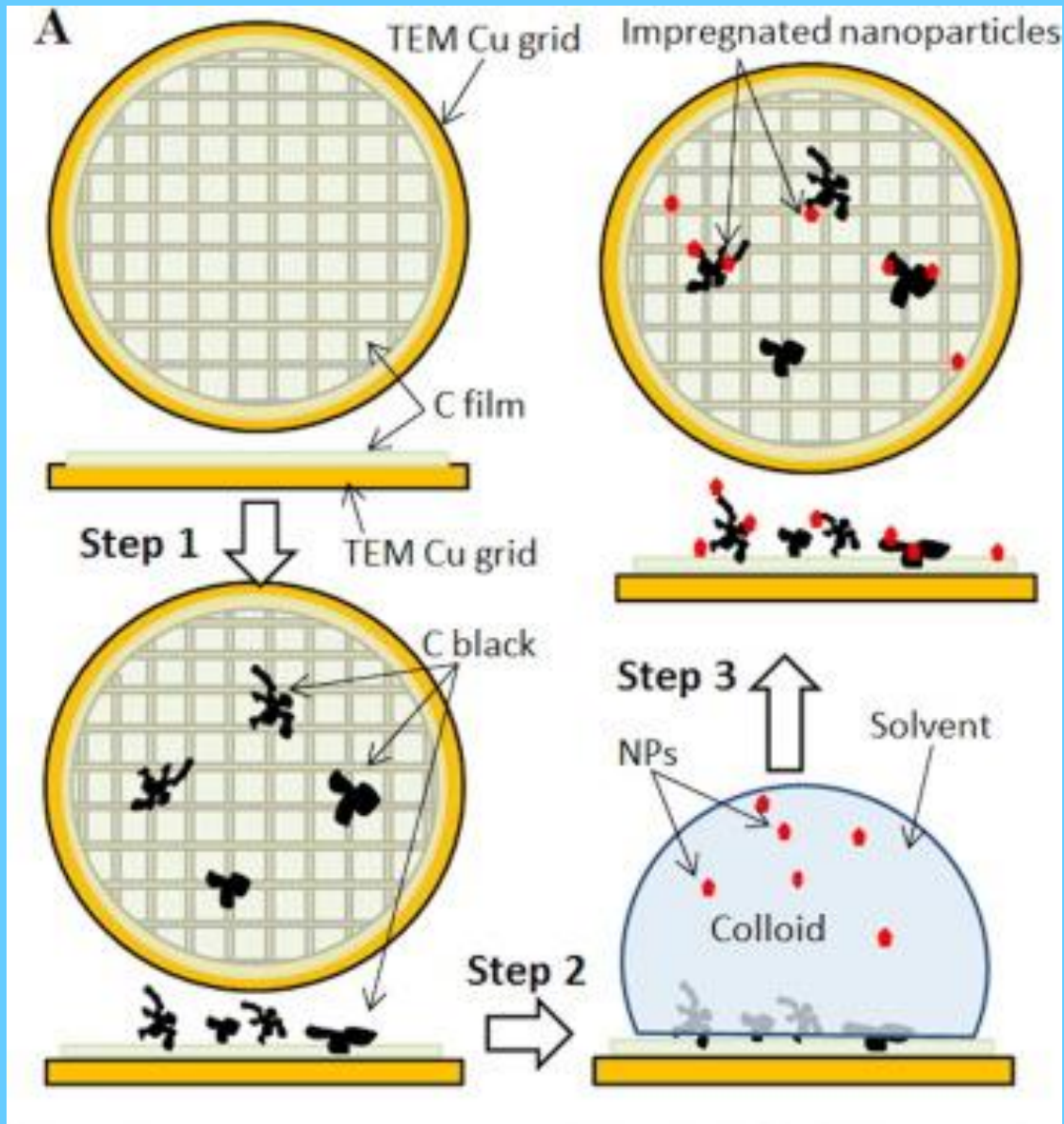
Nanoparticles from solvent



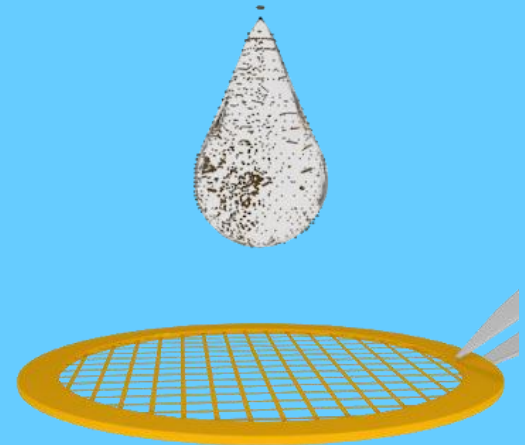
TEM grid



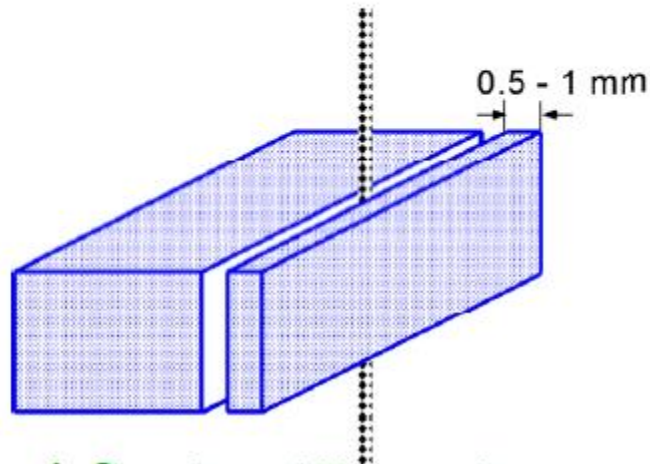
Nanoparticles from solvent



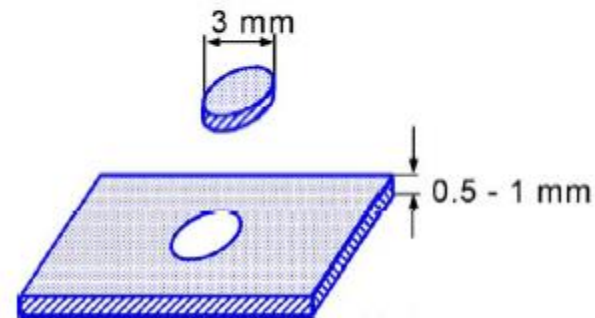
Droplet



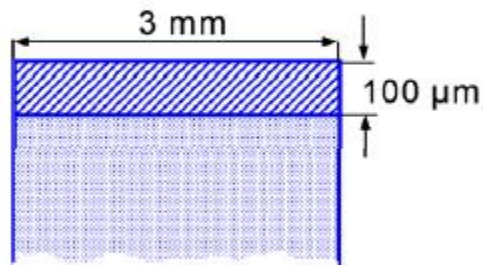
Standard Preparation



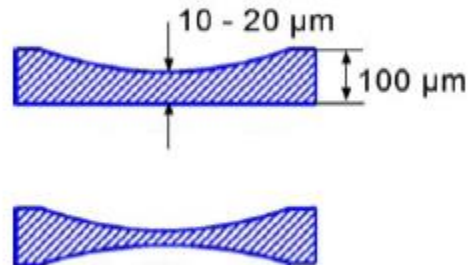
a) first cutting step



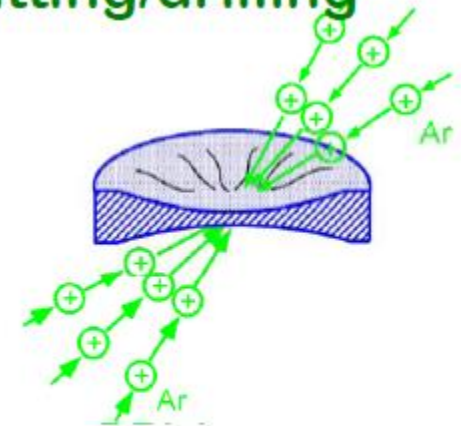
b) ultrasonic cutting/drilling



c) planparallel grinding

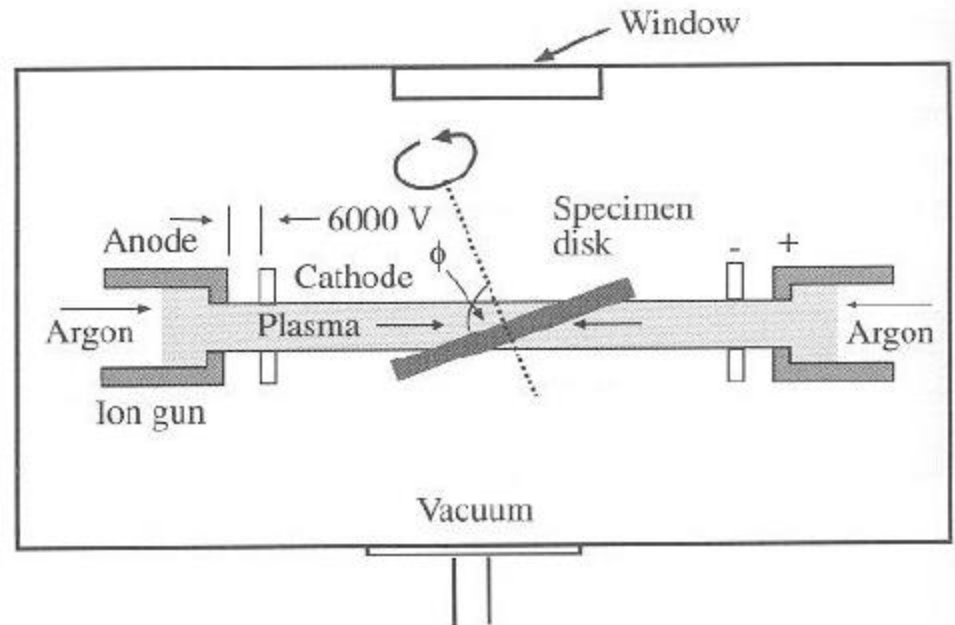


d) dimpling



e) ion-thinning

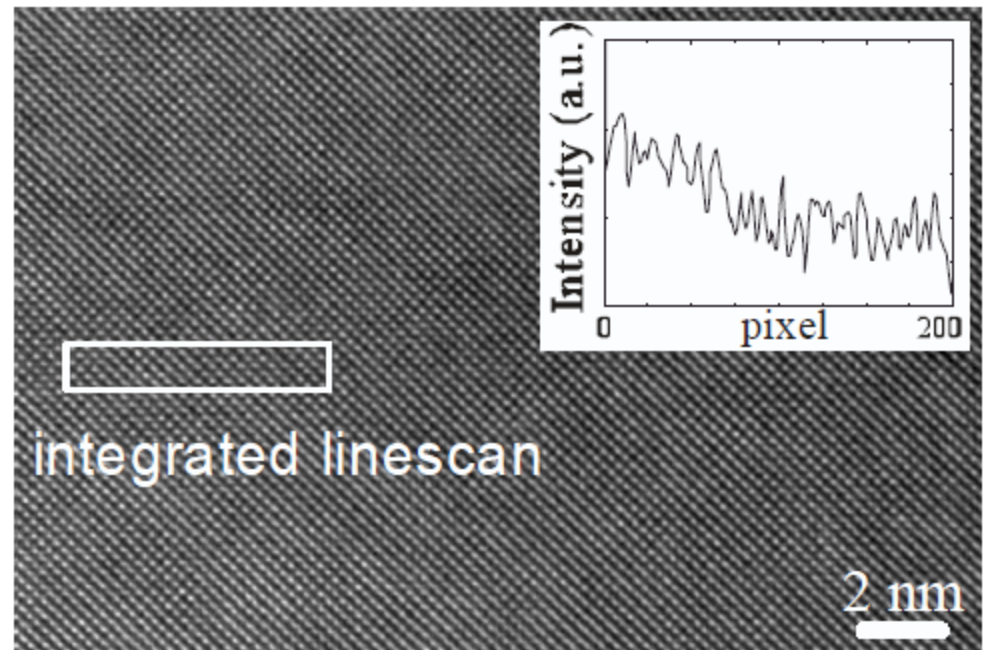
Ion-Milling



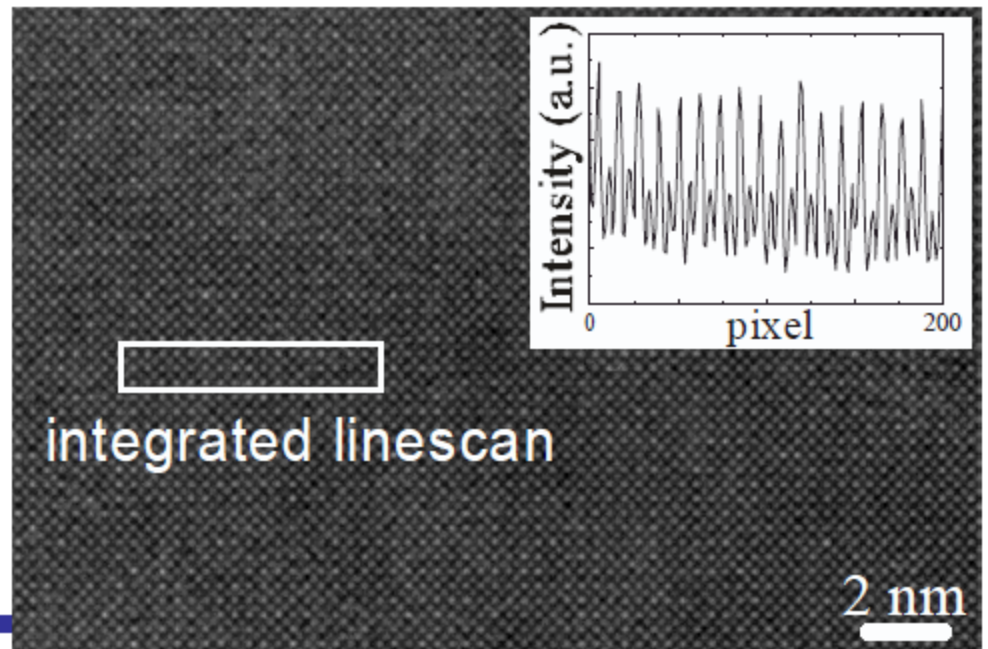
- **Energy:** (~1-7 KV)
- **Current:** (~1-8mA)
- **Angle:** (~2-20°)

Ion-beam Damaging

3.4 keV, 12°

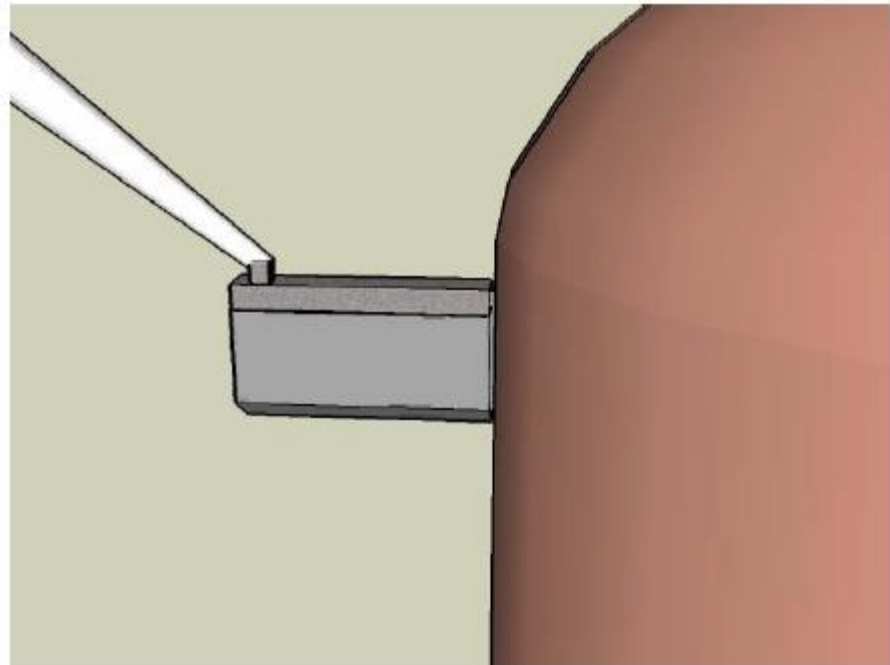


Afterwards:
400V for 30min.



TEM Sample Prep with FIB

- Pt Deposition
- Bulk-Out
- U-Cut
- Lift Out
- Mounting
- Thinning
- Cleaning



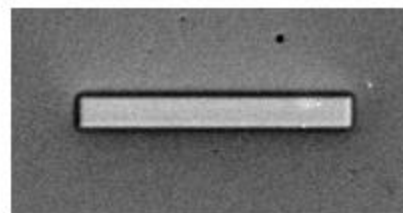
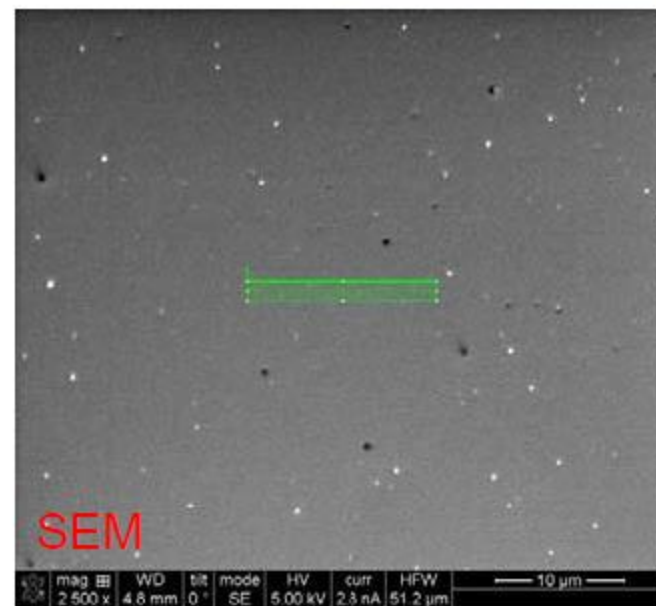
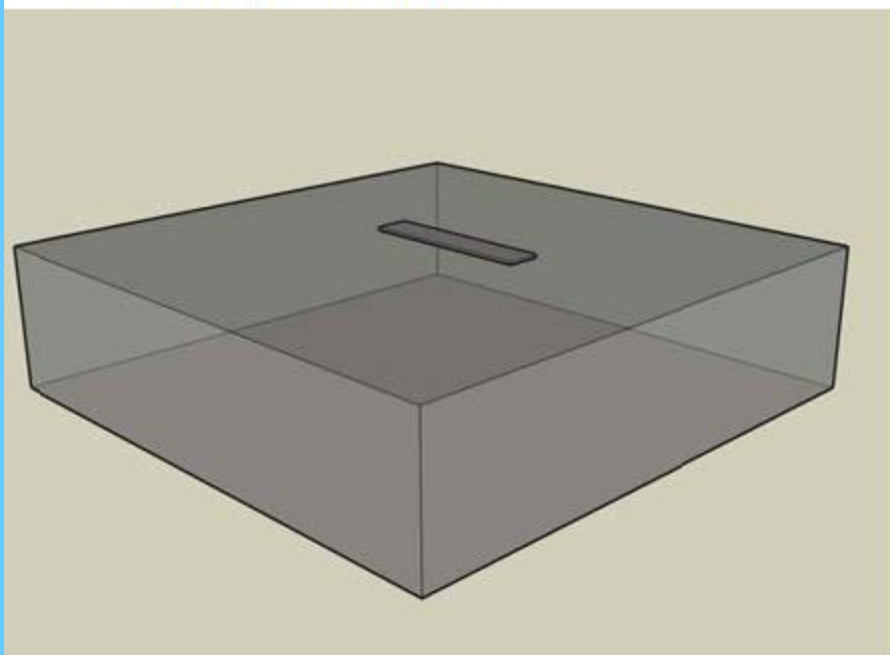
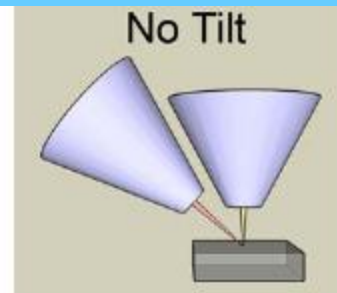
Step-by-step instructions with recipe information
and 3-D rendering of each process

TEM Sample Preparation

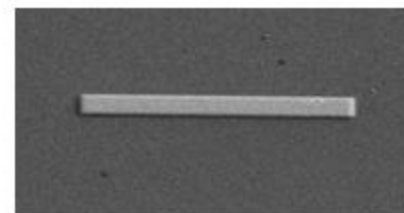
- Sample – (100) Si
- Goal – <50nm thickness with minimal amorphous material for high resolution S/TEM
- Total time for process 1-4+ hours depending on skill level
- Note: Sample was exposed to 30kV Ga beam before sample prep (ie, top 100nm will exhibit ion damage)

E-beam Pt

- 0° Tilt
- Application: Pt e-Dep
- Shape: Rectangle
- 15 μm (X) x 1.5 μm (Y) x 200nm (Z)
- 2-5kV, >1.4nA



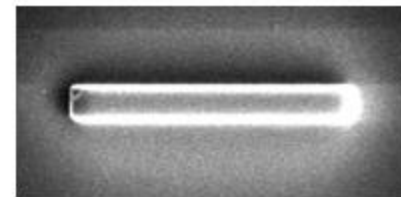
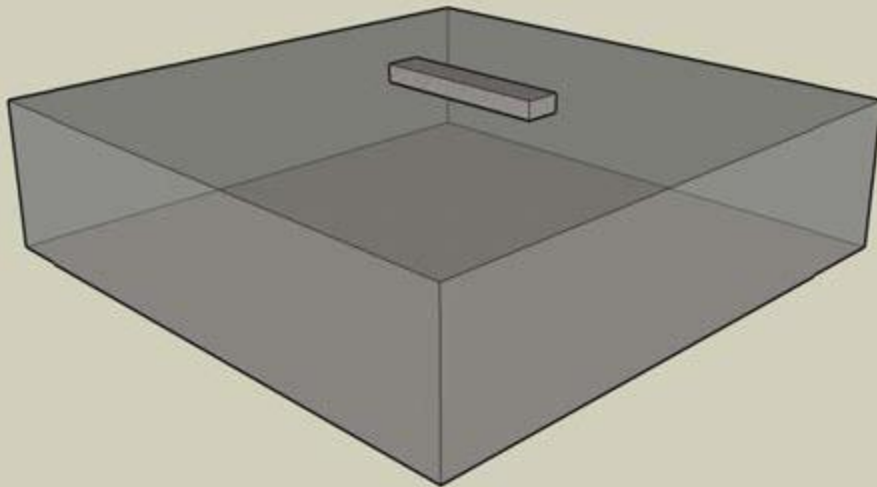
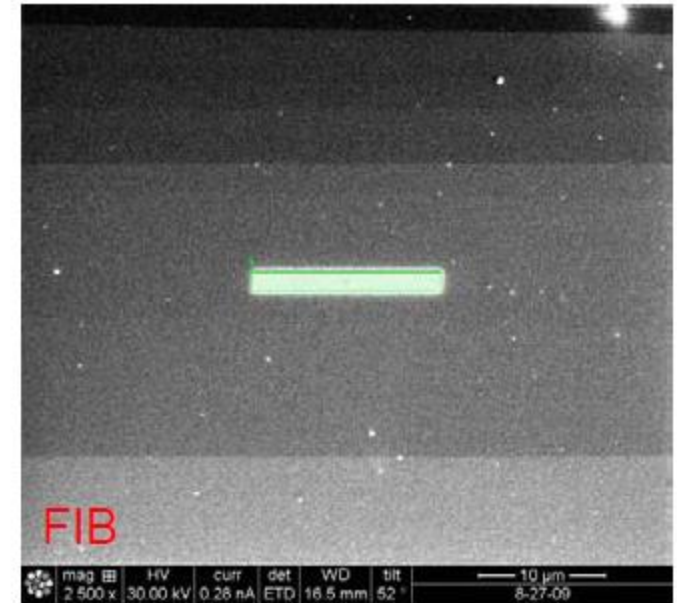
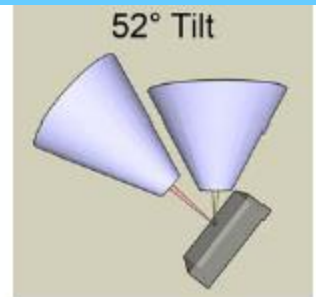
E-beam dep at
0° Tilt (SEM)



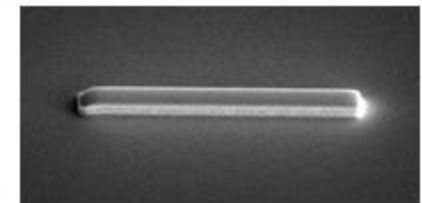
E-beam dep at
52° Tilt (SEM)

Pt Dep (Ion Beam)

- 52° Tilt
- Application: Pt Dep
- Shape: Rectangle
- $15\mu\text{m}$ (X) x $1.5\mu\text{m}$ (Y) x $1.5\mu\text{m}$ (Z)
- 30kV, 93pA-0.28nA



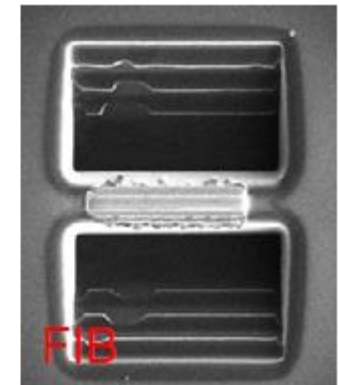
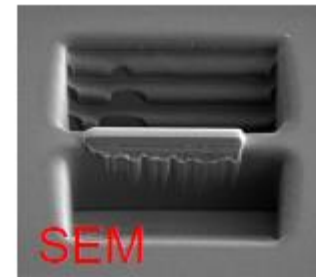
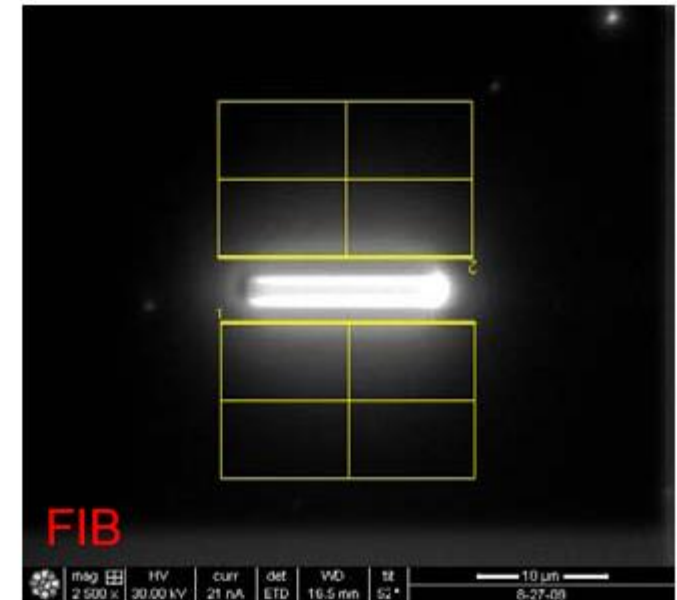
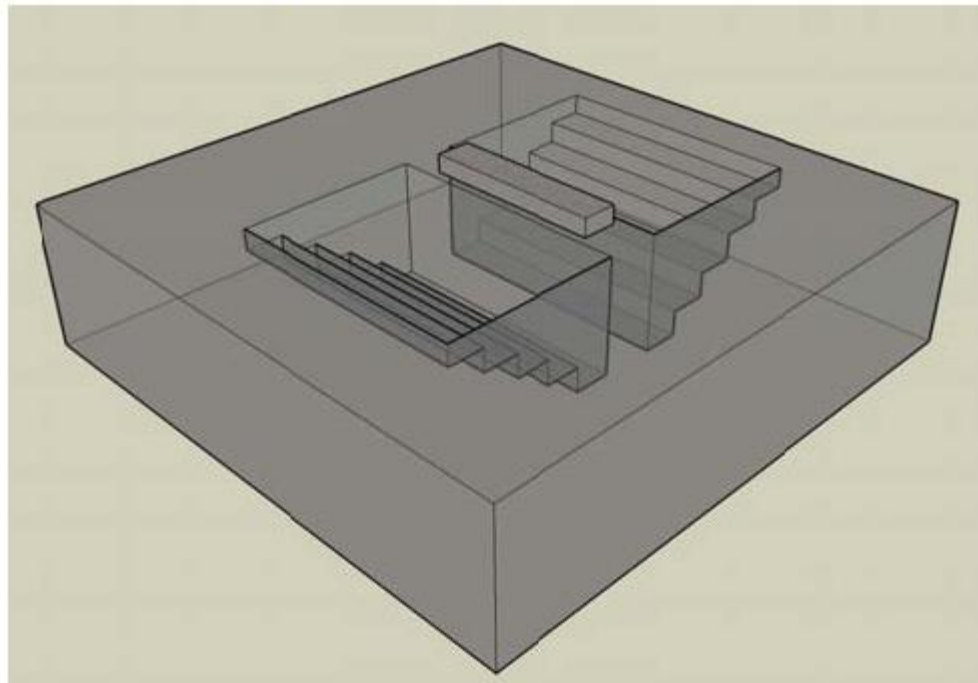
I-beam dep at
52° Tilt (FIB)



I-beam dep at
52° Tilt (SEM)

Bulk-Out (1 of 2)

- 52° Tilt
- Application: Si
- Shape: Regular Cross Section
- $20\mu\text{m}$ (X) x $12\mu\text{m}$ (Y) x $6\mu\text{m}^*$ (Z)
- 30kV, 6.5-21nA

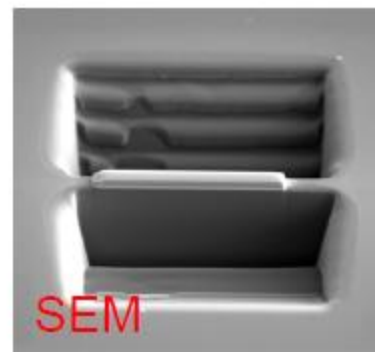
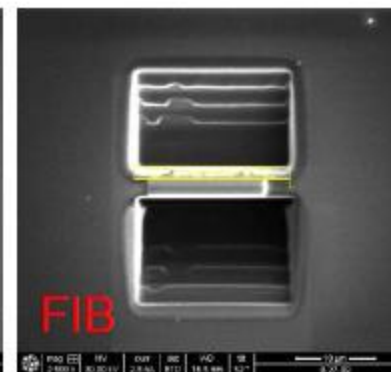
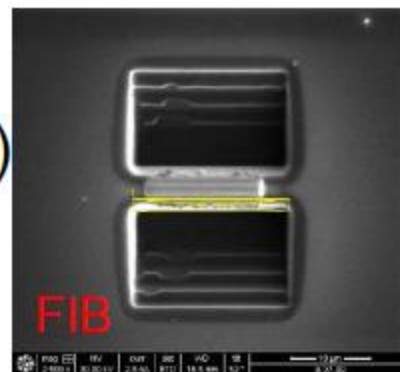
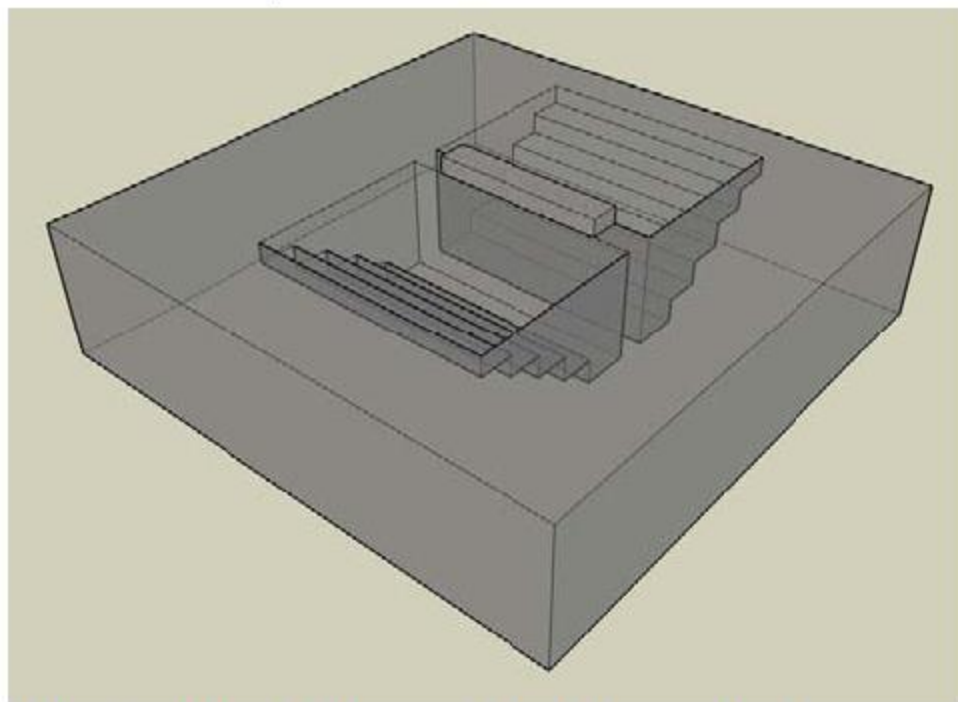


Bulk-Out (1/2)
at 52° Tilt

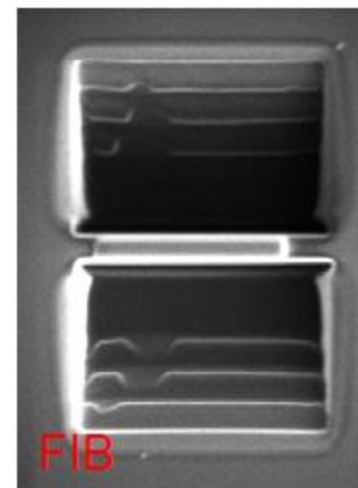
* Actual depth is material dependent

Bulk-Out (2 of 2)

- 52° Tilt
- Application: Si
- Shape: Cleaning Cross Section
- $20\mu\text{m}$ (X) x $1.5\mu\text{m}$ (Y) x $6\mu\text{m}^*$ (Z)
- 30kV, 2.8-6.5nA

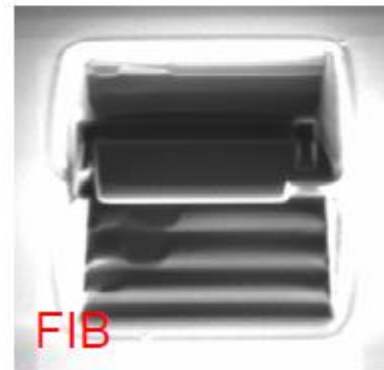
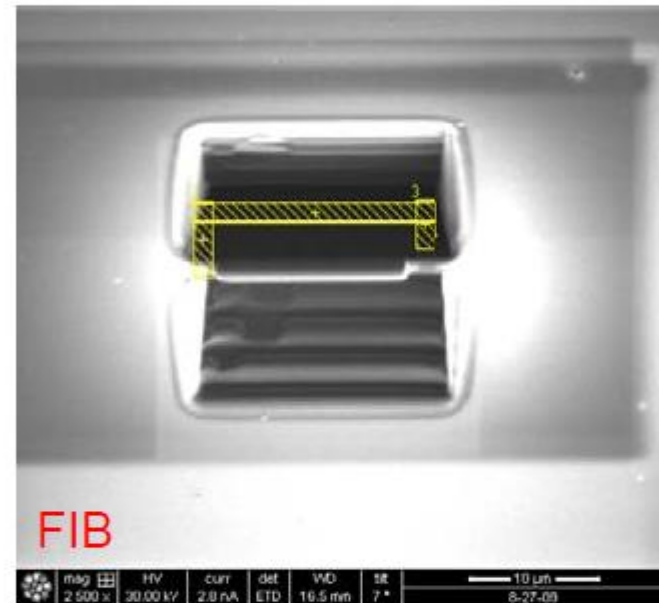
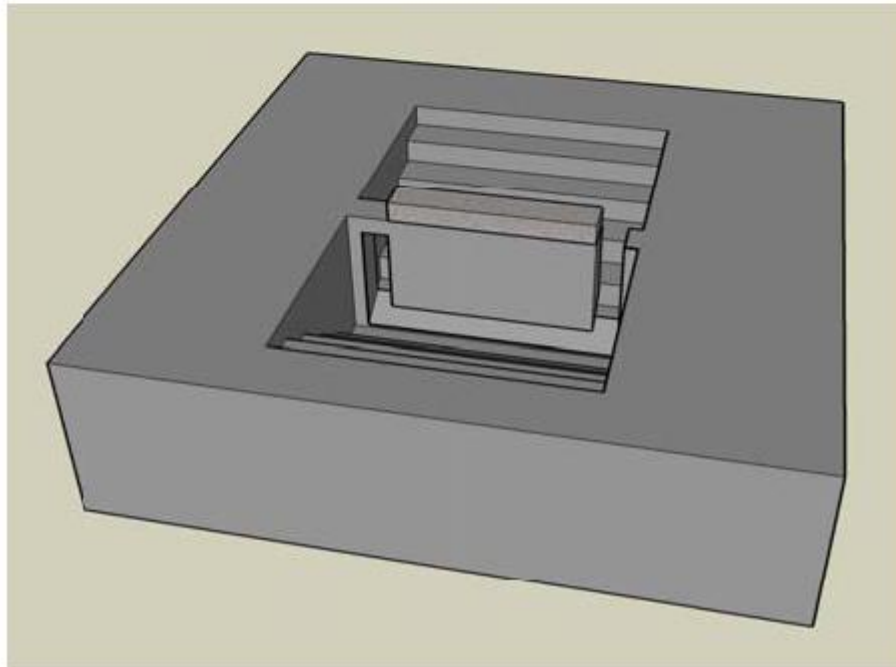
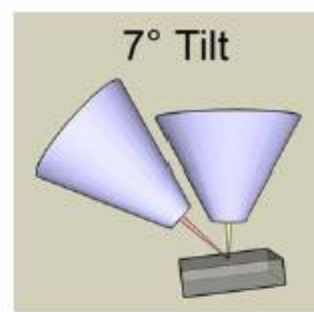


Bulk-Out (2/2)
at 52° Tilt

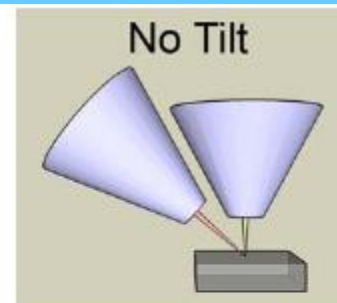


U-Cut

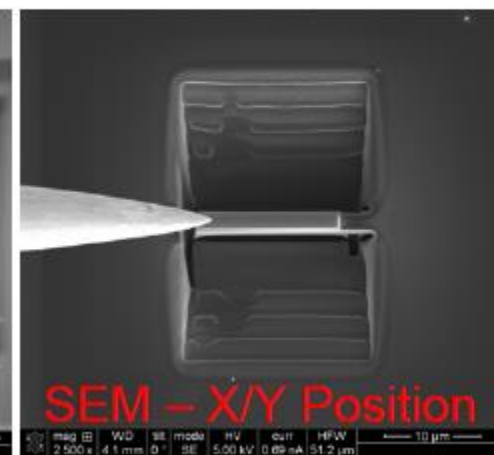
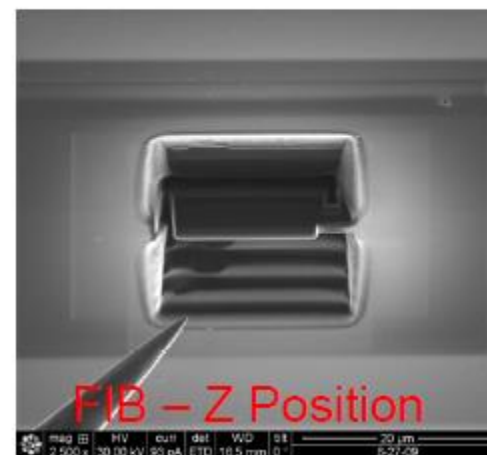
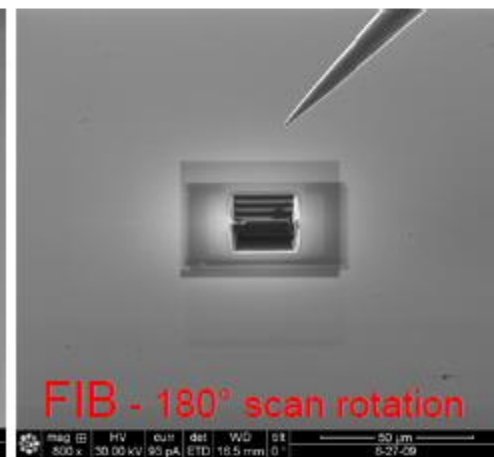
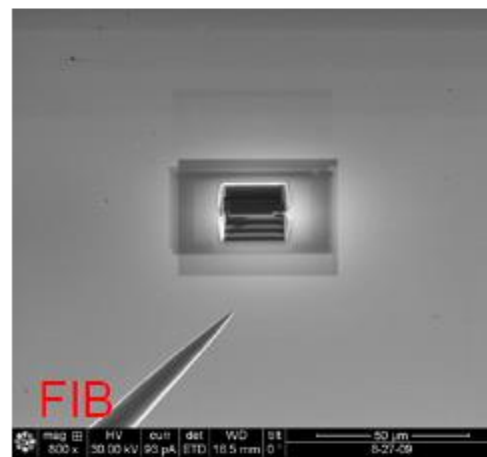
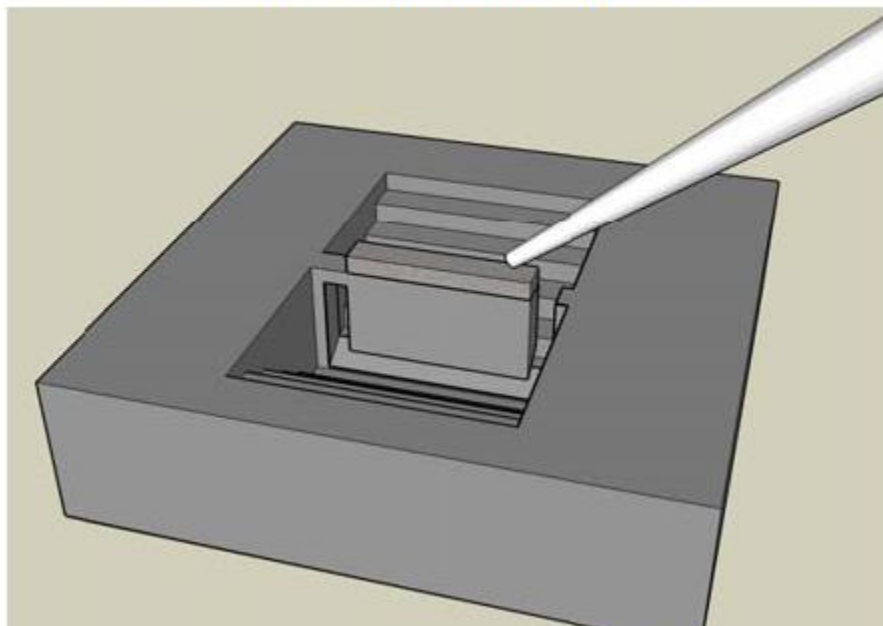
- 7° Tilt
- Application: Si
- Shape: Rectangles
- 1.5 μm wide, overlapping
- 30kV, 0.92-6.5nA



Lift-Out (1 of 4)

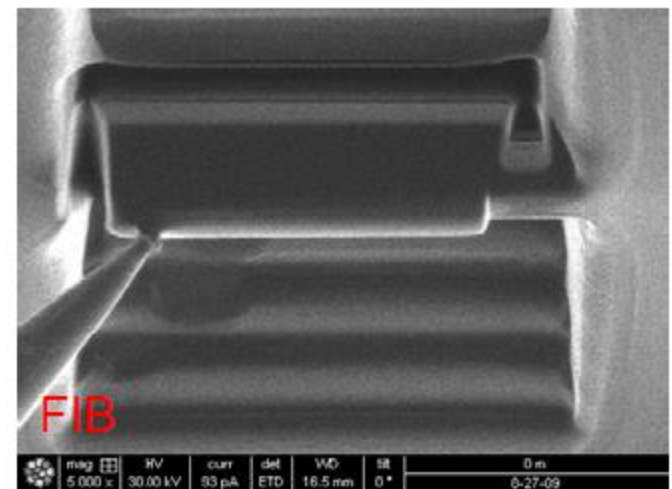
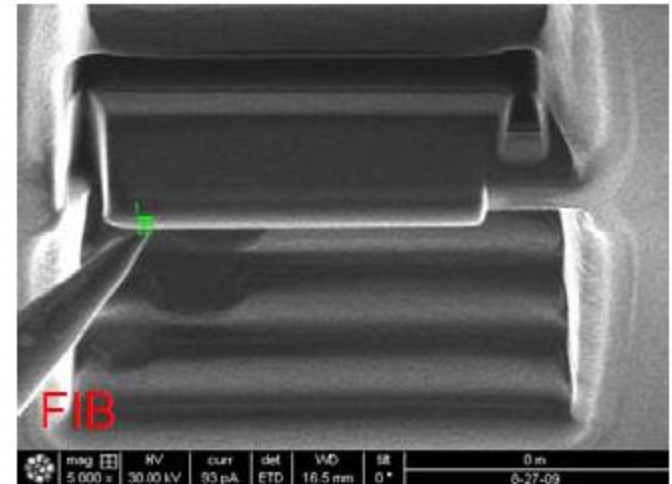
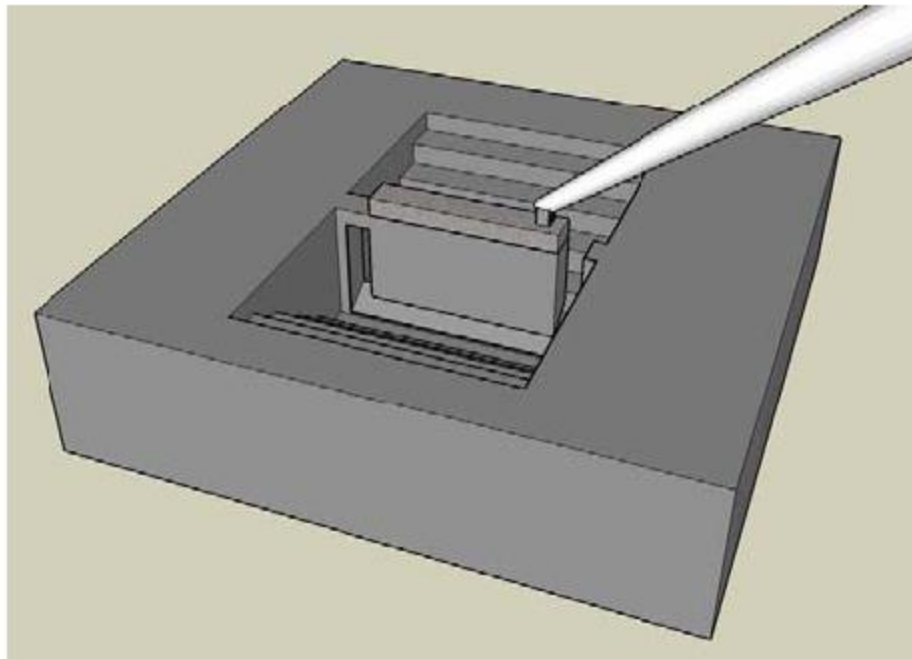
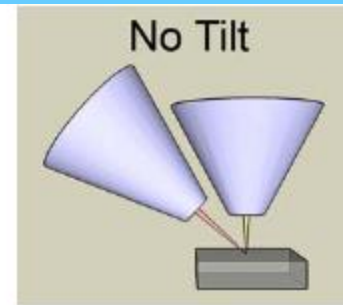


- 0° Tilt
- Insert Omniprobe at Park
- Drive to Eucentric High
- Lower to ~10 μ m from sample
- Insert Pt GIS needle
- Position tip within ~200nm of sample
- 180° scan rotation is helpful



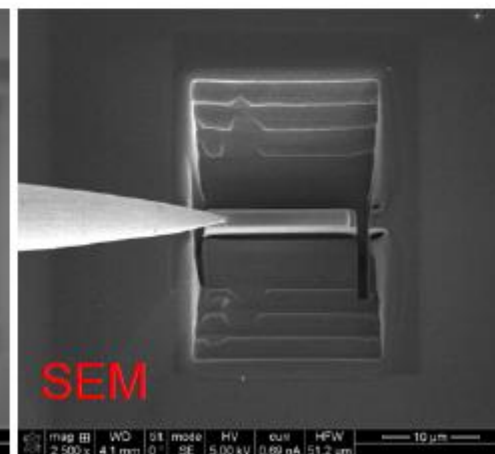
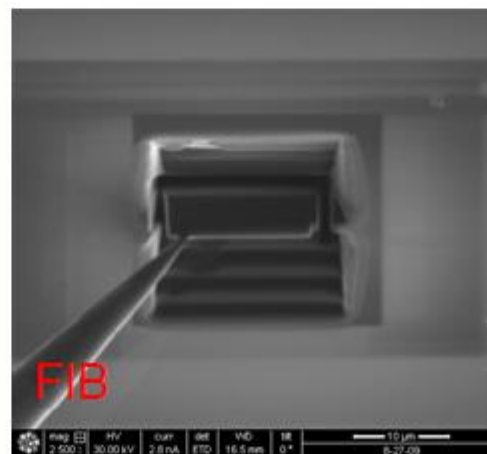
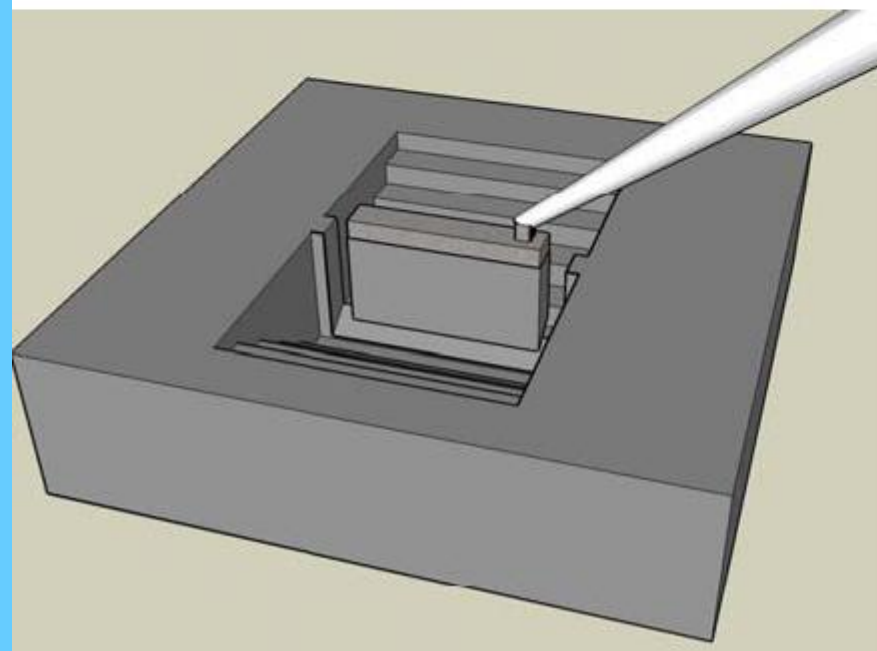
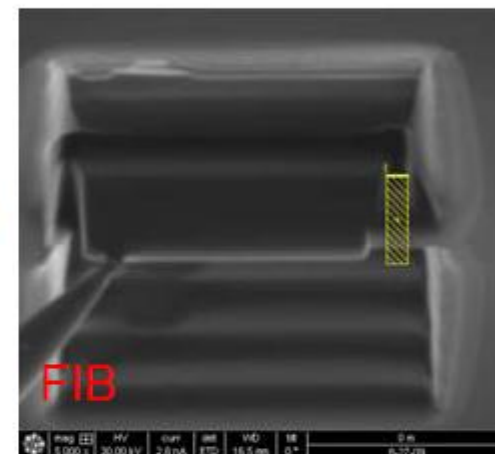
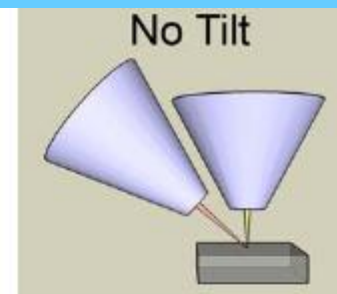
Lift-Out (2 of 4)

- 0° Tilt
- Application: Pt Dep
- Shape: Rectangle
- Size: $<1\mu\text{m}$ (X/Y) x $\sim 200\text{nm}$ (Z)
- 30kV, 93pA



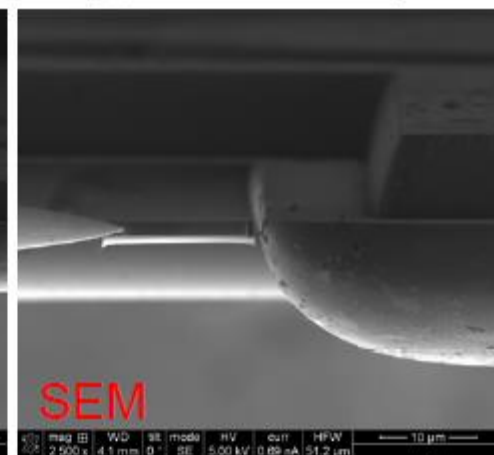
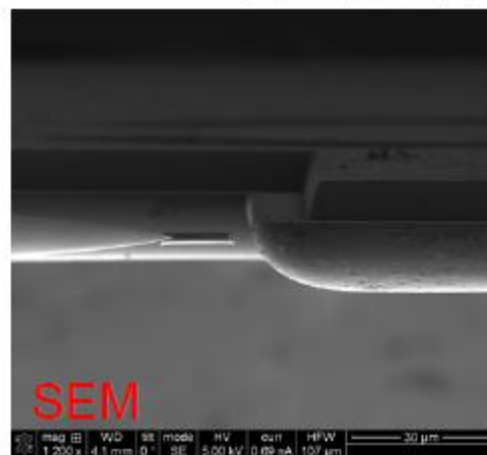
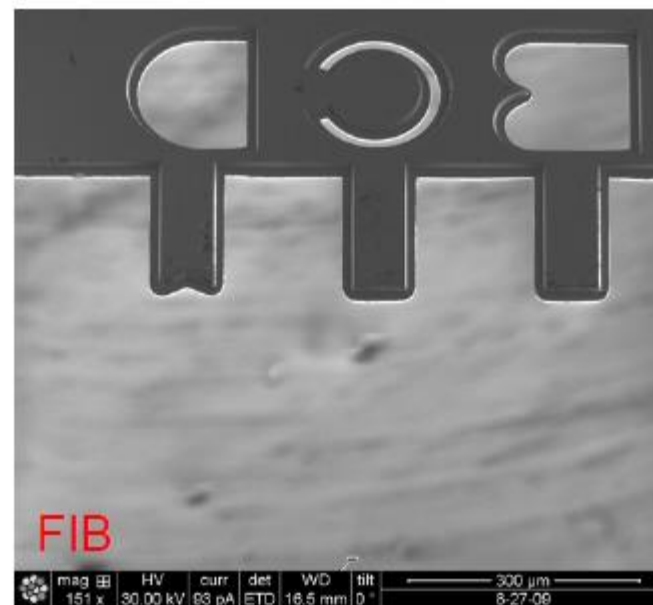
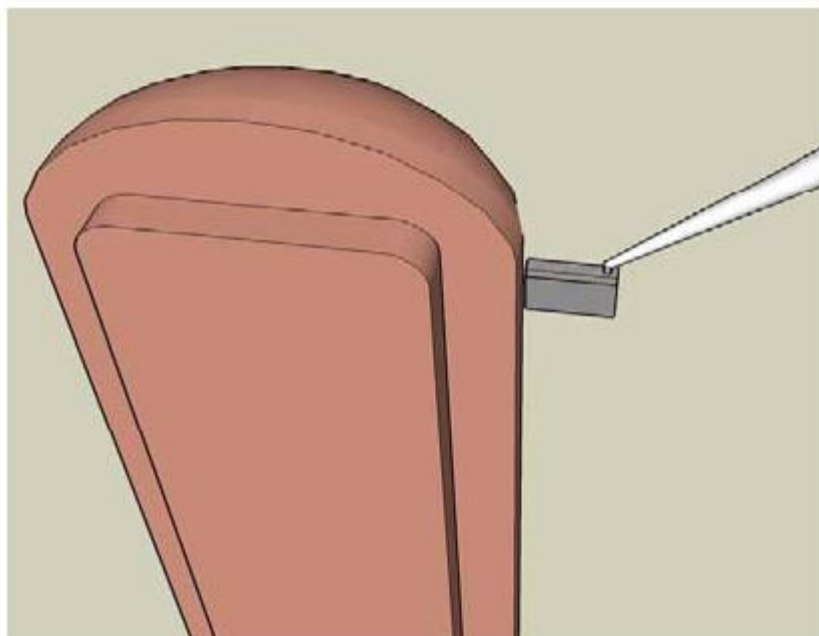
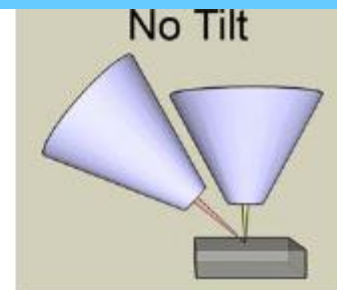
Lift-Out (3 of 4)

- 0° Tilt
- Application: Si
- Shape: Rectangle
- Size: $1.5\mu\text{m}$ (X) x $2\text{--}4\mu\text{m}$ (Y) x $\sim 2\mu\text{m}$ (Z)
- 30kV, 2.8nA



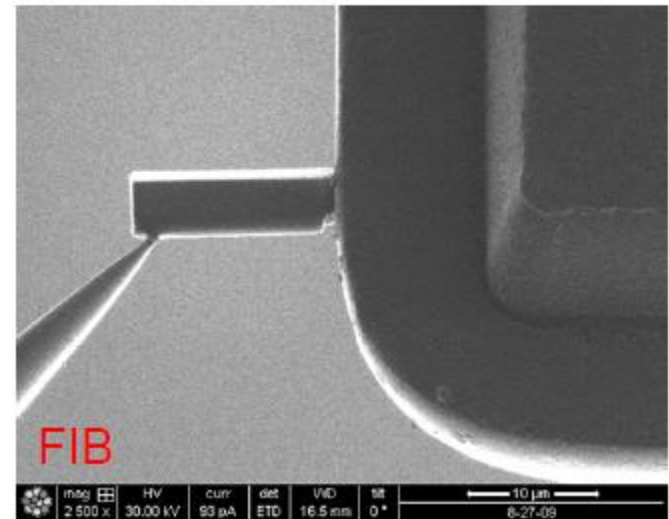
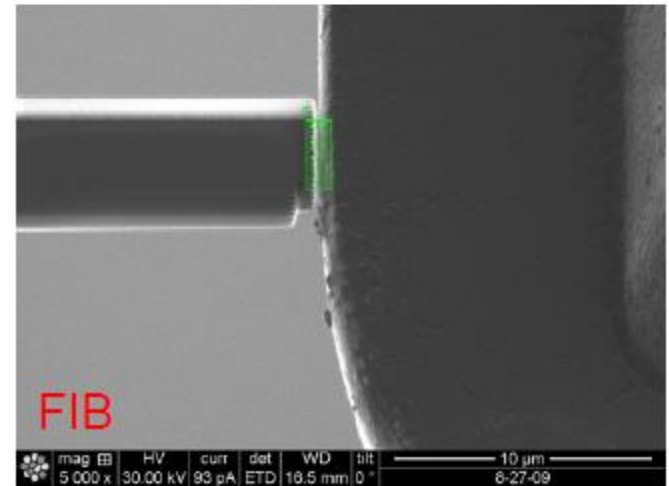
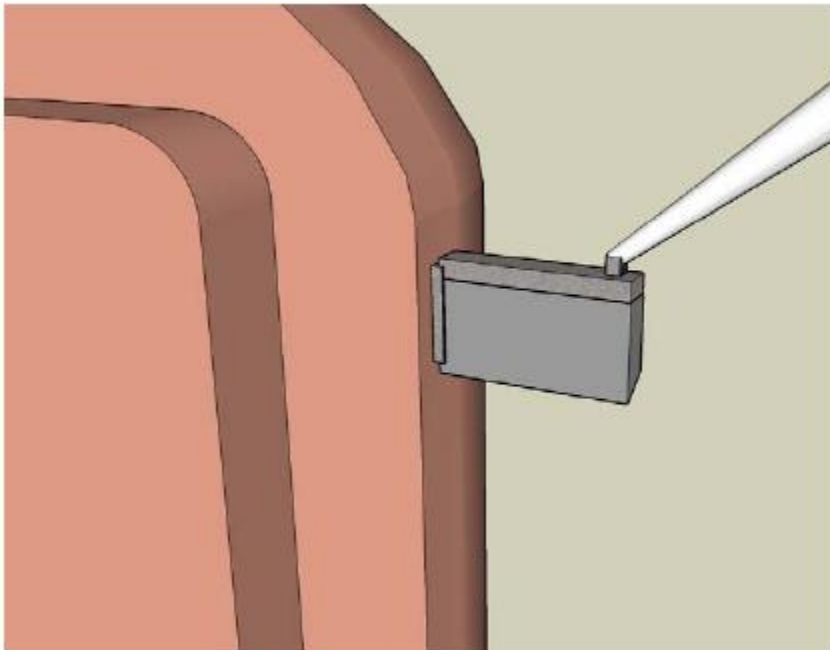
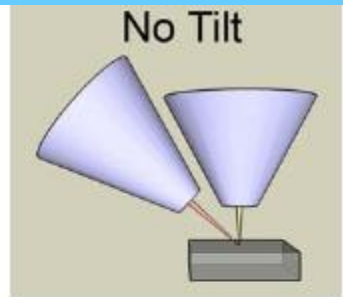
Mounting (1 of 3)

- 0° Tilt
- Reset eucentric height on grid
- Insert Omniprobe at Eucentric High
- Lower to within ~10 μ m of grid
- Insert GIS
- Position sample within ~200nm of grid



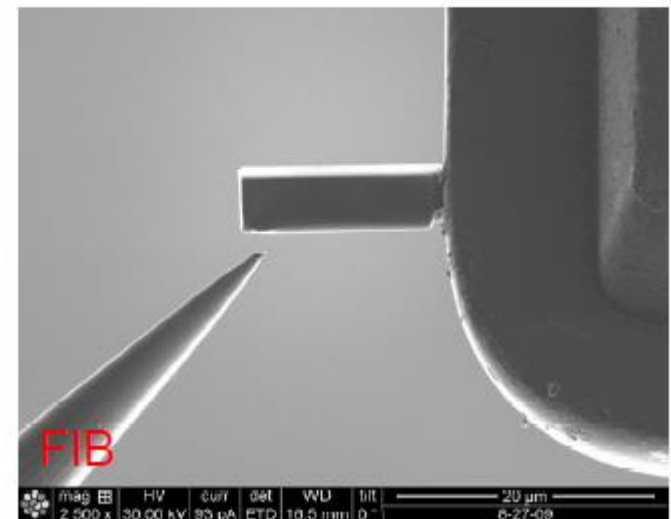
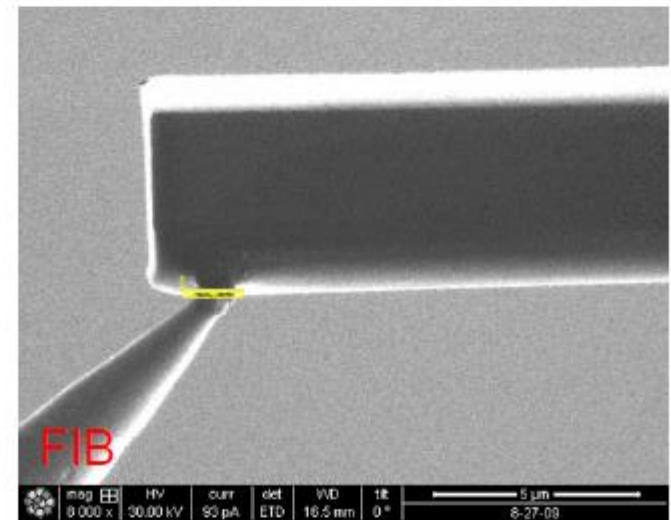
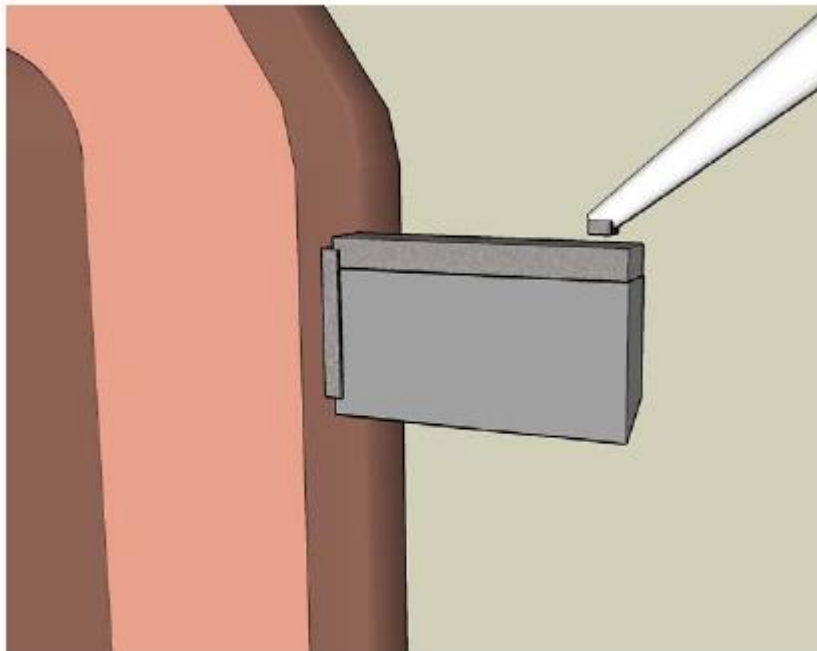
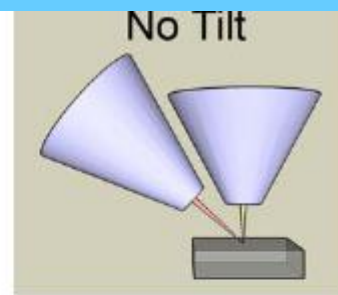
Mounting (2 of 3)

- 0° Tilt
- Application: Pt Dep
- Shape: Rectangle
- Size: $\sim 1\mu\text{m}$ (X) x $3\text{--}5\mu\text{m}$ (Y) x $0.5\mu\text{m}$ (Z)
- 30kV, 28-93pA



Mounting (3 of 3)

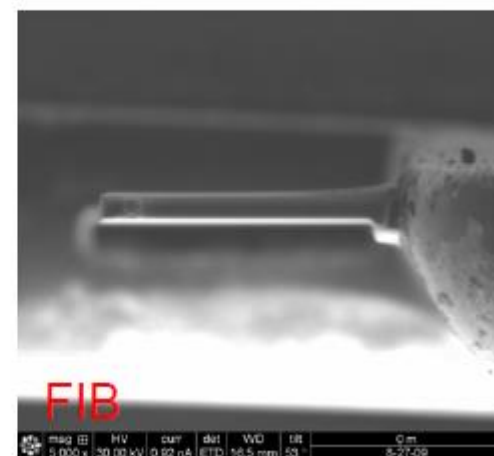
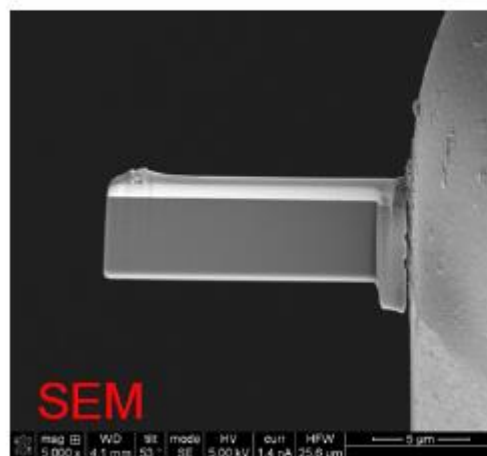
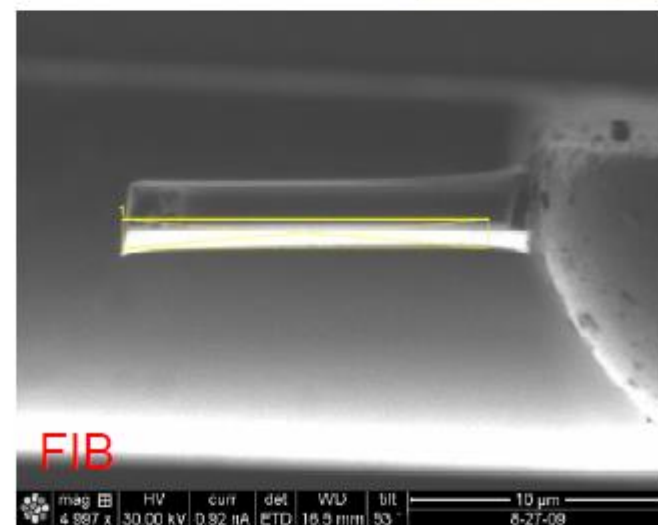
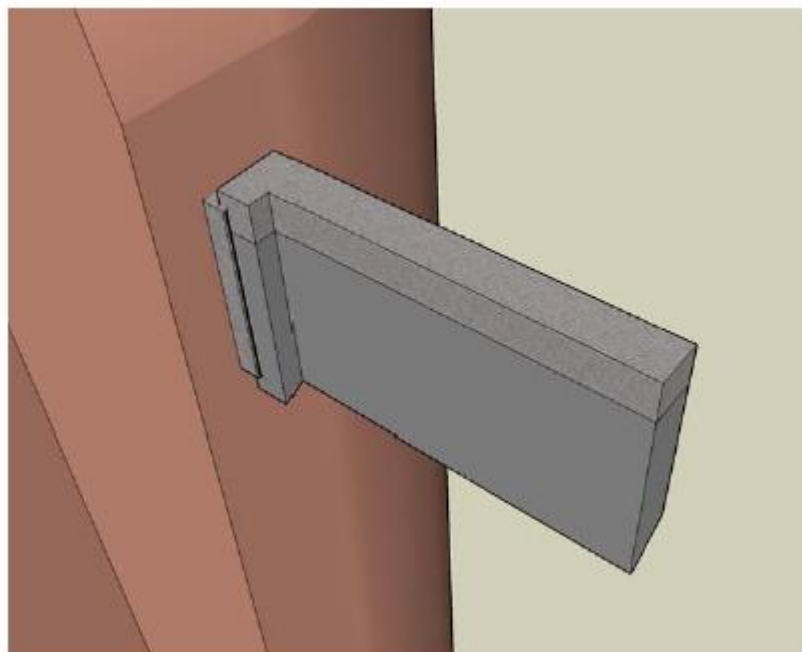
- 0° Tilt
- Application: Si
- Shape: Rectangle
- Size: $\sim 2\mu\text{m}$ (X) x $\sim 200\text{nm}$ (Y) x $\sim 1\mu\text{m}$ (Z)
- 30kV, 93pA-2.8nA



Thinning (1 of 3)

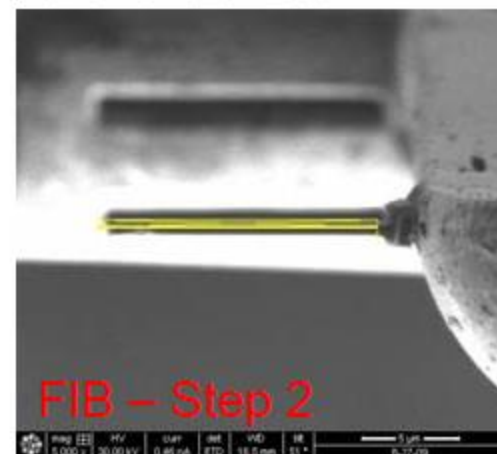
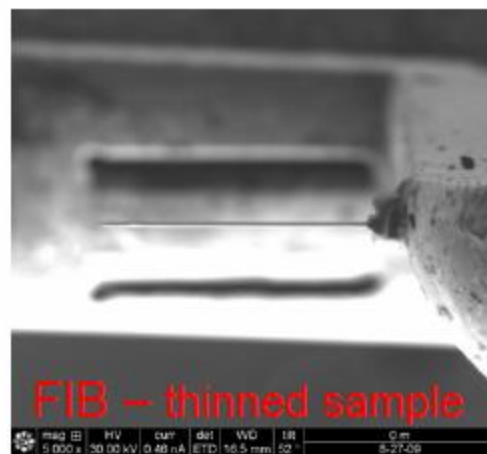
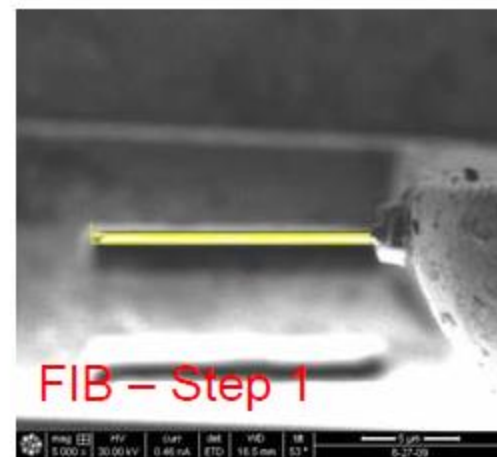
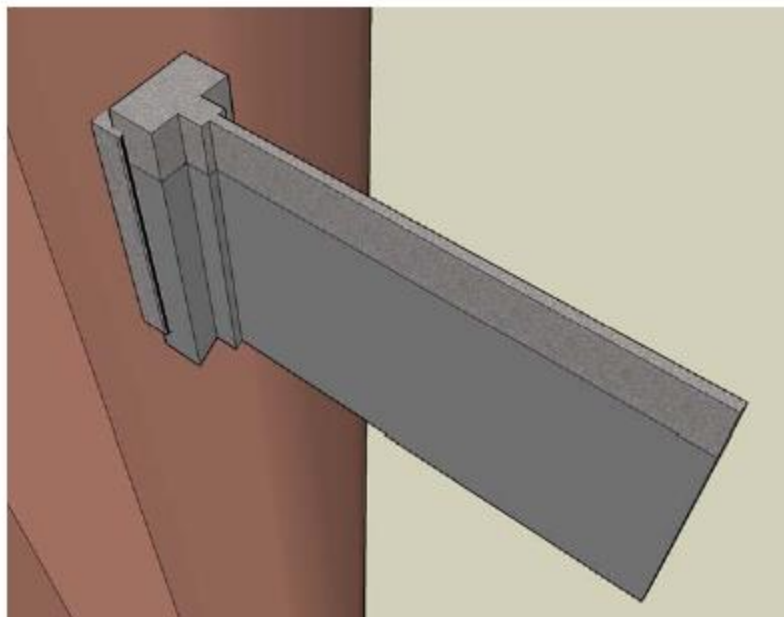


- 53-54° Tilt
- Tilt into sample 1-2°
- Application: Si
- Shape: Cleaning Cross Section
- Size: $\sim 10\mu\text{m}$ (X) x $\sim 500\text{nm}$ (Y) x $\sim 3\mu\text{m}$ (Z)
- 30kV, 0.46-2.8nA

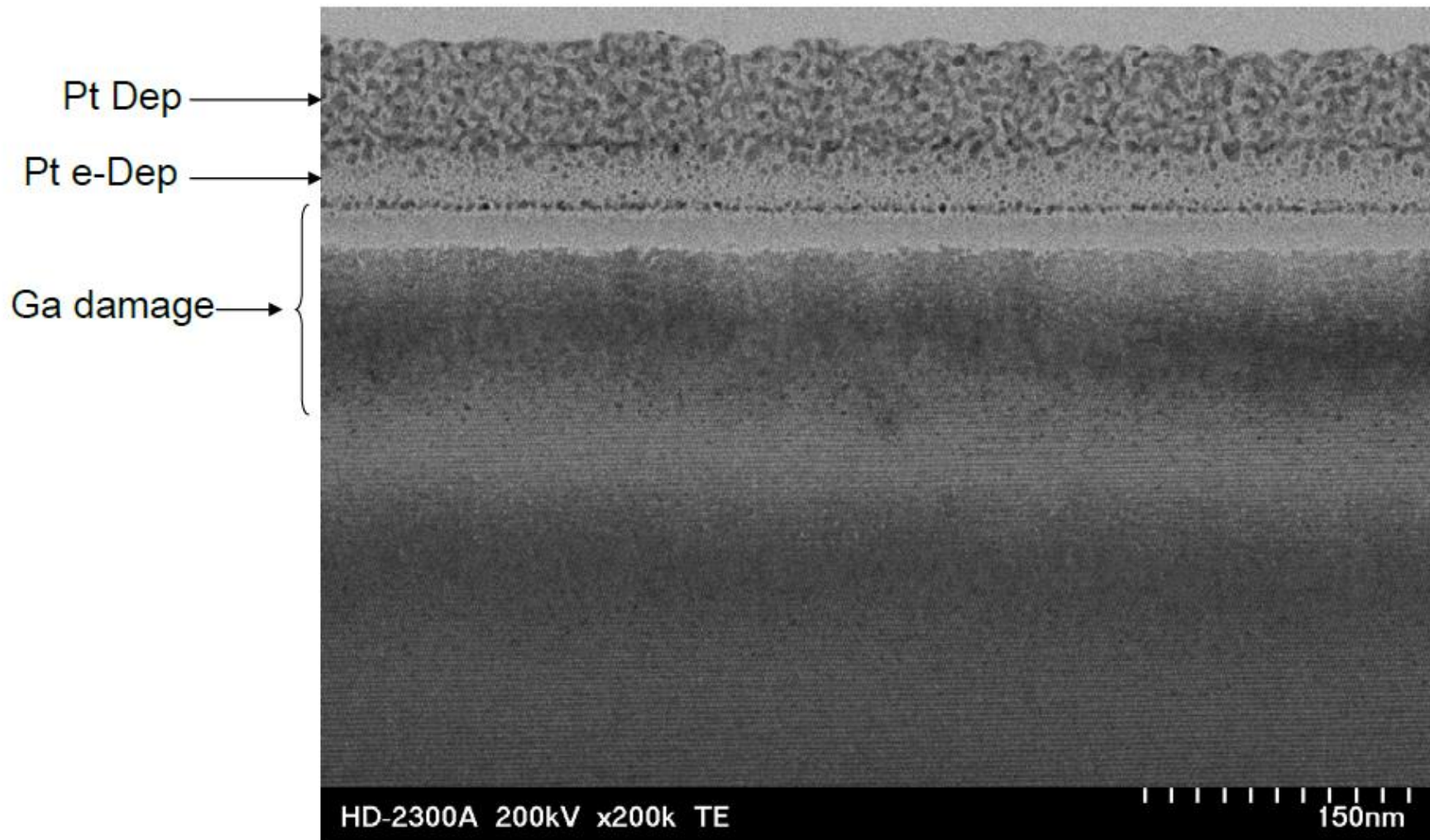


Thinning (3 of 3)

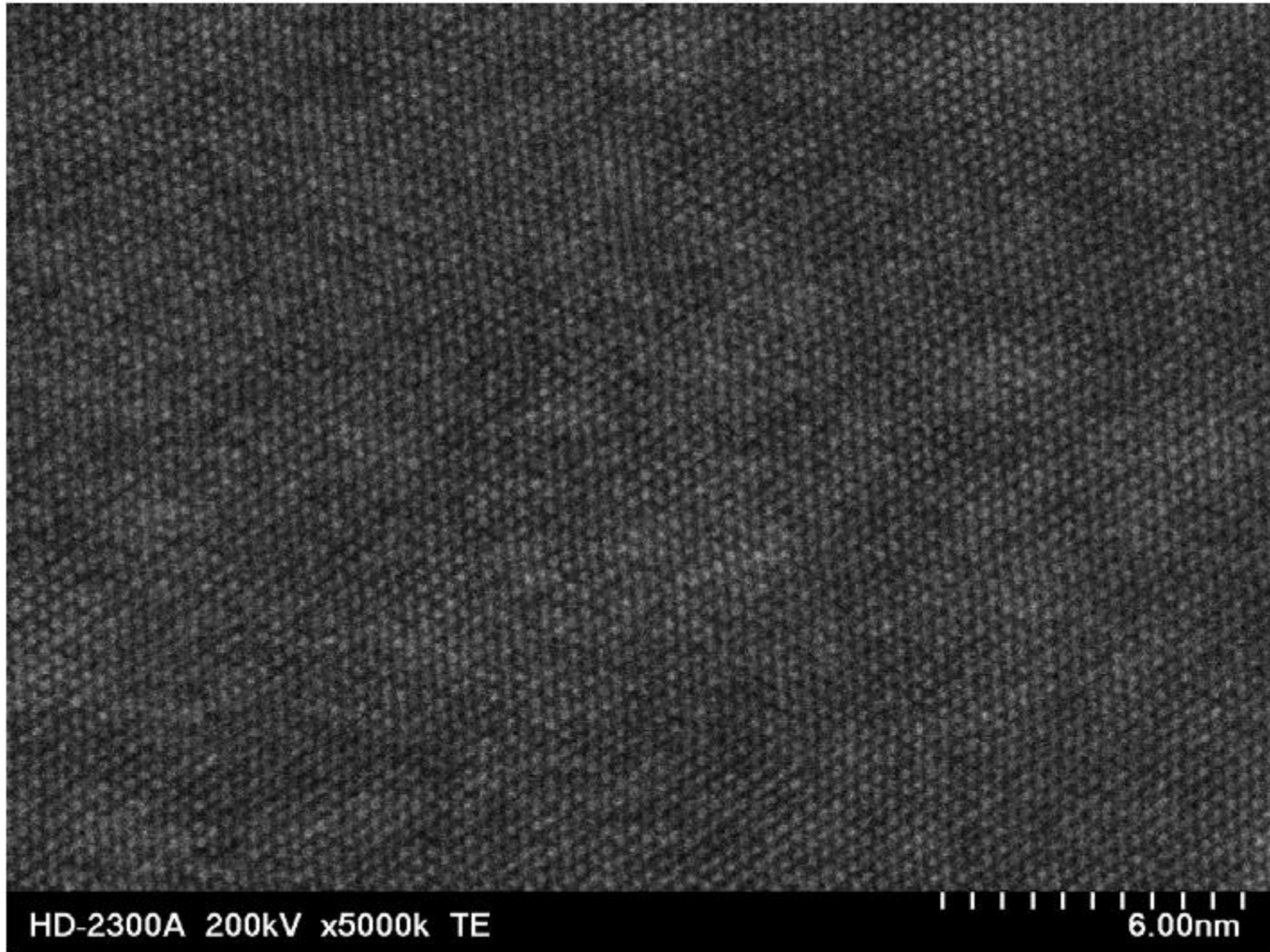
- Repeat thinning steps 1 and 2
- Target final sample thickness +60nm
- Application: Si
- Shape: Cleaning Cross Section
- Size: $\sim 10\mu\text{m}$ (X) x $\sim 200\text{nm}$ (Y) x $\sim 3\mu\text{m}$ (Z)
- 30kV, 93pA-0.46nA



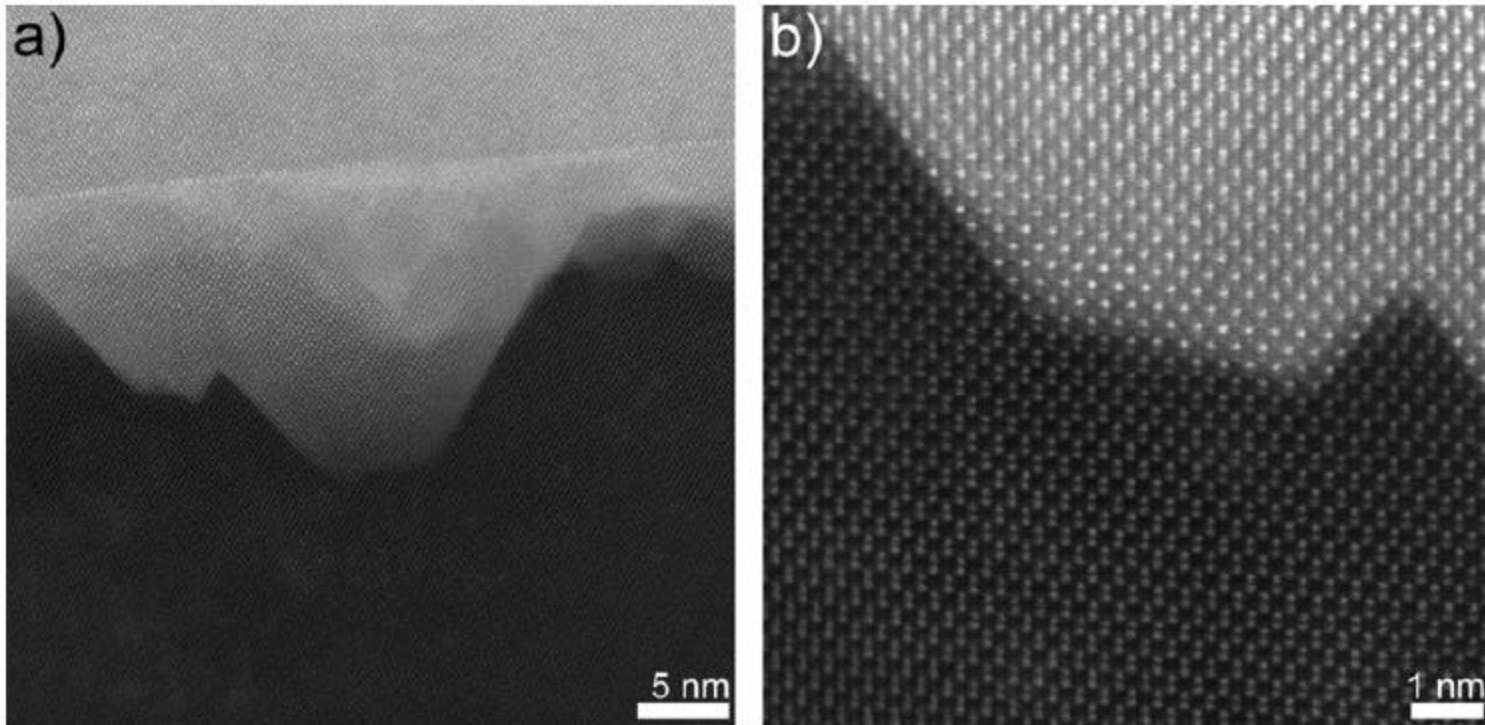
(S)TEM Results (HD2300)



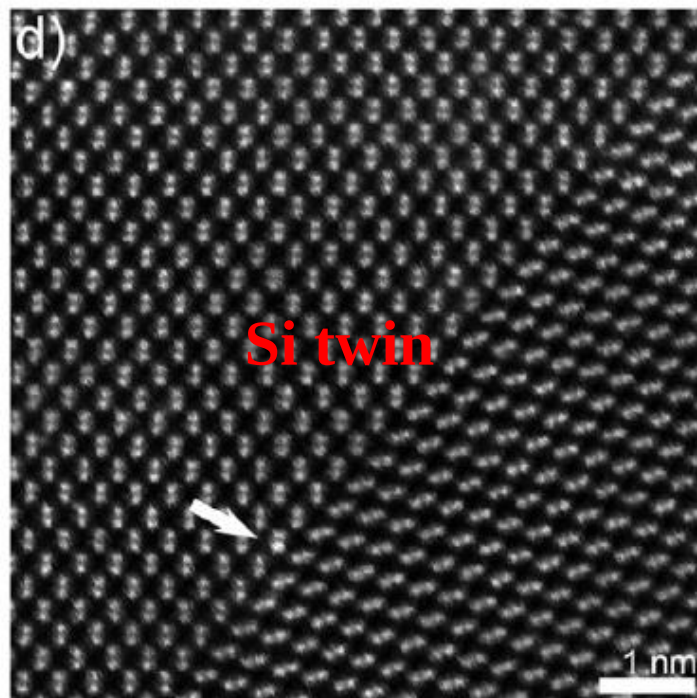
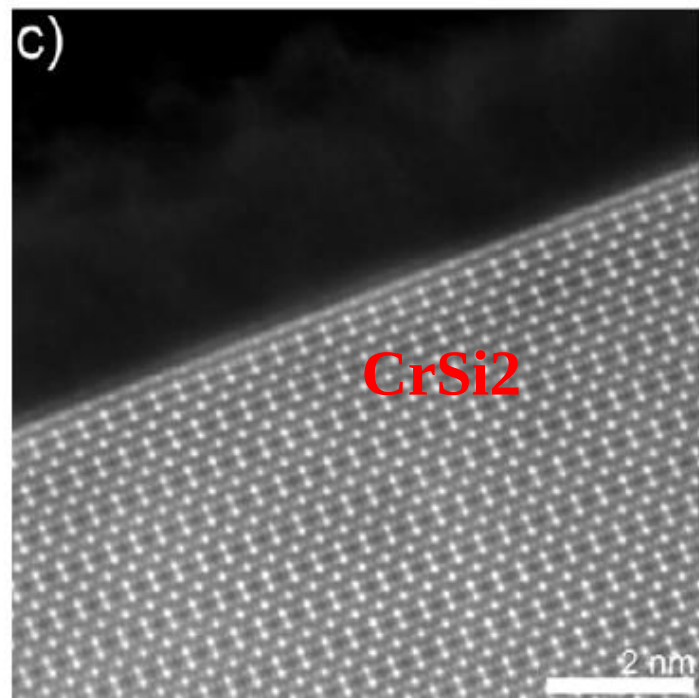
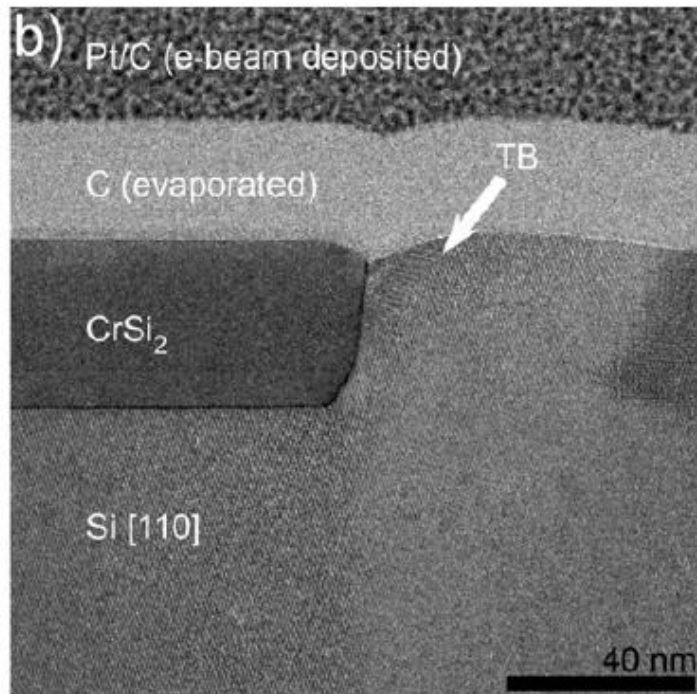
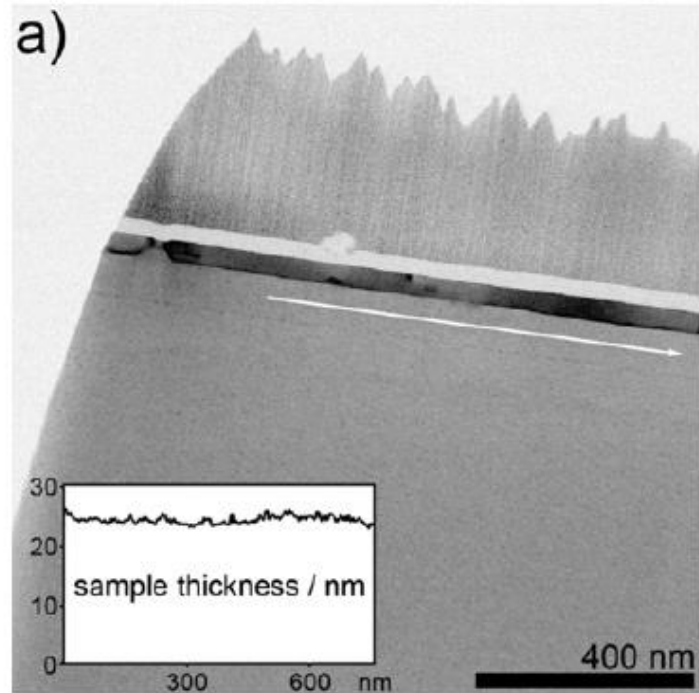
Bright Field STEM



FIB Sample with low voltage cleaning

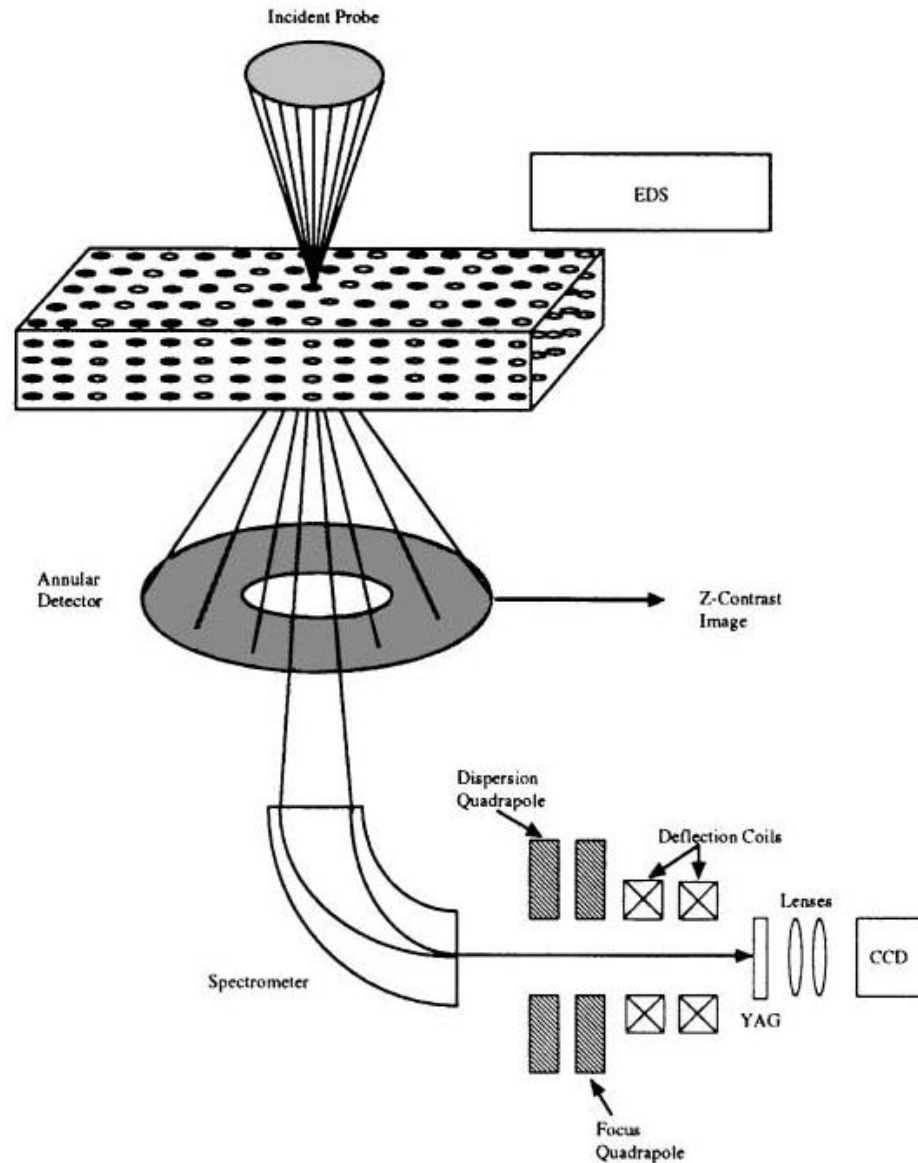


STEM HAADF images of the pyramid features found in backside prepared NiSi_2/Si lamella. The features show sharp interfaces between the NiSi_2 and Si layers.

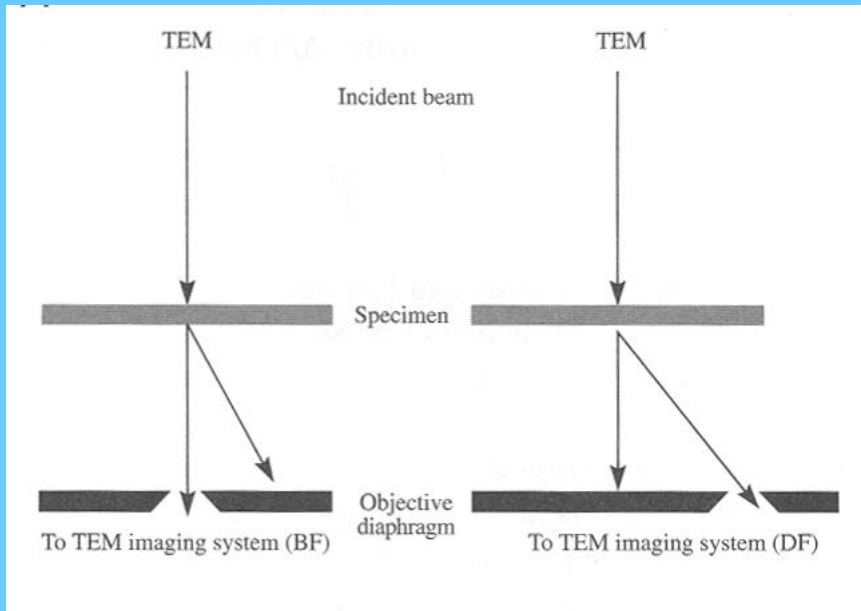


Scanning transmission electron microscopy (STEM)

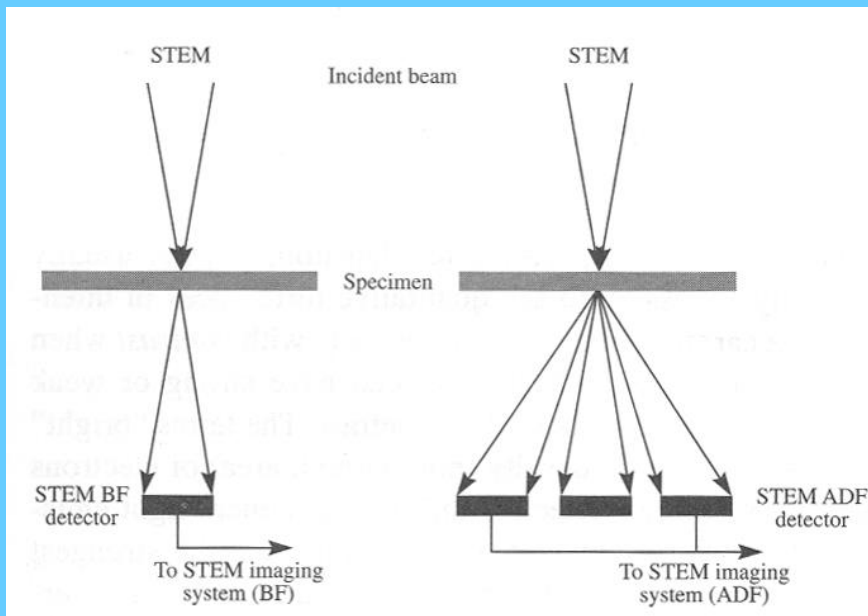
Scanning Transmission Electron Microscopy (STEM)



Scanning Transmission Electron Microscopy (STEM)

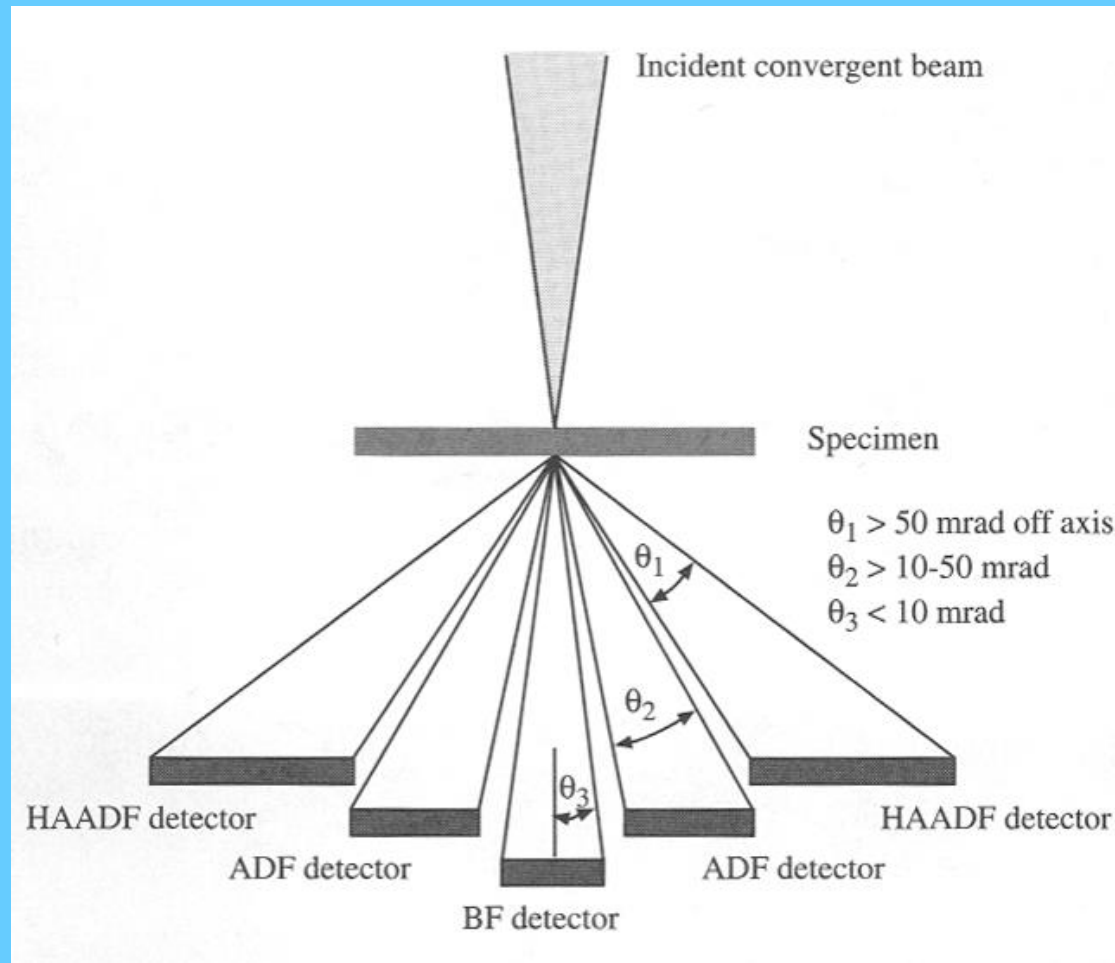


TEM: parallel beam



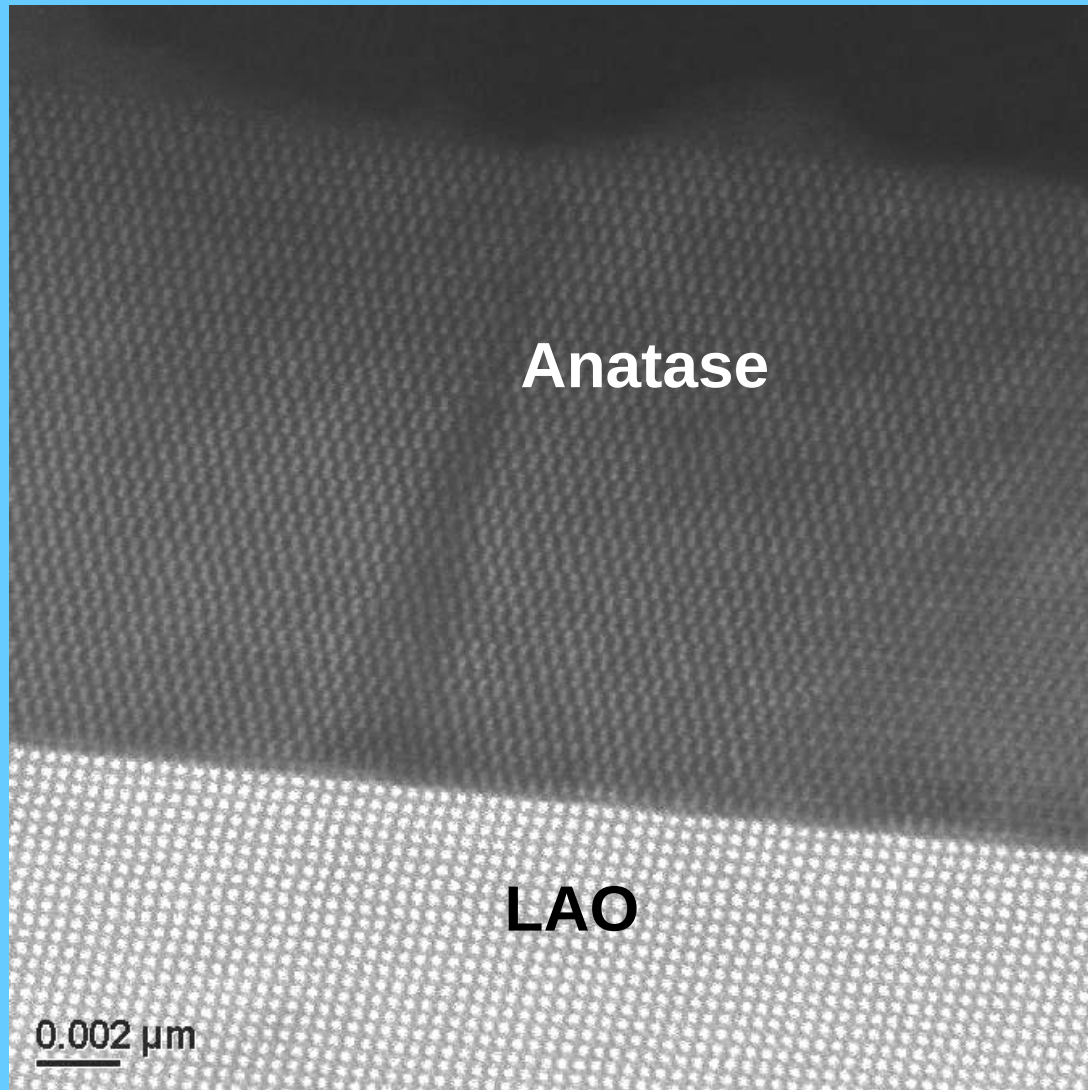
STEM: focused probe

Scanning Transmission Electron Microscopy (STEM)

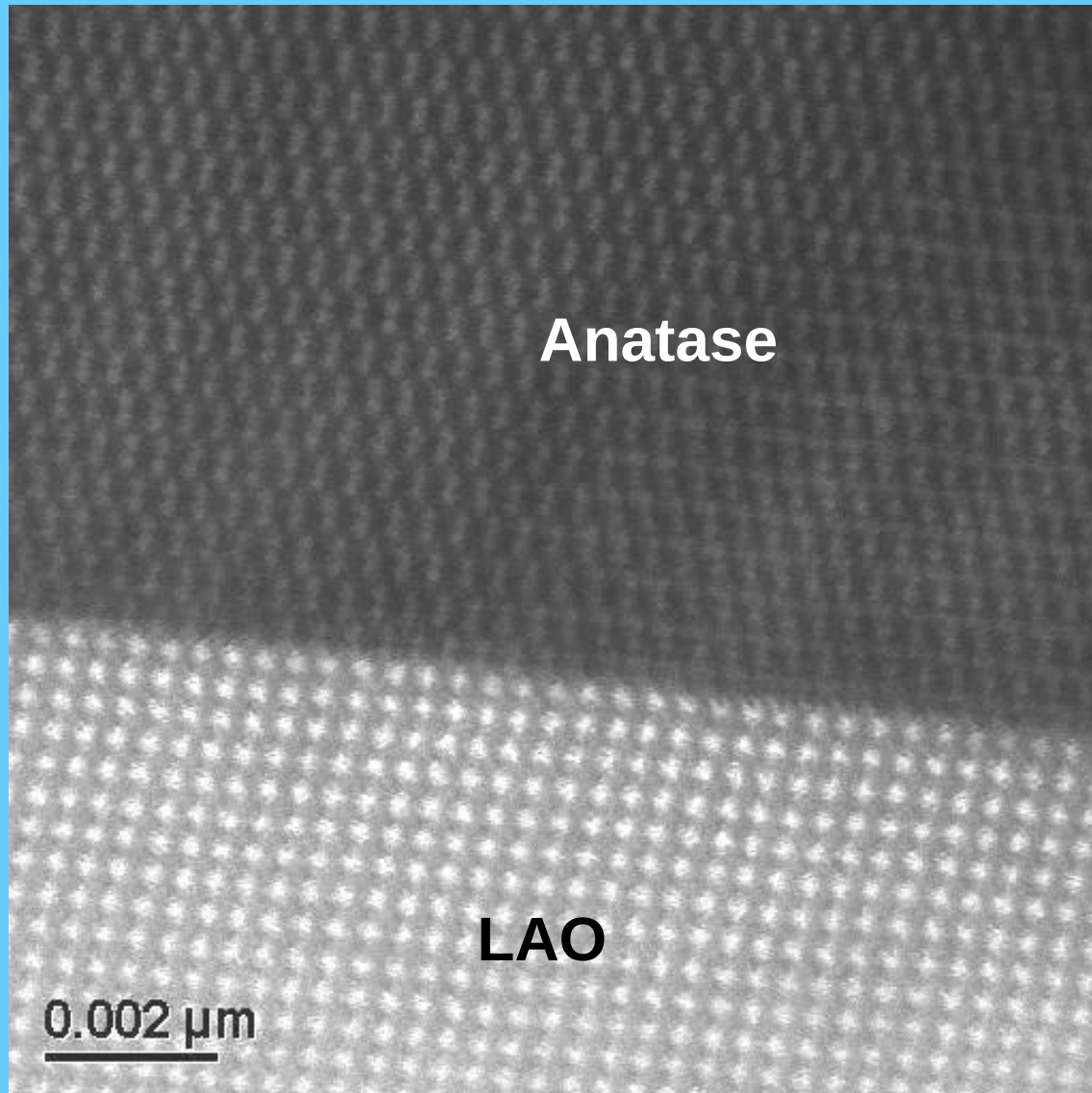


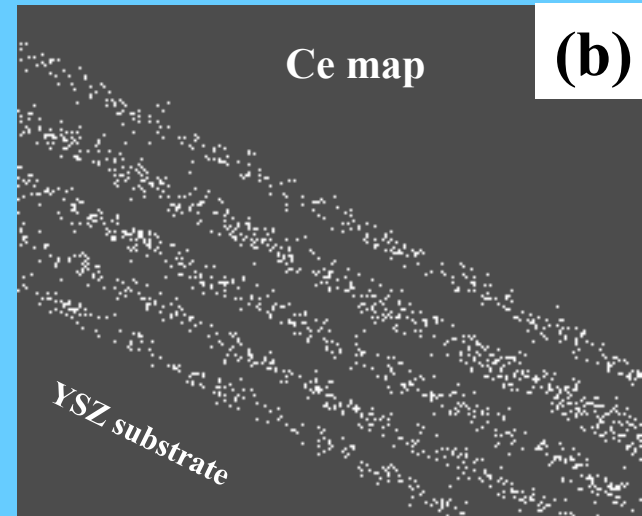
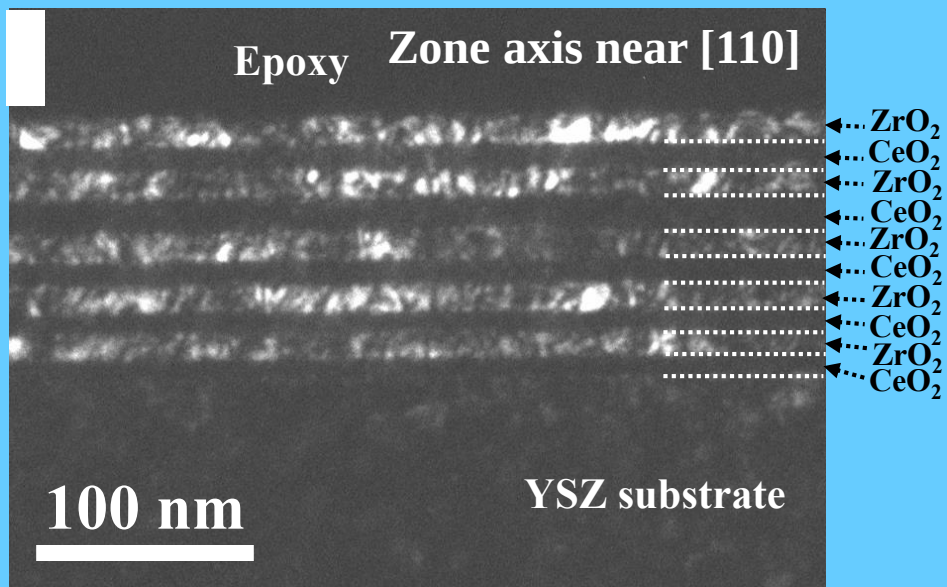
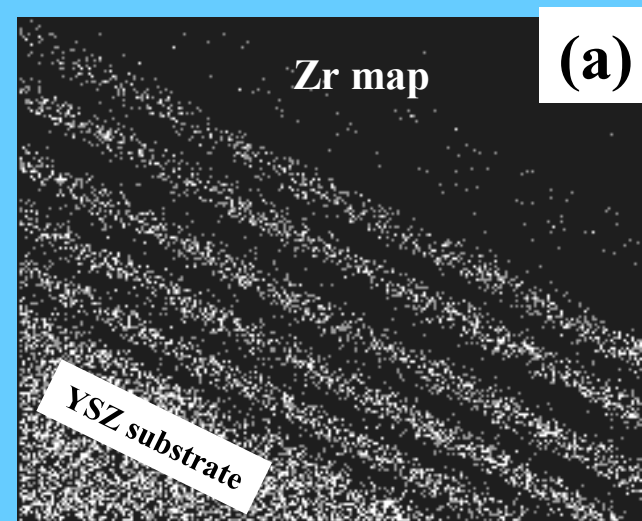
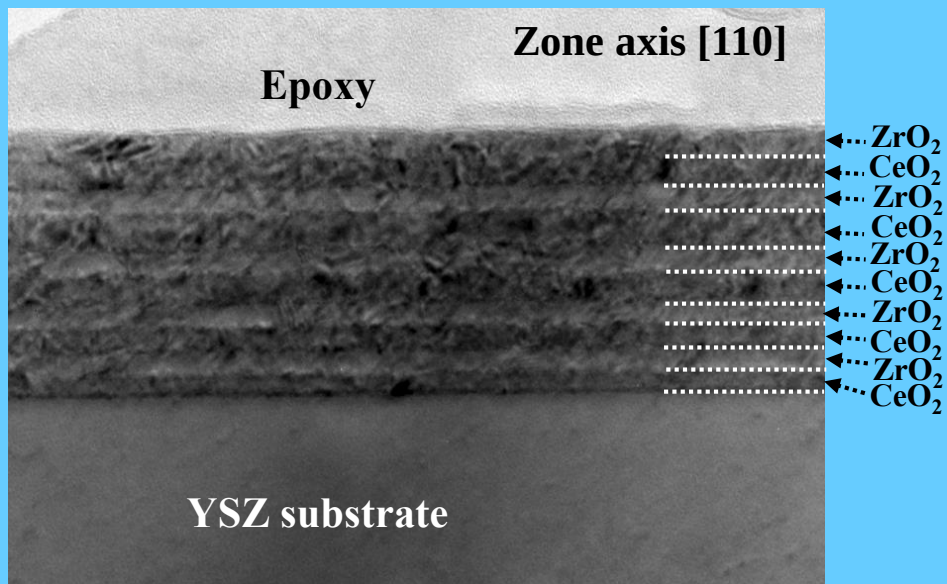
Schematic of the HAADF detector set-up for Z-contrast imaging

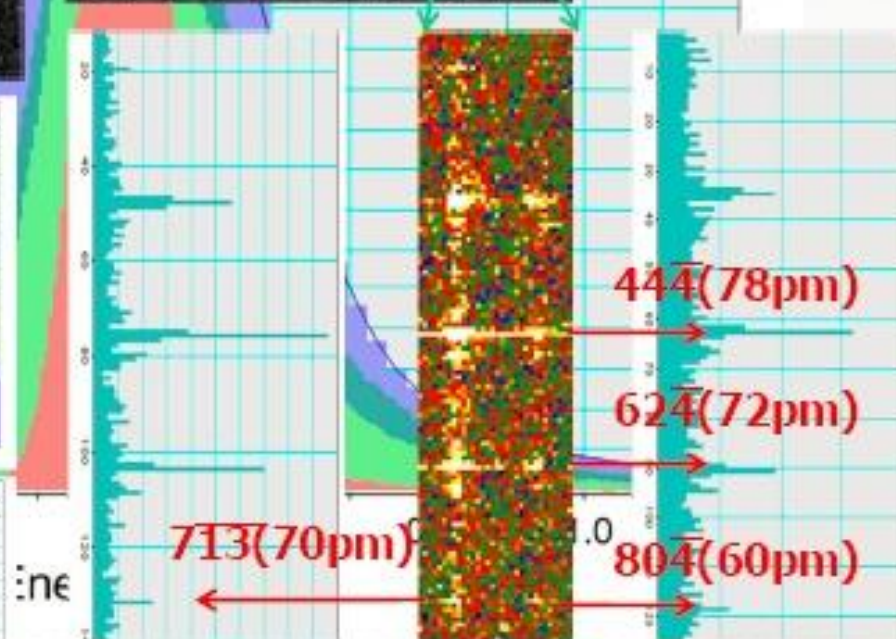
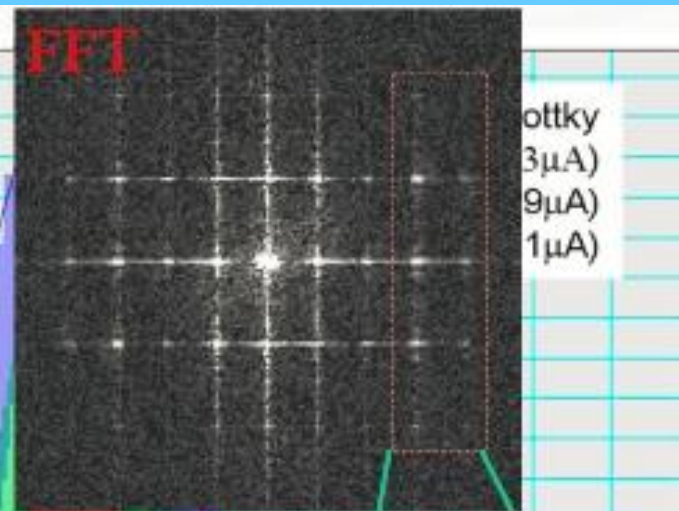
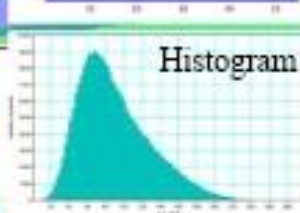
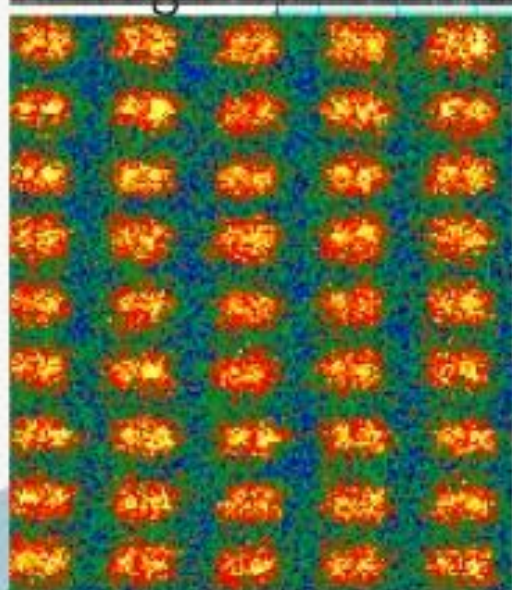
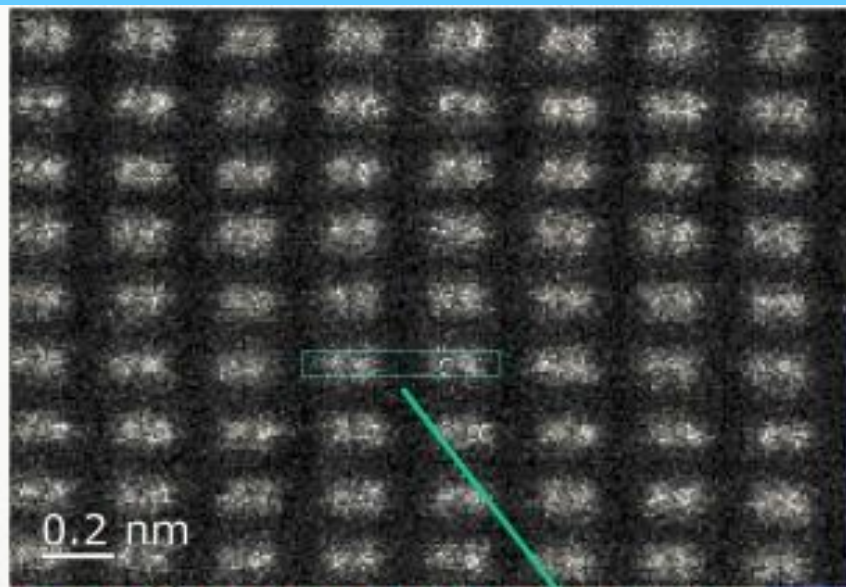
STEM high angle annular dark field image



STEM high angle annular dark field image







Intensity profile from FFT spot

Incident Probe

STEM

Specimen

Annular
Detector

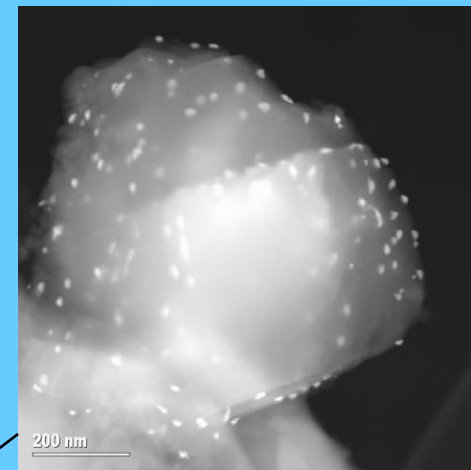
Z-contrast
Image

**EELS
Spectrometer**

EELS

**Z Contrast Imaging
and Analysis at
Atomic Resolution**

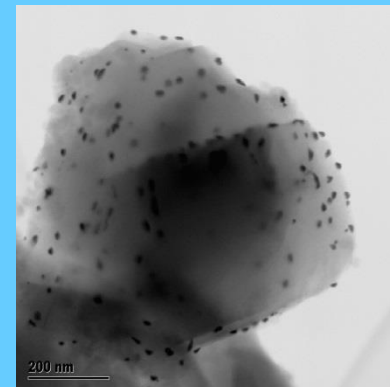
Bright-Field
Image



HAADF

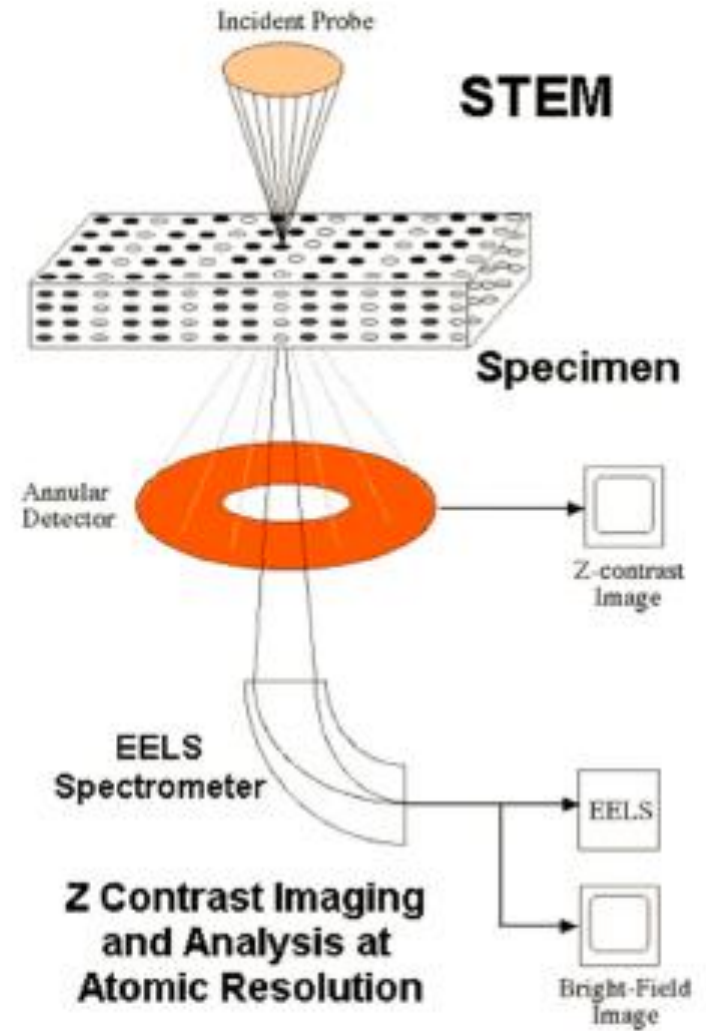
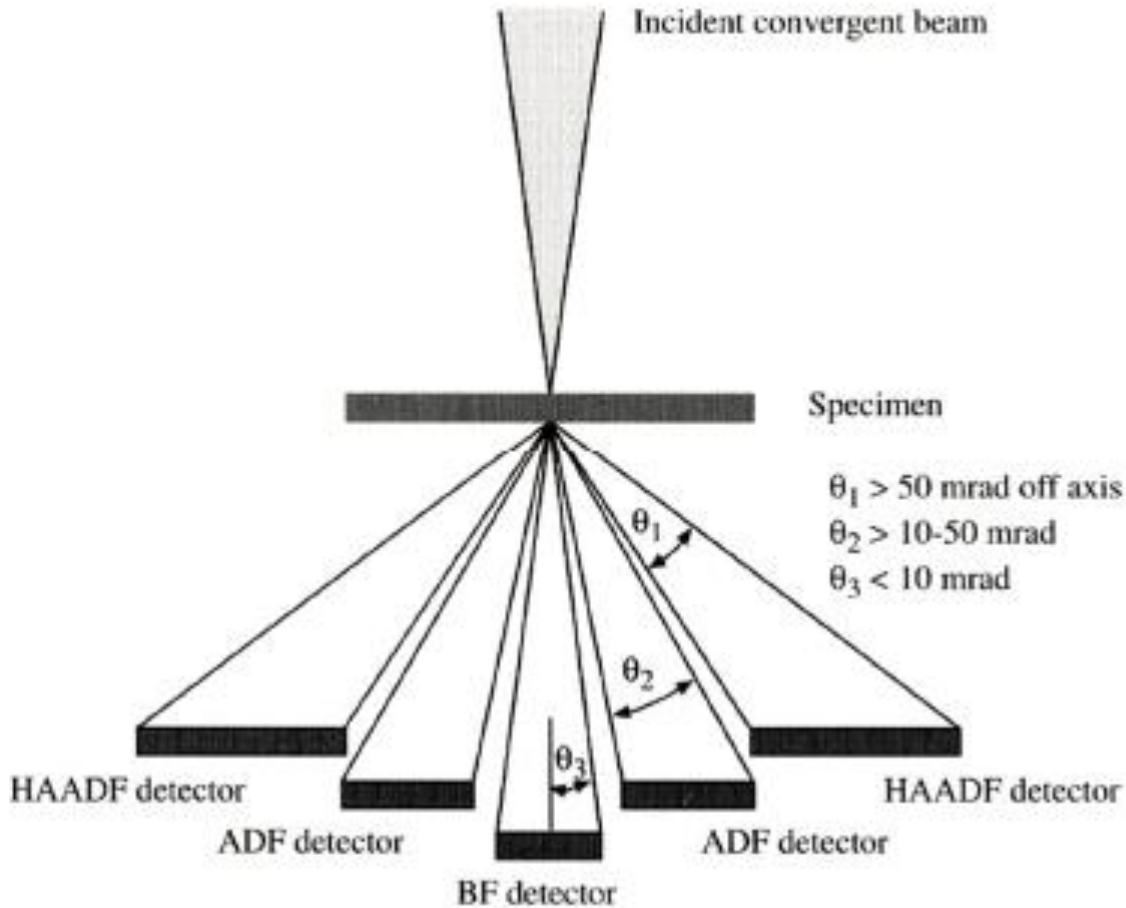


EELS



BF

Detectors

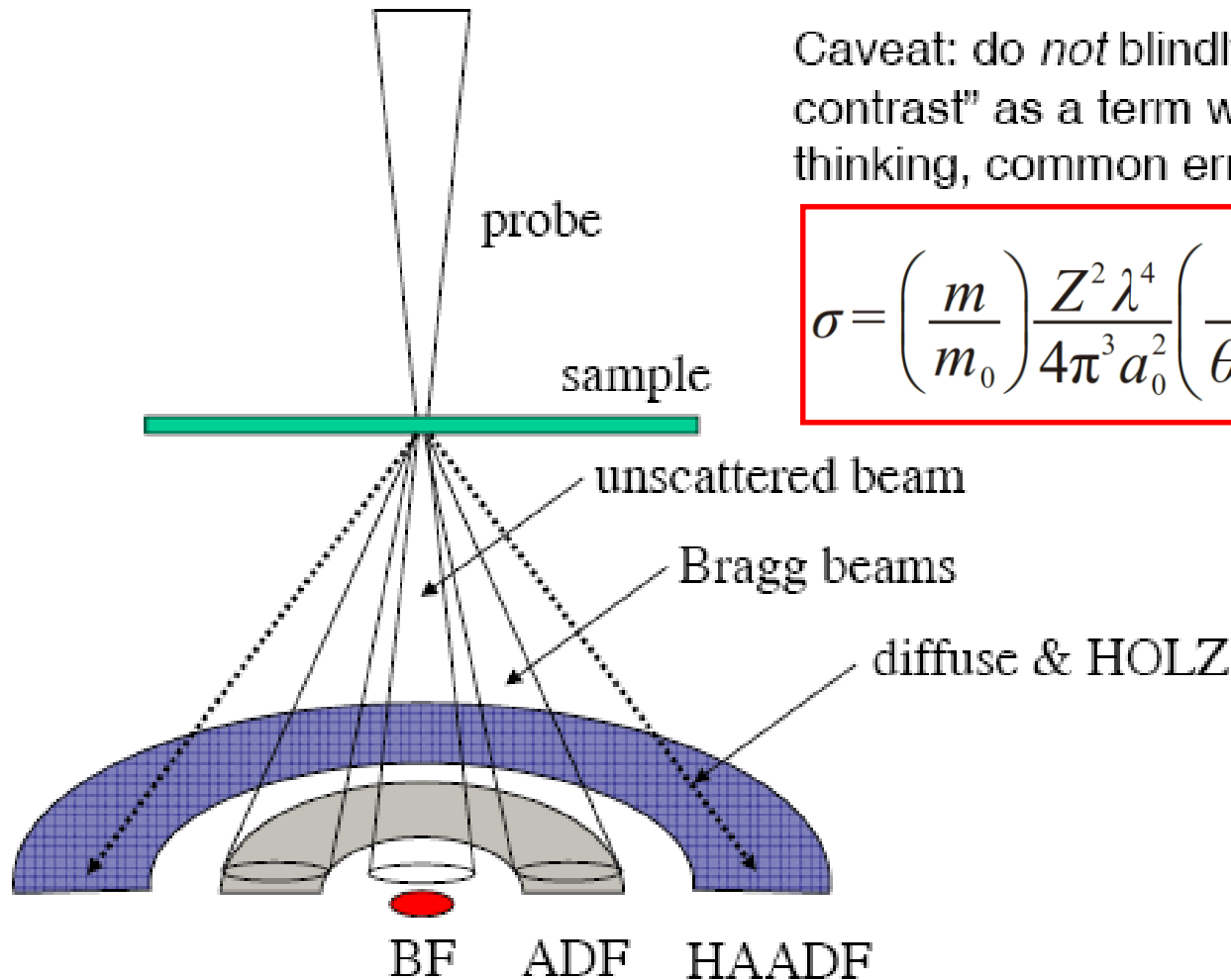


Bright Field & Dark Field

Pick different parts of scattering space with different detectors.

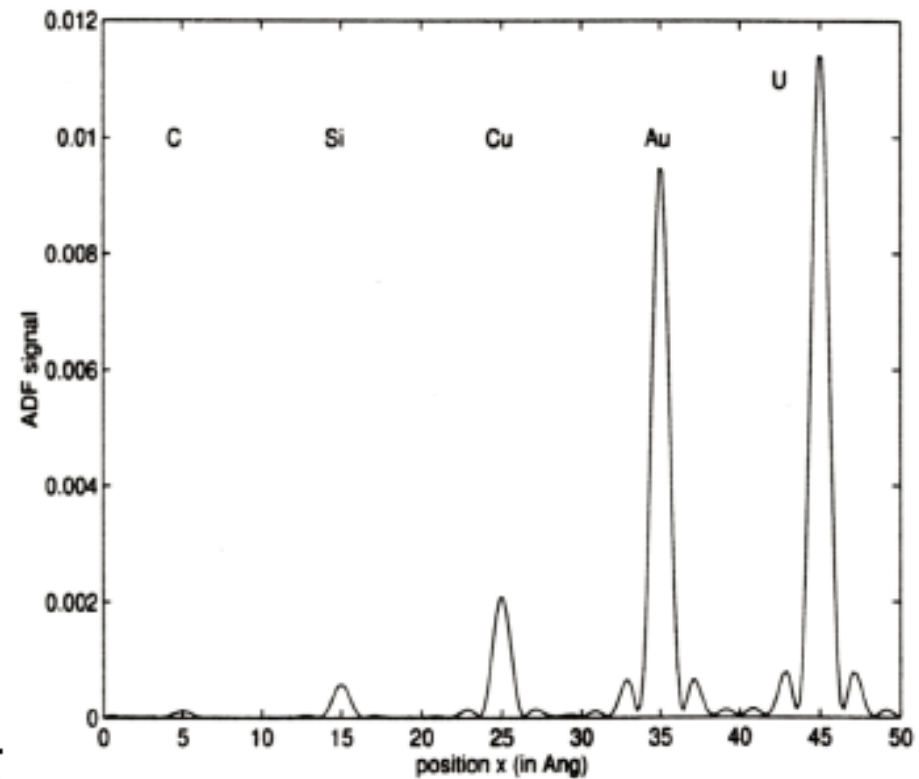
Caveat: do *not* blindly use “Z-contrast” as a term without thinking, common error

$$\sigma = \left(\frac{m}{m_0} \right) \frac{Z^2 \lambda^4}{4\pi^3 a_0^2} \left(\frac{1}{\theta_1^2 + \theta_0^2} - \frac{1}{\theta_2^2 + \theta_0^2} \right)$$



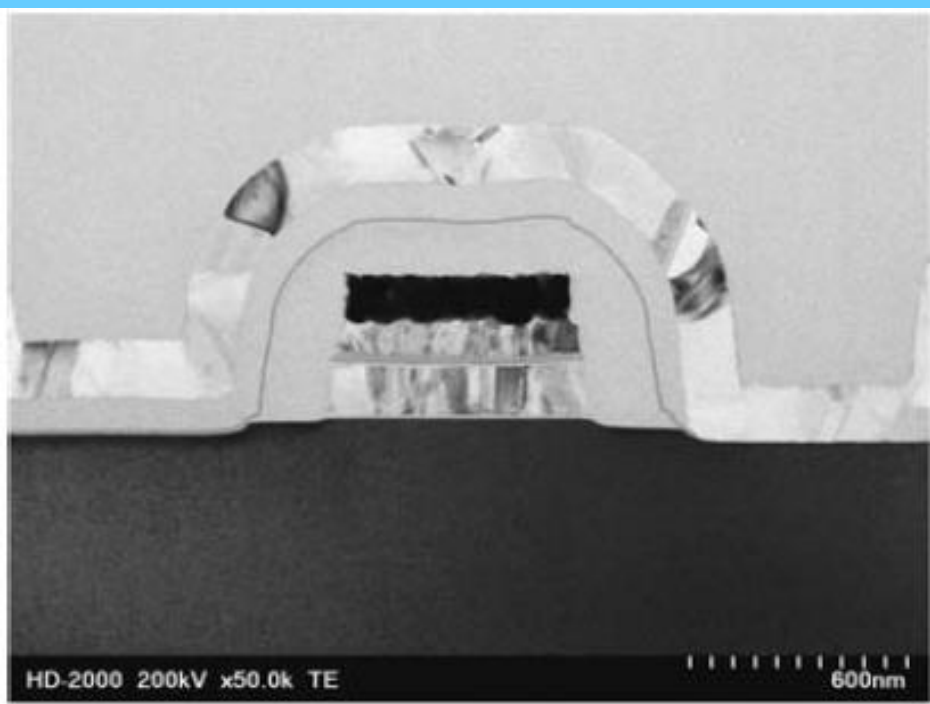
Z-contrast

- Scattering scales as $\sim Z^{1.7}$ for common scattering angles
- Images are readily interpretable
- Images contain some chemical information
- Many problems with this simple picture (good enough for Govt work)

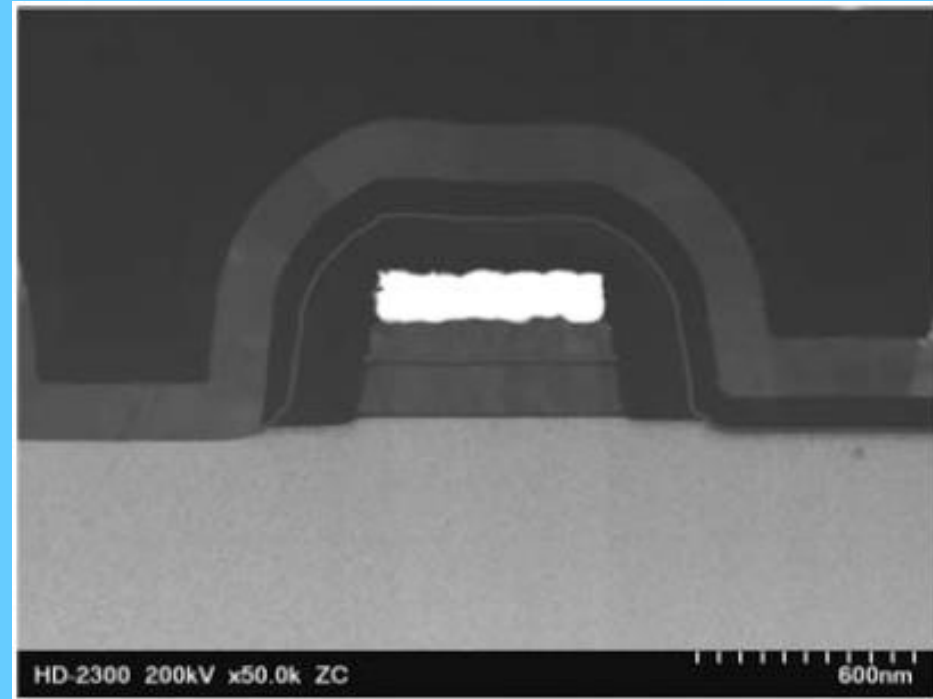


B. J. Kirkland, *Advanced Computing in Electron Microscopy* (Plenum, 1998).

Examples

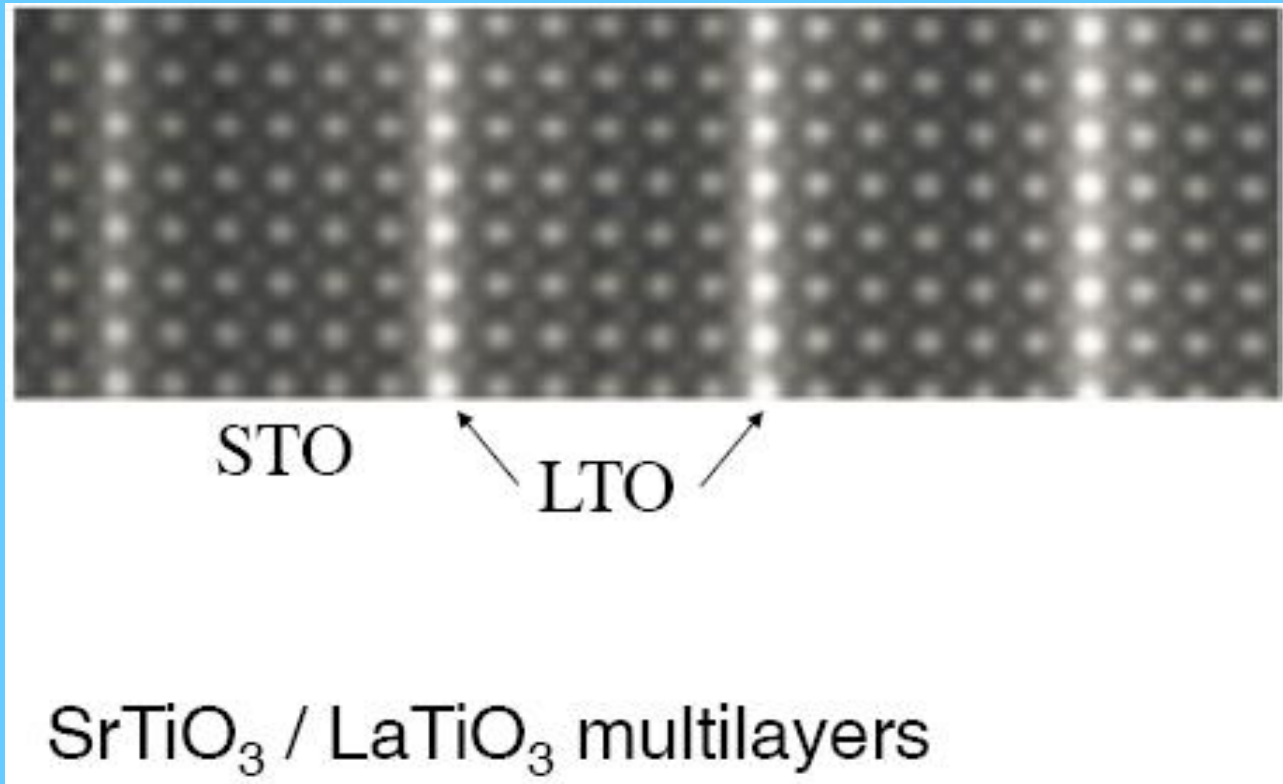


Phase contrast image



Z contrast image

Examples

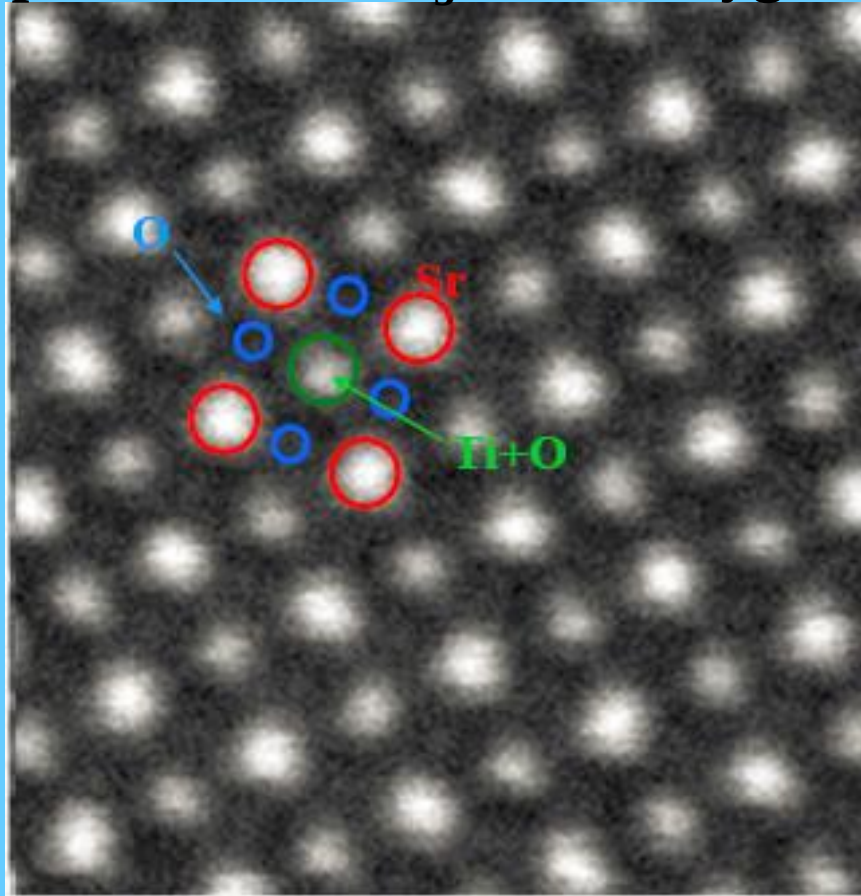


A. Ohtomo Nature 419, 378 (2002)

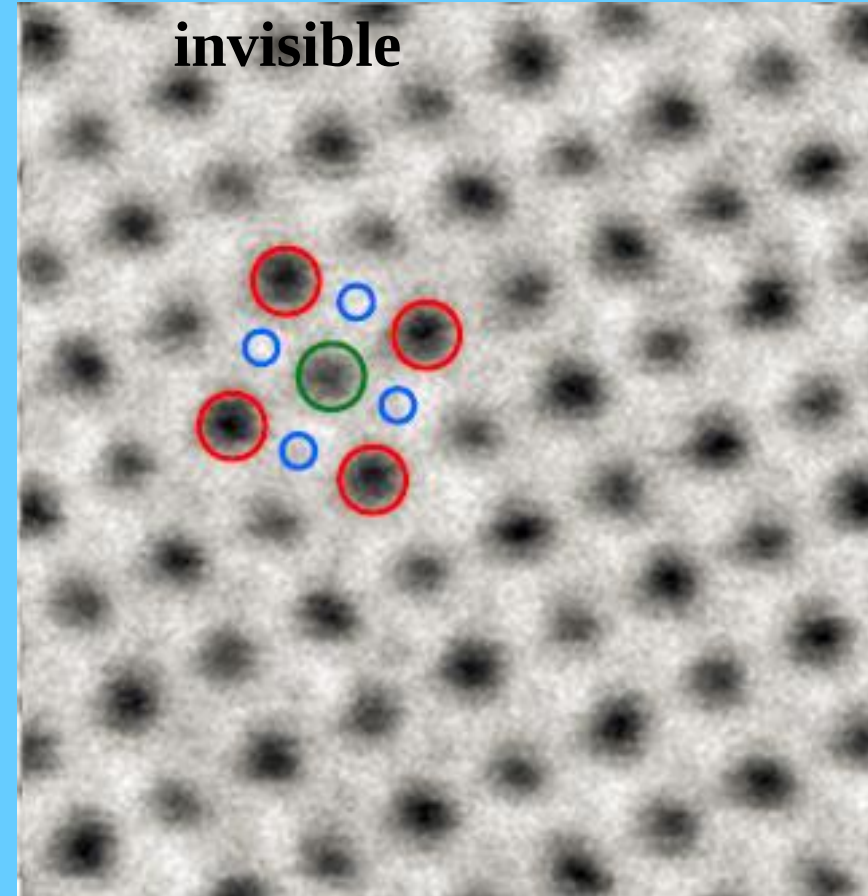
STEM BF image by large angle collection

Specimen: SrTiO_3

Oxygen contrast ; Sometime visible, sometime invisible



HAADF image
(collection semi angle 90-
170mrad)

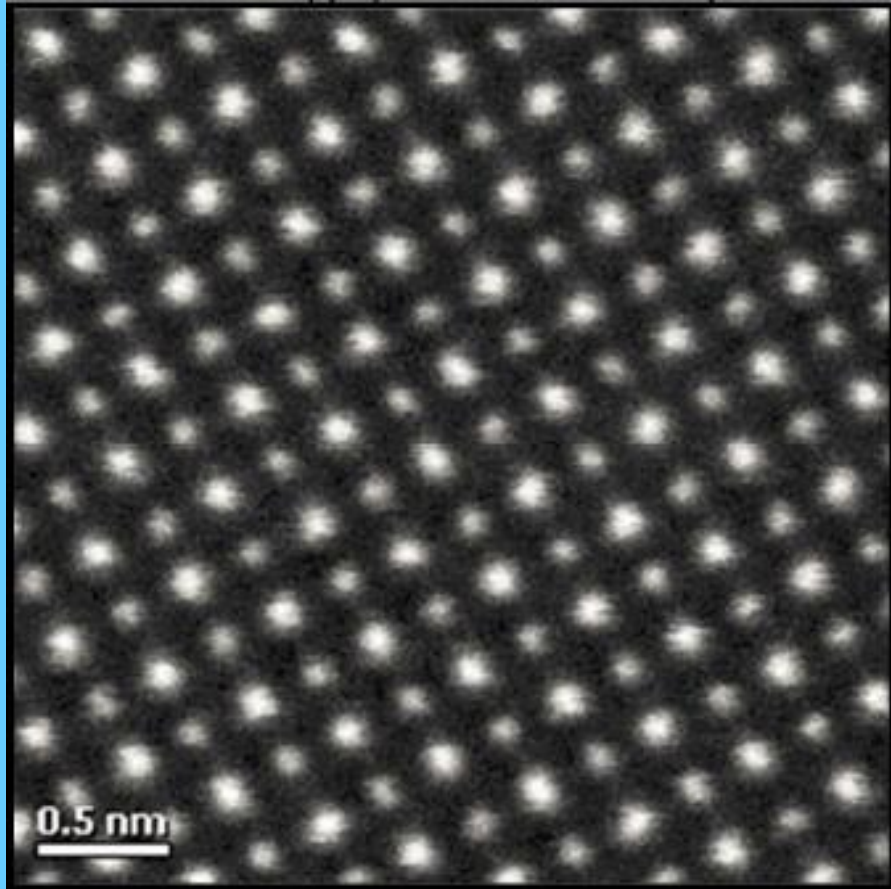


BF image
(collection semi angle 0-
22mrad)

Visualization of O site in SrTiO₃ by ABF microscopy

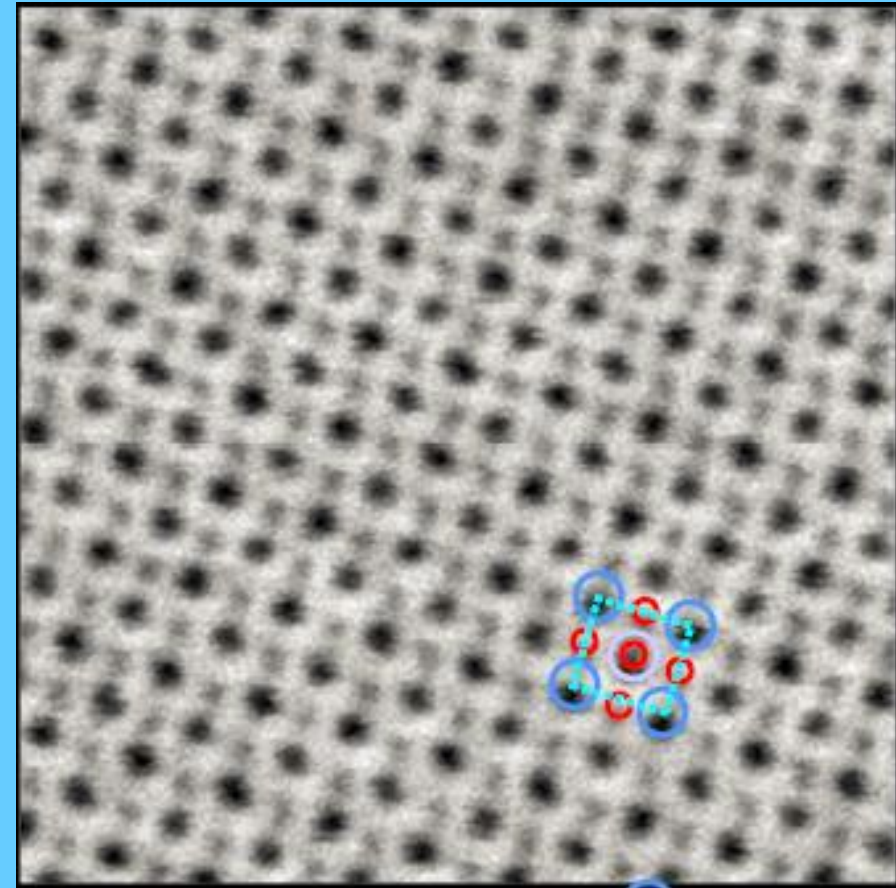
Specimen: SrTiO₃

Oxygen site was visible very clearly



HAADF image(90-170mrad)

$$\sigma \propto Z^2$$

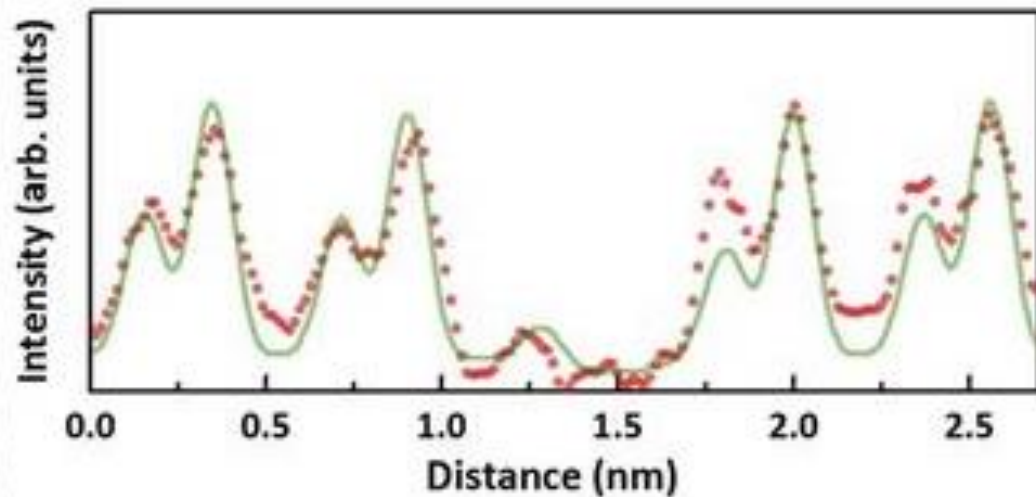
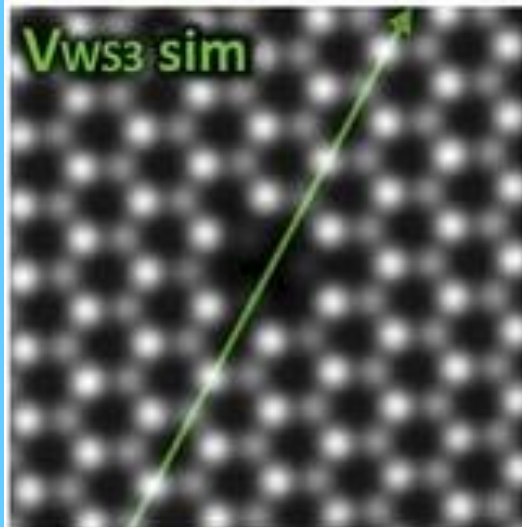
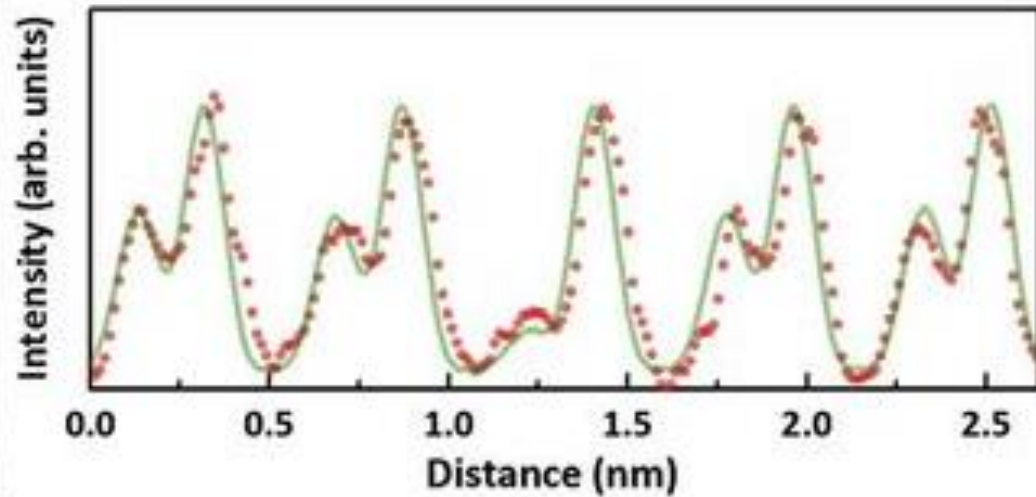
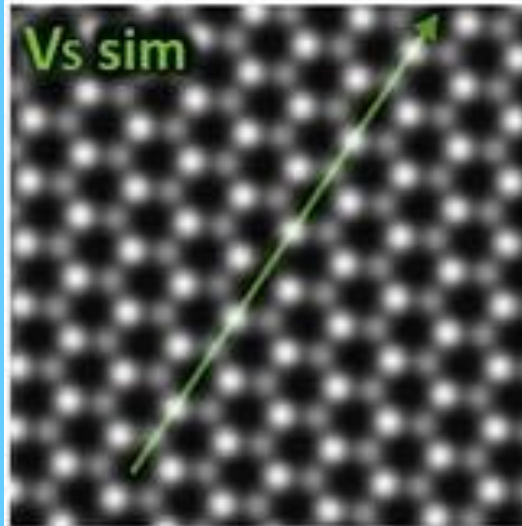


ABF image(11-22mrad)

Oxygen contrast become dark

$$\sigma \propto Z^{1/3}$$

STEM: atomic resolution



- STEM = TEM, but with the optics reversed (if you ignore inelastic scattering)
- BF, DF, HREM can be done in a STEM almost the same as a TEM
- However:
 - The imaging is serial as against parallel, so in general has more noise and lower signals
 - Scan coils and electronics involved, which leads to noise
 - Some types of imaging can be done easier in a STEM than a TEM (they have all been done in both)
 - STEM needs a small source (demagnified further)
 - Some manufacturers have decided to reinvent the wheel

Summary HRTEM versus STEM Imaging

- HRTEM
 - Planer illumination
 - Multi-beam scattering
 - Image contrast
 - Thickness
 - defocus
- Z-contrast
 - Scans a fine probe
 - Electrons are scattered to an annular detector
 - Strength of the scattering $\sim Z$

